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Leveraging EU Funds for Productivity and Growth

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WEIGHTS AND MEASURES

Metric System

ACRONYMS AND ABBREVIATIONS

ATS	Advance Technological Skills	GDP	Gross Domestic Product
BGN	Bulgarian Denominated Lev	GNI	Gross National Income
CAP	Common Agricultural Policy of the EU	GVA	Gross Value Added
CEE	Central and Eastern Europe	ICT	Information Communications Technology
CEEC	Central and Eastern European Countries	IMF	International Monetary Fund
CMO	Common Market Organizations	ISPA	Instrument for Structural Programs for Pre-Accession
CNDP	Complementary National Direct Payments ("top-ups")	MRDPW	Ministry of Regional Development and Public Works
CPI	Consumer Price Index	MTFF	Medium-Term Fiscal Framework
CVT	Continuing Vocational Training	NAVET	National Agencies for Vocational Education and Training
EAGGF	European Agricultural Guidance and Guarantee Fund	NEAA	National Evaluation & Accreditation Agency
EARDF	European Agricultural and Rural Development Fund	NDP	National Development Plan
ECMT	European Confer. of Min. of Transport	NMS	New Member States of the EU
EBRD	European Bank for Reconstruction and Development	PHARE	Poland, Hungary Assistance for the Reconstruction of the Economy
ERDF	European Regional Development Fund	PISA	Program for International Student Assessment
EC	European Commission	PPP	Public Private Partnerships
EIB	European Investment Bank	REA	Road Executive Agency
CNDP	Complementary National Direct Payments ("top-ups")	SAPARD	Special Pre-Accession Program for Agriculture and Rural Development
ERA	European Research Area	SAPS	Single Area Payment Scheme
EUA	European Universities Association	SFA	State Fund Agriculture
EU	European Union	SFPS	Single Farm Payment Scheme
EU-15	The 15 Members of the EU	SME	Small & Medium Scale Enterprise
EUR	Euro (Currency)	TIMSS	Third International Study of Mathematics and Science
FAO	Food and Agriculture Organization	TOT	Training of Trainers
GAO	Gross Agriculture Output	UAA	Utilized Agricultural Area

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EXECUTIVE SUMMARY

1. Bulgaria is facing a dual challenge in the next few years: the first one is to raise employment and productivity to narrow the income gap and facilitate the convergence with the EU. The second one is to program the forthcoming EU grant funds in a manner that promotes income convergence and ensures continued macroeconomic stability. To facilitate income convergence with the EU, efforts to increase employment and productivity are particularly pressing for Bulgaria, because the country has a declining working age population. At the same time, the country has the second lowest labor market participation and the lowest employment rate among Central and Eastern European (CEE) countries, with a large share of employment in unproductive segments of the economy, or in areas where labor productivity is stagnant. The forthcoming EU structural funds, which are designed to help Bulgaria achieve income convergence, provide an unprecedented opportunity for the country to catch up with the EU standards of living, if used wisely. However, given the magnitude of expected resource flows, using them wisely is also a challenge.

2. This report explores the options available for the Bulgarian government to use optimally both the EU grant and Bulgaria's own funds in the pursuit of enhanced economic growth and development. The analysis has selectively focused on four areas—enhancing the skills of the labor force; agriculture; roads; and environmental protection. These areas were selected because they will either receive the bulk of expected EU funds or if used wisely, are most likely maximize the longer term positive impact on productivity and growth. In each area, the report has explored the priorities for funding, and implications of the forthcoming resource flows. In addition, estimates of EU transfers in the first three years of accession have been prepared, both in terms of net transfers to the country (net impact on the balance payments), and the magnitude of funds expected to flow annually through the government budget.

I. FINANCIAL AND BUDGETARY IMPACT OF PROJECTED EU FUNDS:

3. The *net overall financial effect* of expected EU grant funds is projected to be highly positive on Bulgaria, about 3.7 percent of annual GDP on average during 2007-2009, if the country takes all available funds. Experience from EU15 countries indicates that, if effectively used, the EU funds can contribute to economic growth both in the short and long run. In the short run they would create increased domestic demand. In the long term, if invested in the improvement of the physical and human capital with high economic returns, they have the potential of making an important contribution to economic growth through the supply side. However, experience also shows that, if badly used, these resources may have only a short-term positive demand effect on the economy, and in the long term lead to real appreciation of the local currency and reduced competitiveness, aid dependency, and undermine the efforts to improve governance. Funds have proven to be most effective, when put to use under broad and well-defined strategic programs.

4. Beyond the net overall financial effect, the report shows that Bulgaria's *fiscal position* would deteriorate by about two percent of GDP on average during 2007-2009, if the government absorbed all the available funds and there was no restructuring of the fiscal framework. The analysis indicates that the deterioration of the national budget balance would stem from increased expenditures. Taking into account the already high level of public spending, it would be essential to improve the ability of the government to substitute existing expenditures with EU funds--that is, to reorganize the budget in the most efficient way to maximize the absorption of EU funds. The need for restructuring is even more

acute in the light of expanding social spending, related to the rapidly aging population, and high unemployment.

5. The analysis suggests that there is potentially ***ample room in the Bulgarian budget to absorb the expected sharp increase in transfers from the EU by consolidating and increasing efficiency of existing expenditures***, without adversely affecting economic growth. Since the government would like to cap the level of public expenditures as a share of GDP at 40 percent, adding EU funds to the budget without any expenditure restructuring is not an option. That leaves the government essentially with two choices in terms of accommodating the expected EU funds. The first choice is to accommodate them through full fiscal adjustment: through fiscal consolidation equivalent to the projected budgetary impact of EU funds—that is, about two percentage point of GDP. The numbers suggest that the first choice could be achieved solely by rationalizing spending on military, security, health, government employment, subsidies, and social benefits. The second choice is to accommodate the EU funds through a mixture of fiscal adjustment and economic growth. However, from the point of view of increasing the rate of growth, accommodating the funds fully through fiscal adjustment would be preferable in order to make room for, say, further reductions in payroll taxes. Also, the second option would be risky to follow since the government cannot control the rate of growth of GDP.

II. IMPACT AND UTILIZATION OF PROJECTED EU FUNDS BY SECTORS:

6. As mentioned earlier, a large share of the forthcoming EU grant funds is expected to go to support education initiatives, upgrading of roads, implementation of the European common agricultural policy, and environmental standards. In each of these areas, the report analyzed options for utilization of EU funds.

7. **Education:** There is an urgent need to modernize the Bulgarian education system to ensure the country has a highly skilled and productive labor force that can make the catching up with the EU standards of living a reality. This is critical since while the level of education is steadily increasing in EU countries, the participation rate in higher education has been declining Bulgaria. The demographic change and declining age cohorts provide an opportunity to improve the efficiency of public education spending, and ensure it is spent effectively on improving the quality of education and training and increase coverage of the shrinking age cohorts of students.

8. The report recommends that the government does not increase public spending on education as a share of GDP, but seeks efficiency gains within the existing resource envelope. This is advisable, since public spending on education, when adjusted for income per capita, is already on a level comparable to many CEE countries, and spending per student has been steadily increasing, but education outcomes remain well below CEE levels. Provided credible reform is taken, maintaining the existing level of public spending on education as a share of GDP (including possible EU funds) would be warranted. While consolidation of spending is recommended in many other sectors, given the importance of education for promotion of labor productivity, a differential treatment is recommended on education, assuming reforms proceed. Reform of the education system would also build a good base for a more productive use of EU grant funds, and make human resource development programs presented to the EU for funding more credible.

9. The reform measures needed would include gradual consolidation of schools, universities, and vocational colleges in light of declining age cohorts to increase student-teacher ratios and release funds for upgrading of curriculum, textbooks, and facilities. Reforms need to be guided by a leadership group that consults regularly, including across ministries (MES, MLSP, MEE), and that unifies the vision across

ministries with periodic Position Papers that set out not only the vision, but progress to date and the way forward. This group should also lead in garnering support for the reforms by involving the teachers as well as the general public. Explaining the potential benefits of such reforms is crucial for successful implementation. The savings from the reforms should be retained within the sector, and used to finance other quality improvements. To raise the quality of education, there is a need to undertake a curriculum reform, develop better teaching materials, improve teacher training, open up university administration to outside representatives, and establish periodic national student assessments to evaluate the impact of education. At the upper secondary and university levels, it is critical to establish links with the labor markets to ensure that their graduates are equipped with skills required in the market. In a global economy, continuous upgrading of skills of the labor force is increasingly important to stay competitive, and would need to be encouraged.

10. Based on the experience of Ireland, the report presents a series of proposals for possible EU funding to help reform the sector in the areas of: (i) improving the quality of education; (ii) modernizing the educational infrastructure; (iii) improving access for all; (iv) reducing inequalities; (v) promoting social inclusion; (vi) building the capacity of personnel; and (vii) promoting lifelong learning.

11. **Agriculture:** As part of preparations for EU accession, Bulgaria has to gradually adopt an agricultural policy and institutional framework fully compatible with the Common Agricultural Policy (CAP) of the EU. In this context, the government will need to make a series of strategic decisions, which involve the selection of a payment scheme for direct income support measures, the potential topping up of area based payments, payment modalities, eligibility criteria, and transfers among modalities. In making these decisions it should not lose sight of efficiency considerations since only higher productivity and stronger competitiveness can in the long run ensure strong sustainable improvements in the wellbeing of the rural sector.

12. *Selection of a Payment Scheme:* Owing to the complexity of implementing a Single Farm Payment Scheme (SFPS), it is recommended that Bulgaria opts for a Single Area Payment Scheme (SAPS), as was done by almost all new EU member states. It would then have a window of three to five years to prepare for a conversion to SFPS, before penalties for non-conversion kick in.

13. *Topping-Up (Complementary National Direct Payments):* Calculations show that the introduction of SAPS payments to agricultural producers, starting at a level of 25 percent of the EU15 average in 2007, would increase average income of agricultural producers by 19.1 percent already in the first year of accession. This suggests that no top-ups are required to maintain and increase average producer incomes. However, the impact would be uneven across the producers of various agricultural products. Income from tobacco production would decline as current direct payments are replaced with those under the CAP. Producers of dairy would also suffer an initial small loss of income, as would some medium to large-size farmers (50-100 hectares) due to crop-specific effects rather than size effects. The latter, however, would only be a short-term impact, and turn into income increases in the first year after accession. It, therefore, does not justify top-ups. Topping up is expensive, competes with other more productive programs, and establishes a sense of entitlement which makes it politically difficult to remove it later. Given the arguments in this report on keeping the size of the state in check and on focusing all new expenditures on improving productivity, topping-up is not the way forward. A different mechanism of support should be sought if it is necessary for political reasons, perhaps best done through the financing potentially available under Pillar 2.

14. *Payment Modalities:* If nevertheless topping-up is chosen, evidence suggests that the application of specific, coupled CNDPs to limited sub-sectors (tobacco) should be favored over the implementation of broad, sector-spanning top-ups. Moreover, such support should have a sunset clause. Also, since support to specific products introduces distortions into production patterns, the form of CNDPs chosen

should be as simple as possible. Support to particular regions or sub-sectors would best be addressed under Pillar 2 as part of general rural development of a specific region. This would also fit better into strategic considerations of internal convergence among Bulgaria's regions, and with the expected strategy for rural development in the EU in general.

15. *Eligibility Criteria:* According to EU rules, the minimum parcel size for which payments may be requested under SAPS is 0.3 ha, and the minimum holding size eligible for support under SAPS is 1.0 ha. Bulgaria can decide to set the minimum size of eligible area per holding at a level higher than 0.3 ha, as long as it does not exceed one hectare. Fixing the eligibility threshold at the 1.0 hectare level would exclude about 76 percent of all farms, but only 6.6 percent of agricultural land in Bulgaria. Some preference was noted in the Ministry of Agriculture for a mixed threshold model of 0.3 ha for intensive farming and one ha for all others. This is possible, although this report has argued for a single eligibility criterion of one hectare. In making this decisions the authorities will have to weigh carefully the benefits of structural change that would be promoted through consolidation of farms into larger more efficient units, and the reduction of costs stemming from not having to administer the SAPS for very small farms versus the adjustment costs for the farmers that would retire.

16. *Transfers among Budgetary Allocations:* If Bulgaria decides to top up, there are two ways to do it. It can top up the direct payments coming from the EU from its own resources, or it can top them up by using up to 20 percent of the monies available under Pillar 2, but it must provide counterpart-financing of these diversions from Pillar 2 funding from its own resources. The Pillar 2 route is obviously fiscally less demanding, but it would take funds away from activities that potentially have a much larger impact on the competitiveness of Bulgaria's agriculture than direct payments to farmers.

17. *Efficiency Considerations:* Direct payment schemes such as SAPS can provide some level of income stability, and help mitigate farmers' exposure to income risk by guaranteeing a minimum income from farming. Direct support schemes, however, cannot substitute for the need to raise the productivity and improve the competitiveness of Bulgarian agriculture. Productivity growth and improved competitiveness remain the only sustainable solution to agricultural income problems. Agricultural support schemes targeting specific sectoral adjustment needs, such as farm consolidation, productivity or competitiveness-enhancing farm-level investments, and diversification to off-farm activities is the way forward. These can be tailored to the specific needs of the different regions, and it can all be done under Pillar 2.

18. In addition, direct income support payments may reduce the incentives for farmers to restructure, consolidate, and modernize. Direct income payments may reduce incentives for consolidation, because they guarantee a minimum income from farming, and therefore reduce the incentive for unprofitable or older farmers to sell or lease their assets to more productive entrepreneurs. SAPS payments and their effects on land prices or land rental rates may also make land consolidation more difficult, in particular in a farming environment in which agricultural credit is not well developed. Evidence from EU member states indicates that support payments granted on a per hectare basis lead to significant increases in prices for buying and especially leasing agricultural land. This will significantly disfavor tenants, slow the consolidation toward more viable farm sizes, and would in effect support land owners rather than farmers. These trade-offs between direct income support payments, and the need for higher productivity and competitiveness point to the need to clearly separate rural social measures (such as measures to reduce social tensions and provide social protection in the rural areas) from the major instruments of agricultural income support.

19. *Road network:* Recent surveys have shown that only one-third of the Bulgarian national road network is in a good condition, and the road capacity around the larger urban centers is not sufficient to satisfy the growing traffic demand. Furthermore, a considerable portion of the core national road network

still needs to be upgraded to European technical and safety standards. Bulgaria has committed to completing the upgrade of its main road network to EU standards by 2014, with a number of roads upgraded by 2008.

20. The Government would need to give high priority to the rehabilitation and maintenance of the national road network, while at the same time continuing to construct select motorways and other high capacity roads. All proposed investments will be subject to the minimum (five percent) economic return criterion of the EU. The maintenance of the existing road assets, including clearing the large backlog of road rehabilitation and repair of critical bridges, could be largely financed from Bulgaria's own resources. These investments generally have high economic returns. Rehabilitation of the core network would also encompass improving the bearing capacity of road pavements together with improvements to road geometry and riding quality in order to align these with EU standards. The measures would be expected to lead to lower transport costs and higher road safety standards in Bulgaria. Financing of motorways, however, would go beyond the currently planned budget envelopes for road transport to the tune of 1-2 percent of GDP per year.

21. The report provides four different scenarios for optimization and financing of investments in additional class I roads and motorways. The scenarios suggest how to maximize the use of available EU Cohesion funds, which could be utilized either for construction of new motorways, or for improvements to Class I roads that form part of the TEN-T corridor. Previous economic analyses have indicated that the Trakia and Struma proposed motorway projects would have the highest economic rate of return among motorway projects. These two motorways are also assessed to have the highest potential for PPP financing. However, since PPP financing would generate long term contingent liabilities for the government if state guarantees are required, some scenarios presented include and some exclude the PPP as an option.

22. **Environment:** Meeting the environmental part of the *acquis communautaire* will increase demands on budgetary resources in Bulgaria over the next decade, and total spending on environmental protection is projected to increase by about 50 percent in 2004-2008. The spending plans for the water and waste sectors are, however, inconsistent with the overall budget envelope for environment, and significantly exceed the expected budget allocations. Also, there are several problems with the details of water and wastewater sector financing plans that would need to be addressed. Perhaps the most serious is the unrealistic nature of assumptions regarding municipal financing. These are predicated on increases in tariffs that are overly optimistic. In case the tariff increases are not realized, programs will either fall behind their schedule, or the state will have to make up the shortfall. This would cause expenditure in this category of the consolidated budget to go over its planned limit.

23. It is also quite likely that the level of funds anticipated from the EU will not be forthcoming in full. Although MOEW has increased its pipeline of prepared projects for such funding, and these should be sufficient to absorb at least the 2007-2008 funds, the pipeline will have to be substantially increased for further expected amounts to be absorbed. As a result, the commitment made by the government to the EU over the medium term may not be realized, and there is a need to prioritize the undertaking of investments.

24. In sum, though undesirable, one can expect some delay in Bulgaria in the implementation of environmental programs agreed with the EU. However, Spain, Portugal, Greece, and the recent new EU member states have also implemented their environmental programs on a delayed schedule, owing to large financing requirements.

1. BULGARIA'S EU ACCESSION AND AVAILABILITY OF EU FINANCING

1.1 Bulgaria is close to joining the European Union. With accession there could be a potentially large net transfer of grant resources from the EU to Bulgaria. The aim of these transfers will be to facilitate Bulgaria's income convergence with the rest of the EU. Given the much lower standard of living in Bulgaria, the EU has committed a considerable amount of assistance to the country after its expected accession in 2007.¹ In 2004, the EU committed EUR 4.6 billion in 2004 prices to Bulgaria for the period 2007-2009. Since then the figure has been increased to reflect increased funding for rural development.

1.2 It is an opportunity that does not come along too often for a country. Hence, it is natural that the Bulgarian government is focusing on how to optimize the use of the projected grant transfers from the EU. The government wishes to do that in the context of optimizing all its public expenditures, since, while the EU grant funds themselves may well be optimally used, they may provide room for other expenditures from Bulgaria's own resources to be allocated sub-optimally. Hence, the problem is one of optimizing the entire public expenditure envelope, regardless of the source of the funds.

1.3 The opportunity of extra funds from the EU comes with a huge challenge: namely, to ensure that the EU grant funds together with own funds are used in a manner that helped Bulgaria converge to the EU standard of living. At roughly 30 percent of the average GDP per capita for EU25 (in PPP terms), Bulgaria's starting position is not enviable. It also faces a demographic challenge. Bulgaria's population is declining at almost one percent per year, its labor participation rate of 51 percent and its employment rate of 46 percent are among the lowest in Europe,² and the productivity of its labor force is low. Analysis in the recent Country Economic Memorandum on Bulgaria (World Bank 2005) shows that assuming a rise of its participation rate from 50 to 70 percent by 2015 and that GDP per capita can be made to grow at 5.5 percent per year for the next twenty years, Bulgaria will reach two thirds of the EU25 average GDP by 2025. The problem is that while there are many countries that manage to grow at those rates for a decade, the number that achieves that for two consecutive decades drops off precipitously, and three decades of such rapid growth in a row is a rare event. In the face of such odds, errors will be costly. Hence the huge challenge.

1.4 The magnitude of the remaining challenge may well sound as a discouraging message after Bulgaria has done so well in the past few years. As Table 1.1 shows, GDP growth has been in the 4-5 percent per year range for the past 7 years. Its pace has moved into the 5-6 percent range in 2004, and seems to have kept up so far during 2005. The unemployment rate has been falling, and as of November 2005 was approaching ten percent, although this should be tempered by the realization that this is in the context of very low labor participation and employment rates. Private investment has been growing strongly, in part driven by foreign direct investment. The budget is in balance, the underlying structural budget is also in a good shape, primary budget is strongly positive, reserves are strong, and public debt has fallen precipitously relative to GDP and is now well within the Maastricht bounds. As a result, Bulgaria's sovereign debt ratings have strengthened to investment grade. Perhaps the one dark cloud on the immediate horizon is the current account deficit, which is currently running at over 14 percent of GDP on an annual basis. This could present a problem, if foreign direct investment inflows were to slow down sharply. For this reason the authorities need to continue to be vigilant to protect the growth momentum that has been achieved through prudent fiscal policy.

¹ Assuming Bulgaria joins the EU in 2007, as is currently planned. For the purposes of the analysis, it is assumed throughout the report that accession takes place in 2007.

² National Statistical Institute, Labor Force Survey for the third quarter of 2005.

1.5 The huge challenge faced by Bulgaria is thus one of sustaining rapid growth for some decades into the future in the face of a difficult demographic, labor participation and labor productivity situation. To get a perspective on the magnitude of the challenge, one need to consider the fact that there were on average about four Bulgarians supported by each employee under a labor contract in the private sector in 2004, or, put another way; 1.37 million employees under contract in the private sector were supporting 7.76 million Bulgarians. The group supported includes all the children, some 740,000 workers in the public sector (of whom 467,000 were employed in the general government), pensioners, and about half of the working age population that is not formally employed for one reason or another. Even adding the employees employed in public enterprises to those in the private sector would not change the picture substantially. This lopsidedness shows how weak the private sector is in Bulgaria. Hence, it is essential that strong measures are taken and sustained to help the employment in the private sector grow, and that the productivity of all workers, private and public, is raised rapidly and continuously for many years to come. The expected EU grant funds, if used wisely, can help the government to do that.

Table 1.1: Bulgaria: Selected Economic Indicators

	1996	1997	1998	1999	2000	2001	2002	2003	2004
	(Percent)								
Real GDP growth	-9.4	-5.6	4.0	2.3	5.4	4.1	4.9	4.5	5.6
CPI inflation (p.a.)	121.6	1058.4	18.7	2.6	10.3	7.4	5.8	2.3	6.1
Lending interest rate, nominal, e.o.p. a/	481.3	13.8	13.4	12.4	12.2	13.2	10.2	9.5	8.7
Deposit interest rate, nominal, e.o.p. b/	211.9	3.0	3.3	3.3	3.3	3.4	3.2	3.2	3.0
Unemployment rate, annual average	11.1	14.0	12.1	13.8	18.1	18.1	17.7	14.3	12.7
	(Percent of GDP)								
Total revenues and grants c/	36.6	35.6	37.7	38.5	38.5	38.6	38.4	40.0	41.2
Total expenditure and net lending c/	51.8	35.9	36.4	38.3	39.1	38.4	39.0	40.0	39.5
Interest payment c/	20.1	8.3	4.3	3.8	4.0	3.7	2.2	2.1	1.8
Primary balance c/	4.9	8.0	5.6	4.0	3.4	3.9	1.6	2.1	3.5
Overall fiscal balance c/	-15.2	-0.3	1.3	0.2	-0.6	0.2	-0.6	0.0	1.7
Public and publicly-guaranteed. debt c/ d/	302.0	105.0	83.0	86.8	77.1	69.9	56.3	48.4	40.9
Gross Domestic Investment	8.1	9.9	16.9	17.9	18.3	20.7	19.8	21.7	23.5
Gross Domestic Savings	13.5	14.5	17.1	12.1	12.9	13.1	13.2	12.0	13.2
External current account balance d/	1.7	10.1	-0.5	-5.0	-5.6	-7.2	-5.3	-9.3	-8.5
Foreign Direct Investment, net d/	1.4	4.9	4.2	6.2	7.9	5.9	5.6	10.4	9.2
Workers' remittances d/	2.0	1.4	1.5	1.9	2.2	3.1	3.2	3.5	4.2
External debt d/	106.2	99.9	83.2	89.2	86.9	78.6	65.1	60.7	63.0
Official reserves (mos. of imports, G&S)	0.9	4.5	5.6	6.3	5.3	5.2	5.6	5.7	6.1
Domestic credit	108.7	20.8	15.6	15.3	17.8	20.2	23.7	29.8	36.2
Broad money	74.8	34.1	29.6	31.7	36.8	41.7	42.9	48.1	53.7
	(annual percentage change)								
Merchandise export growth, real	-5.2	8.9	-8.6	-3.7	20.6	11.5	11.6	11.8	12.0
Merchandise import growth, real	-11.4	2.7	11.1	11.0	19.6	15.3	6.9	18.0	11.9

Sources: NSI, MoF, BNB, Employment Agency, World Bank and IMF databases; Poverty Assessment (1995, 1997, and 2001)

a/ Refers to interest rate on short-term credits in Leva.

b/ Refers to interest rate on time deposits in Leva.

c/ Data according to GFS Manual 2001 with the exception of 1989 (GFS 19'86).

d/ Since 1999 external current account balance, external debt and public and publicly guaranteed debt are calculated from Euro values.

I. OBJECTIVE OF THE REPORT

1.6 This report aims to assist the Bulgarian government in exploring their options for the optimal use of both expected EU grants and Bulgaria's own funds in the pursuit of enhanced economic growth and development. The analysis focuses on the EU commitments to Bulgaria in the first three years of accession, estimates the projected net EU grant flows to the country and the budget, and highlights options to accommodate them into the budget (Chapters 1-2). In addition, the report explores the priorities for funding, and implications of the forthcoming resource flows in four areas—enhancing the skills of the labor force; agriculture; roads; and environmental protection (Chapter 3-6). These areas were selected because they will either receive the bulk of forthcoming EU funds or, if used wisely, are most likely to maximize the longer term positive impact on productivity growth.

II. PROJECTED NET EU GRANT FLOWS TO THE COUNTRY AND BUDGETARY IMPLICATIONS

1.7 Upon accession to the EU, Bulgaria is expected to receive significant grant resources from the EU. A major portion of these resources will come from three funds—European Regional Development Fund, European Social Fund, and Cohesion Fund. Transfers from these funds will need to be channeled in a way that facilitates three Cs – **convergence** (both external and internal), regional **competitiveness** and employment, and European territorial **cooperation**. In addition there will be substantial support to agriculture through market measures, direct payments, and in support of Bulgaria's rural development policy. There will also be EU support for internal policies, restructuring of the nuclear facilities, institution building, and administration. Finally, there will be some initial support to the budget (a sort of advance payment) to smooth out the cash-flow path since it will take a while to scale up the disbursement of the various funds, and since funds for direct payments to farmers are disbursed only retroactively. Box 1.1 summarizes the broad policy objectives supported through the various EU financial instruments.

Box 1.1: Summary of Broad Policy Objectives and EU Grant Support Instruments for Bulgaria

Common Agricultural Policy

- ▶ **Market Measures** include export subsidies and expenditures to stabilize and regulate agricultural markets.
- ▶ **Direct Payments** intend to replace the national system for agricultural subsidies with the aim of improving the competitiveness of farmers.
- ▶ Allocations for **Rural Development** are intended to improve agricultural efficiency, enhance environment in the countryside, promote greater diversification of the rural economy and its quality of life, and support the bottom up local development strategy.

Structural Actions

- ▶ **Structural Funds** aim to reduce disparities in infrastructure and human capital endowment between regions, and improve the environment. As incremental budget resources they are subject to the principle of additionality (that is, they cannot substitute national expenditures).
- ▶ **Cohesion Fund** aims to strengthen the economic and social cohesion at the EU level by financing groups of projects in the fields of environmental infrastructure and trans-European transport networks.

Internal Policies

- ▶ **Existing Policies** resources finance mainly expenditures on education, culture, social affairs, energy, environment and research, related to the implementation of the acquis.
- ▶ **Nuclear Safety** resources aim to assist the country in decommissioning installations at the Kozloduy nuclear power plant and to address its consequences, covering measures for environmental upgrading, modernization

of the conventional energy production, transmission and distribution sectors, and measures to improve energy efficiency, use of renewable energy sources, and security of energy supply.

- The transition facility for **Institution Building** is intended for the establishment of adequate administrative and judicial systems, and for strengthening the country's ability to implement the Acquis.

Administration

- Administrative expenditures cover only accession-related expenditures, such as increased interpretation and translation needs. No administrative costs are specifically allocated for Bulgaria.

Compensation

- Budgetary compensations are earmarked to avoid worsening the country's fiscal stance.

Summary of EU Financial Instruments

Financial Instrument	Objectives
Regional Development Policy	
European Regional Development Fund (ERDF) European Social Fund (ESF) Cohesion Fund	1. Convergence
European Regional Development (ERDF) European Social Fund (ESF)	2. Regional competitiveness and employment Regional Level National Level (European Employment Strategy)
European Regional Development Fund (ERDF)	European territorial cooperation
Common Agricultural Policy	
The European Agricultural Guidance and Guarantee Fund (EAGGF)	1. Provides direct income support 2. Supports market measures
The European Agricultural Rural Development Fund (EARDF)	1. Improves competitiveness of farming and forestry 2. Provides support for the environment and the countryside 3. Provides diversification of the rural economy and its quality of life 4. Provides use of bottom up local development Strategies

1.8 The projected EU grant funds have two distinct effects: first, there is the overall effect on the balance of payments of the country; and second, there is the effect on the national budget. The distinction between net flows to the country and net flows to the budget is significant. Flows to the country could potentially have an impact on the country akin to those of the so called Dutch disease. Macroeconomic policy must therefore manage them in a way that does not add to inflationary pressures. If they can be channeled to activities that develop the supply side of the economy, they will help speed up convergence of Bulgaria to the EU25 GDP per capita, which is the key objective. Net transfers to the budget are part of these overall net flows to the country. If added to the budget without any expenditure restructuring, they would by definition increase the level of public expenditures and worsen the budget balance by the amount of the net transfer. However, if absorbed in the budget through restructuring of existing expenditures, concerns on macroeconomic policy management would lessen and there might be other potentially beneficial effects, as will be discussed later in the report.

1.9 The calculation of net transfers to the country is straightforward, although with some initial complexities because the direct payments to farmers are made retroactively. By contrast, the estimation

of the budgetary impact on a cash basis has to make allowances for the way the different EU funds are incorporated (or not) into the budget. In the calculation of the net budgetary effect, several considerations were taken into account: (i) Bulgaria's contributions to the EU budget; (ii) co-financing requirements, assuming that all scheduled payments will actually be made; (iii) requirement for pre-financing of the Common Agricultural Policy (CAP)-related direct payments to the agricultural producers³; (iv) the fact that there is no additionality requirement for the Cohesion Fund; and (v) potential savings from agricultural subsidies paid prior to accession, if there is no topping-up of the CAP direct payments. To capture the macroeconomic implications, pre-financing of CAP direct payments to agricultural producers and structural actions are included in the calculation of the net effect on the national budget.⁴ It is worthwhile noting, however, that when determining the formal compliance with the Maastricht budget criteria, pre-financing of CAP direct payments will not be taken into account in the budget deficit calculation, since the ESA-95 accounting standards are on an accrual basis. The impact of this delay on the budget on a cash basis is softened considerably by a budget advance from the EU (reflected in the "compensation" line in Table 1.2 below).

1.10 Further, EU accession has additional direct and indirect effects on the revenue side of the budget, owing to the alignment of the national legislation with the *acquis communautaire* and the second-round positive impacts of EU accession on trade integration, investment and growth. In this report only direct effects on the budget have been estimated and the impact on the budget balance was calculated on a cash basis, assuming that all EU related public expenditures would be recorded in the consolidated fiscal program.

1.11 Table 1.2 below presents the updated appropriations for EU commitments for Bulgaria for the period 2007-2009 in constant 2004 prices. This is the budget envelope that is expected to become available for Bulgaria from the EU. However the amounts actually paid out in a given period of time may be smaller, because projects may extend over several years, and absorption capacity of the country may be limited. In other words actual disbursements (or appropriations for payments) may be much lower due to the time that it takes to prepare, agree to, and implement the various programs. There are also substantial disbursements (not shown in Table 1.2) that will be made from the previously committed pre-accession funds. Finally, Bulgaria will have to pay its dues to the EU budget, which amount to just over one percent of its projected GNI.

³ Expenditures for direct payments charged to the EU budget in year N cover expenditures of the paying agencies between October 16 of year N-1 and October 15 of year N. This means that direct payments for the calendar year N are actually paid in the budget year N+1 and expenditures under this heading will likely have to be pre-financed out of the national budget in the first year of accession (see footnote 5). In addition, due to the phased annual increases of direct payments to the Bulgarian farmers, there will be some additional, but small cash-flow implications for the budget in subsequent years.

⁴ Technically, these funds will not be taken into account in the budget deficit computation, and the budget deficit deterioration shown in Table 2 decreases to an average of 1.3 percent of GDP in 2007-09.

Table 1.2: EU Financial Package for Bulgaria, € millions, 2004 Prices

Appropriations for Commitments	2007	2008	2009	2007-2009
<i>Common Agricultural Policy</i>				
Market measures	132	129	127	388
Direct payments	0	198	233	431
Rural development	231	312	397	940
Total CAP	363	639	757	1759
<i>Structural Actions</i>				
Structural Funds	311	438	577	1326
Cohesion Fund	180	253	334	767
Total Structural Actions ¹⁾	491	691	911	2093
<i>Internal Policies</i>				
Existing policies	88	89	91	268
Nuclear safety	70	70	70	210
Institution building ²⁾	10	7	3	20
Total Internal Policies	168	166	164	498
<i>Administration</i> ³⁾				
<i>Compensation</i>	122	59	59	240
Total Commitment Appropriations	1144	1555	1891	4590

Source: Financial Package for Bulgaria and Romania, indicative allocations 2007-2009 based on Commission Communication of 10 February 2004 and EU Accession Treaty for Bulgaria and Romania of April 2005.

Notes: ¹⁾ The Accession Treaty was negotiated on the basis of the 2000-06 acquis, where rural development is financed under two funding sources: the EAGGF Guarantee Section, and the EAGGF Guidance Section (as part of the Structural Funds). Future rural development allocations for the 2007-13 periods will all be financed from a single fund, the new European Agricultural Fund for Rural Development (EAFRD), but they are derived from the two 2000-06 funding sources. Thus we have subtracted the funds available under the EAGGF Guidance from the Structural Funds and added them to the resources available for Rural development. ²⁾ Institution building appropriations for commitments are calculated on the basis of a population key for Bulgaria and Romania. ³⁾ No administrative costs are specifically allocated for Bulgaria. Therefore, no commitment and payment appropriations under this heading are included.

1.12 Net transfers of EU grant resources to Bulgaria as a country are estimated to be on average 3.7 percent of the projected GDP in 2007-2009 (see Table 1.3). The portion of these transfers that flows to the national budget is estimated to worsen the country's fiscal position by about 1.6 percent of GDP on average during 2007-09, if all available funds are absorbed and the expected EU accession is not accompanied by a restructuring of the fiscal framework. Table 1.4 (on which Tables 1.3) presents the complete picture of expected flows between the EU and Bulgaria (both inflows and outflows) in relation to projected GDP. The net transfer to the country starts at 2.6 percent of GDP and builds up to 4.2 percent of GDP in the course of the first three years of accession. The impact on the budget on a cash basis on the other hand, is front-loaded, with the budget balance projected to deteriorate by 2.0 percent in 2007, ceteris paribus, with the impact diminishing to 1.3 percent of GDP by 2009, despite an increase in the overall flows over time. The need to mobilize national resources is particularly high in the initial year of accession, owing to the required pre-financing of direct payments to farmers from the national budget and the required co-financing.⁵

⁵ Direct payments to the farmers may or may not add to the deterioration of the fiscal balance, depending on whether they are paid out in December 2007 or in early 2008. The payment scheme for these subsidies is as follows: first, agricultural producers apply for such assistance by end-April 2007. Then their applications are processed, and the

Table 1.3: Summary of Inflows and Outflows as % of GDP to Bulgaria and the National Budget, at Current prices, Average for 2007-09	Bulgaria	National Budget
Post-accession appropriations for payments o/w Budgetary compensation	3.6%	0.3%
Pre-accession appropriations for payments	1.3%	
Total inflow of funds	4.9%	0.3%
Own Resources	1.2%	1.2%
Co-financing		0.8%
Pre-financing of direct payments		0.4%
Possible substitution of expenditures under the Cohesion fund		-0.2%
Possible budget savings from agricultural subsidies if there are no top-ups		-0.2%
Total outflow of funds	1.2%	1.9%
Net fiscal effect	3.7%	-1.6%

national authorities make actual payments to the farmers in the period between December 1, 2007 and June 30, 2008. The EU will reimburse the government with a 2.5 month delay. If the direct payment system runs smoothly-- that is, if there are no problems with the eligibility criteria and the applications are good -- payments to the farmers may be made already in December 2007. In that case, the EU will not be able to reimburse the funds before early 2008, which will put pressure on the 2007 national budget on a cash basis. The calculations in this report are based on this scenario in order to follow a more conservative budgeting approach, and to give the Bulgarian government some flexibility for short-term adjustments in fiscal policy.

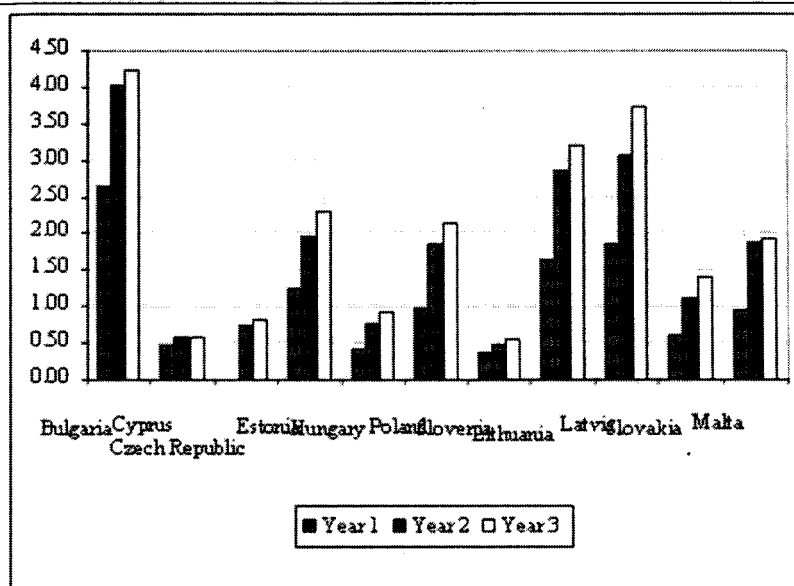
Table 1.4: Summary Table of the Net Impact for Bulgaria and the Net Budgetary Effect of EU Accession

	Disbursements		Budget revenue		Budget expenditure (without co-financing)		Co-financing		Substitution in expenditures		Net impact on the budget	
	2007	2008	2007	2008	2007	2008	2007	2008	2007	2008	2007	2008
Common Agricultural Policy												
Market measures	140	140	140	140	140	140					0	0
Direct payments	0	214	0	214	214	257			40	44	-174	1
Rural development	110	197	110	197	110	197	27	49			-27	-49
Total CAP	250	551	250	551	464	594	27	49	40	44	-202	-49
Structural Actions												
Structural Funds	155	277	155	277	155	277	27	49			-27	-49
Cohesion Fund	6	49	6	49	6	49	1	9	6	49	5	40
Total Regional Policy	161	327	161	327	161	327	28	58	6	49	-23	-9
Internal Policies												
Existing policies	34	58	34	58	34	58					0	0
Nuclear safety	0	76	0	76	0	76					0	0
Institution building	4	5	4	5	4	5					0	0
Total Internal Policies	37	140	37	140	37	140					0	0
Administration												
Compensation	129	64	129	64							129	64
Total	577	1081	577	1081	662	1060	56	107	46	93	-95	7
PHARE	148	120	148	120	148	120	49	40			-49	-40
ISPA	145	132	145	132	145	132	48	44			-48	-44
SAPARD	85	83	85	83	85	83	28	28			-28	-28
Pre-accession aid	379	336	379	336	379	336	126	112			-126	-112
Traditional own resources	24	25	-24	-25							-24	-25
Resource based on VAT	36	34	-36	-34							-36	-34
Resource based on GNI	207	229			207	229					-207	-229
Reserve	1	1			1	1					-1	-1
Great Britain Adjustment	24	31			24	31					-24	-31
Bulgarian contribution to the EU budget	292	319	-60	-59	232	260					-292	-319
Net position with the EU	664	1098										
% GDP	2.6%	4.6%										
Net impact on the budget												
% GDP			896	1358	1273	1656	182	219	46	93	-513	-424
			3.6%	5.0%	5.1%	6.1%	0.7%	0.8%	0.2%	0.3%	-2.0%	-1.6%
Net impact on the budget, excluding pre-accession funds												
% GDP			518	1022	894	1320	182	219	46	93	-513	-424
			2.1%	3.8%	3.6%	4.8%	0.7%	0.8%	0.2%	0.3%	-2.0%	-1.6%
Net impact on the budget, excluding (i) direct payments under CAP, (ii) structural actions and (iii) pre-accession funds												
% GDP			357	481	519	736	182	219	46	93	-299	-381
			1.4%	1.8%	2.1%	2.7%	0.7%	0.8%	0.2%	0.3%	-1.2%	-1.4%

1.13 The projected net flows as summarized in Table 1.3 cover all financial transactions between Bulgaria and the EU, that is, all inflows and outflows, both pre and post-accession, for the country as a whole and through the lens of the budget only, and on an actual disbursement (“payments appropriations”) basis.⁶ The estimates of grant flows are based on the official Financial Package for the Accession Negotiations of Bulgaria and Romania of 2004, and updated with information from the Accession Treaty for Bulgaria and Romania. In addition, the report uses the latest available information and data obtained from the national authorities in Bulgaria and the EU regarding expected disbursements under the pre-accession instruments and projections of Bulgarian GDP and GNI. Finally, the recently adopted changes in the 2007-2013 financial perspective regarding EU funds’ attribution and co-financing rates have also been taken into account.

1.14 The expected net transfer to Bulgaria will be significantly higher than to the states that joined the EU in May 2004. This can be seen from Figure 1.1, which, based on Hallet (2004), compares the net transfers for the first three years of accession to Bulgaria with those to the eight new EU member states. The net transfer to Bulgaria as a percentage of GDP is of the same order of magnitude as that to Latvia and Lithuania. All other new EU member states received smaller transfers. The larger net payments to Bulgaria (as a percentage of GDP) are justified by the low income level compared to EU15 and the new EU member states.

Figure 1.1: Net Transfers of EU Grant Funds to New EU Member States in the First Three Years of Accession, Percent of GDP



Source: Hallet (2004), own estimations

1.15 However, payments under the Structural and Cohesion Funds, and Rural Development as well as the remaining disbursements of the pre-accession aid are conditional on project implementation, and therefore are not automatically transferred to the country. The actual utilization of resources will depend on how the various projects are set up and executed.

⁶ The details underlying Tables 1.2-1.4 are reported in a separate paper prepared for this report (Vincellette and Vassileva, 2005).

III. EFFECTIVE USE OF EU GRANT FUNDS: LESSONS FROM EU COUNTRIES

1.16 Experience of EU15 shows that structural and cohesion funds can play an important role in accelerating the catch-up process (see Meth-Cohn and Shields (2005)), although success depends on a number of factors. Meth-Coh and Shields offer four lessons on how to use the EU funds effectively:

- a. ***Develop a clear strategic vision at all levels.*** In this respect the development of the National Development Plan aims at the alignment of strategies at national, regional, and local levels to ensure that funds have a cumulative and long-term impact on growth, rather than being allocated on one-off projects.
- b. ***Encourage public-private partnerships*** to ensure high-quality and commercially viable projects that at the same time promote public sector goals. Cooperation between the private sector and the public bodies allows the public sector to overcome its budget constraints and focus more on the profitability and viability of the projects. Private companies, at the same time, gain possibilities to share risk and take on more business.
- c. ***Obtain involvement in the projects at the municipal level.*** The EU may expect a simpler, more transparent and decentralized system from 2007 onwards. Higher participation of local governments is considered crucial for management of the projects, and municipalities could get actively involved in seeking private funds to co-finance EU funds.
- d. ***Take into account the inherent inflexibility of EU funds.*** EU regulations and national authorities impose tight restrictions on the use of funds under the structural actions. With the exception of the grants available for small and medium-size enterprises, private companies generally benefit from EU funds only indirectly, for example through public-private partnerships.

1.17 In the context of these four recommendations, Bulgaria faces important challenges related to effective utilization of the EU funds. These are related to the improvement of the budget procedures, restructuring of public spending to accommodate EU-related expenditures, and enhancing of the institutional capacity in the country:

1.18 *First*, at the national level, there is an acute *need for reform of budget procedures* to maximize the absorption capacity for the EU funds. The ongoing introduction of a program budgeting approach is a positive step in this direction. However, significant further efforts would need to be made to make the use of EU funds as efficient as possible.

1.19 *Second*, Table 1.2 indicates that the deterioration of the national budget balance would stem from increased expenditures, rather than from lowered revenues. Taking into account the already high share of the government in the economy (government expenditures-to-GDP ratio amounted to 40 percent in 2004, as will be discussed in Chapter 2), it is essential to *improve the ability of the government to substitute existing expenditures with EU funds*. Fiscal space needs to be made (i) for expanding EU-related capital expenditures (and the national co-financing of these expenditures), (ii) for liquidity reasons (like pre-financing of direct payments to farmers in the first year of accession) and (iii) for the national contribution to the EU budget. The need for restructuring is even more acute in the light of expanding social spending, related to the rapidly aging population, and high unemployment.

1.20 *Third*, actual EU transfers to the national budget depend on the *absorption and institutional capacity*, that is, on the extent to which a member-state is capable of spending the structural fund resources allocated to it in an efficient and effective way. As stated in the UNDP's "Assessment of the

Municipal and District Capacities for Absorption of the EU Structural and Cohesion Funds” (2004), absorption capacity has three main dimensions: macroeconomic situation (the overall level of development), co-financing capability, and administrative capacity. Preliminary assessments reveal an institutional capacity that varies significantly among the Bulgarian line ministries. If the skills of their personnel are not upgraded, the absorption of funds might suffer.

1.21 Unfortunately, since experience of new EU member states with EU funds is so recent, there is not much reliable information regarding problems surrounding absorption and lessons to be learned yet. Funds that flow more or less “automatically”, such as advance payments under the structural and rural development funds, market measures in agriculture and direct payments to farmers, and funds for internal policies, have been transferred according to the expected timetable to the new member states (European Commission 2005b). The judgment as to the efficiency of absorption of structural and cohesion funds, however, is still out. The recent European Commission report (see EC 2005c) is rather hopeful, given that the amount of project applications has exceeded the resources available for programs in new EU member states in 2004, but applications are not the same as successful implementation. The latter will depend greatly on how Bulgaria addresses some of the potential problems with absorption discussed earlier in this section.

1.22 If effectively used, EU funds should contribute to economic growth both in the short and in the long run. In the short run they would create higher domestic demand. In the long term, if invested in the physical and human capital improvements in the country, they have the potential of making an important contribution to economic growth from the supply side. This demand vs. supply side impact distinction may be particularly important in the case of funds for agriculture. Pillar 1 spending in agriculture (market support measures and direct payments to farmers) has its impact mainly on the demand side. Pillar 2 spending in agriculture (rural development) has the potential of making a permanent contribution of strengthening the supply side of the economy.

1.23 EU funds have proven to be *most effective when put to use under broad, well-defined strategic programs*. Perhaps the biggest task facing Bulgaria today is to develop such a program (for example in the form of the National Development Plan), and then back it with real money through a well-thought out Medium Term Fiscal Framework (MTFF) for 2007-09 and the corresponding annual budgets that reflect the MTFF. Moreover, the MTFF would need to focus on the EU funds that could engender growth-oriented activities, while containing and, if possible, reducing the public expenditure envelope relative to GDP over time.

1.24 The next chapter will explore the fiscal challenge Bulgaria is facing, and options to make space in the national budget for the expected transfers from the EU.

2. POLICIES FOR GROWTH AND FISCAL SUSTAINABILITY

2.1 The government has three choices to accommodate the expected EU grant funds. The first one would be to add the funds to the budget without any fiscal adjustment, and let public expenditures increase as a share of GDP. This would initially worsen the budget balance. The second option would be to accommodate them through full fiscal adjustment equivalent to the projected budgetary impact of EU funds—that is, about two percentage points of GDP. The third option would be to accommodate the EU funds over time through a mixture of fiscal adjustment and economic growth.

2.2 Of these options, accommodating the expected EU funds to the budget through full fiscal adjustment would be recommended both from the point of view of increasing growth in the long run and to make space for, say, further reductions in payroll taxes. The first option is ruled out, since the Bulgarian government has expressed a clear intent to keep their public expenditures at their current level of 40 percent of GDP or below. Following the third option would be risky, since GDP growth cannot be controlled by the government, and budget plans should be made on the basis of prudent assumptions about revenue growth.

2.3 This chapter explores the fiscal challenge presented by the second option: whether full fiscal adjustment would be feasible. It also begins to explore options to make space in the national budget for the expected EU funds. The chapter, after presenting the main outlines of Bulgaria's national budget (by sector/function), goes on to analyze what the appropriate size of Bulgaria's government might be given its level of development, and where the savings to reduce the size of government might be found.

I. PUBLIC EXPENDITURES IN BULGARIA

2.4 Table 2.1 presents the medium-term consolidated state budget of Bulgaria by function (sector), highlighting relative magnitudes of public expenditures in the various sectors and providing a framework for adjustments needed to accommodate EU transfers. Figures for 2006 refer to the current year's programmed budget. Figures for 2007-08 are projections, and, as such, they are not forecasts but rather illustrations of what the budget envelopes for the various sectors might be in the next two years, barring major changes. They give a sense of the magnitude of the challenge of fitting in the expected transfers from the EU.

2.5 The challenge for the Government will be to balance the needs in each of the sectors against the intention to keep the overall budget envelope in check when the EU funds arrive. Four sectors that are likely to receive a major share of the expected grants from the EU are human resource development, agriculture, road transport, and the environment. As will be seen later in the report, the recommendation for human resource development will be that the prospective EU funds for human resource development should be fitted inside the current envelope of 4.2 percent of GDP, but in a context of accelerated reforms in the education sector. In road transport, the current envelope of 1.1 percent of GDP (road transport is in Chapter VIII—Economic Activities) would be able to cover an expanded program of maintenance, rehabilitation and limited new construction, but would not be able to pay for new motorways. In agriculture, whose budget envelope is about 1.6 percent of GDP (including State Fund Agriculture and the Tobacco Fund), any effort to top up direct payments from own resources would likely call for an expansion in its budget. Projected needs in environment (water, sewerage and solid waste) are substantial compared to the current budget of about 1.0 percent of GDP (part of Chapter VI in Table 2.1). If it is not possible to fit the EU funds inside these budget envelopes, then expenditure cuts in other sectors would be required. Given all this, it is apparent that some restructuring of the budget will be inevitable.

Table 2.1: Consolidated State Budget by Function (in BGN million)

	2003	2004	2005	2006	2007	2008
BGN mln.	Reported	Reported	Program	Program	Projection	Projection
I. General Government Services	1097.8	1115.2	1232	1546.1	1374.1	1419.9
<i>% of GDP</i>	<i>3.2%</i>	<i>2.9%</i>	<i>3.0%</i>	<i>3.4%</i>	<i>2.8%</i>	<i>2.7%</i>
II. Defense and Security	1787.7	1946.4	2086.8	2345.2	2440	2667.7
<i>% of GDP</i>	<i>5.2%</i>	<i>5.1%</i>	<i>5.0%</i>	<i>5.1%</i>	<i>5.0%</i>	<i>5.0%</i>
III. Education	1504.7	1652.4	1798.9	1899.6	2094.3	2272
<i>% of GDP</i>	<i>4.4%</i>	<i>4.3%</i>	<i>4.4%</i>	<i>4.2%</i>	<i>4.3%</i>	<i>4.3%</i>
IV. Health Care	1697.7	1769.1	1777.7	1997.4	2109.5	2289.1
<i>% of GDP</i>	<i>4.9%</i>	<i>4.7%</i>	<i>4.3%</i>	<i>4.4%</i>	<i>4.3%</i>	<i>4.3%</i>
V. Social Security	4805.2	5238.4	5596.5	6169.3	6431	6766.2
<i>% of GDP</i>	<i>13.9%</i>	<i>13.8%</i>	<i>13.5%</i>	<i>13.5%</i>	<i>13.2%</i>	<i>12.8%</i>
VI. Housing, Construction, Public Works, Public Utilities and Protection of the Environment	497.3	586.3	946.8	981.0	1106	1209.8
<i>% of GDP</i>	<i>1.4%</i>	<i>1.5%</i>	<i>2.3%</i>	<i>2.2%</i>	<i>2.3%</i>	<i>2.3%</i>
VII. Recreation, Culture, Religious Activities	286.1	303.7	286.1	330.7	318.3	335.6
<i>% of GDP</i>	<i>0.8%</i>	<i>0.8%</i>	<i>0.7%</i>	<i>0.7%</i>	<i>0.7%</i>	<i>0.6%</i>
VIII. Economic Activities and Services	1621.8	1838.5	1786.4	2100.6	2032.3	2291.8
<i>% of GDP</i>	<i>4.7%</i>	<i>4.8%</i>	<i>4.3%</i>	<i>4.6%</i>	<i>4.2%</i>	<i>4.3%</i>
IX. Expenditures Unclassified Within the Other Functions	770.1	748.8	1314.3	1100.5	1212.1	1179.2
<i>% of GDP</i>	<i>2.2%</i>	<i>2.0%</i>	<i>3.2%</i>	<i>2.4%</i>	<i>2.5%</i>	<i>2.2%</i>
TOTAL EXPENDITURES	14068.5	15198.7	16461.5	182588.0	18853.4	20158.6
<i>% of GDP</i>	<i>40.7%</i>	<i>40.0%</i>	<i>39.8%</i>	<i>40.0%</i>	<i>38.7%</i>	<i>38.0%</i>

Source: Government of Bulgaria (May 2005), Budget Projections for the period 2007-08; and Budget for 2006

II. HOW LARGE A GOVERNMENT?

2.6 When viewed through the lens of “dependency” ratios mentioned in Chapter 1--each employee in the formal private sector supporting at least three other Bulgarians--it becomes clear why the public sector needs to be made more efficient and probably smaller relative to GDP than it is now. A restrained public sector would reduce the burden on the private sector and thus help make it stronger. At the moment, public expenditures in Bulgaria are about 40 percent of GDP. It is a level that the Bulgarian authorities would like to sustain and preferably reduce, while keeping the budget in balance. This would allow them to retain some fiscal flexibility in the face of potential negative shocks, the most threatening of which in the short run may well come from the side of the current account, although pressures from politically powerful sectors for increased expenditures could also play their part.

2.7 Public spending at 40 percent of GDP implies a heavy tax burden on households and private business, as high taxes and contributions are practically unavoidable at such a spending level. This is a concern because high taxes generally distort economic incentives and transfer resources from the private

sector to the public sector, which is typically less efficient, and thereby reduce the growth potential of a country.⁷ While some studies of relatively advanced economies have found no impact of taxation on growth, most find a negative effect and no study has shown positive growth effects of high taxation.⁸

2.8 Not all taxes have equally distortionary effects, however. High marginal tax rates on income and profit are more distortionary than proportional taxes on consumption such as the value-added tax. For this reason tax reforms often aim to increase economic efficiency by broadening the tax base, while lowering marginal rates on consumption and saving, or by replacing income taxes with proportional sales taxes.

Table 2.2: Social Security Contribution Rates in Europe

	Old Age, Disability, Survivors			All Social Sec. Programs		
	Insured Person	Employer	Total	Insured Person	Employer	Total
Albania	9.50	29.90	39.40	9.50	39.50	49.00
Bulgaria	21.75	7.25	29.00	25.00	17.70	42.70
Croatia	20.00	0.00	20.00	20.00	17.20	37.20
Czech Rep.	6.50	21.50	28.00	12.50	37.00	49.50
Estonia	2.00	33.00	35.00	2.00	33.00	35.00
Hungary	8.50	18.00	26.50	13.50	32.00	45.50
Latvia	2.00	18.00	20.00	..	..	24.32
Lithuania	2.50	23.40	25.90	3.00	28.00	31.00
Poland	16.26	16.26	32.52	26.96	19.68	46.64
Romania	11.66	23.34	35.00	19.66	35.34	55.00
Slovakia	7.00	19.00	26.00	9.40	25.60	35.00
Slovenia	15.50	8.85	24.35	22.10	15.90	38.00
Austria	10.25	12.55	22.80	17.15	24.95	42.10
Belgium	7.50	8.86	16.36	13.07	17.92	30.99
Finland	4.60	22.75	27.35	6.10	25.36	31.46
France	6.65	9.80	16.45	15.45	33.86	49.31
Germany	9.75	9.75	19.50	20.00	21.33	41.33
Greece	6.67	13.33	20.00	11.55	24.10	35.65
Ireland	8.00	10.75	18.75	8.00	10.75	18.75
Italy	8.89	23.81	32.70	8.89	30.90	39.79
Netherlands	19.15	8.90	28.05	39.45	17.20	56.65
Portugal	11.00	23.75	34.75	11.00	23.75	34.75
Spain	4.70	23.60	28.30	6.25	31.58	37.83
Sweden	7.00	11.91	18.91	7.00	25.87	32.87
United Kingdom	11.00	12.80	23.80	11.00	12.80	23.80
Average	9.53	16.44	25.98	14.11	25.05	38.57

Source: United States Social Security Administration (2004).

⁷ Under the traditional neoclassical theory of economic growth, in the long run only the level of GDP is affected by taxation. By contrast, in endogenous growth theory both the level and long-run growth rate can be impacted by tax policy. For a short review of this literature, see V. Tanzi and H.H. Zee, "Fiscal Policy and Long-Run Growth", *IMF Staff Papers*, Vol. 44, No. 2, June 1997. An empirical analysis of the importance of expenditure composition and tax structure is provided by "M. Bleaney, et al., "Testing the Endogenous Growth Model: Public Expenditure, Taxation, and Growth over the Long Run", *Canadian Journal of Economics*, Vol. 34, No. 1, February 2001.

⁸ See A. Afonso, et al, "Quality of Public Finances and Growth", *European Central Bank Working Paper No. 438*, February 2005.

2.9 In Bulgaria, high social insurance taxes suggest that both the supply of and demand for human and capital resources may be adversely affected. Table 2.2 shows that social insurance contribution rates in Bulgaria are high compared to most other European countries. Bulgarian workers currently pay about 43 percent of gross wages to fund social programs. These rates may need to be reduced, even beyond current plans, to support faster productivity growth and job creation. Bulgarian government is planning to lower pension contributions by six percentage points in 2006, and shift an additional percentage point of gross pay to the mandatory private pension fund, while general revenues would be used to offset the decreased funding of the public pension pillar from payroll contributions. This may, however, necessitate increasing revenues by either increasing tax rates or improving collections. To the extent that this reintroduces a burden on workers and employers in the private sector, it is not promoting job creation.

2.10 Ultimately, the determination of the size of government boils down to estimating the marginal cost of raising the necessary public funds, and comparing it with the social return of each particular project financed with public funds. If a project does not yield a social return greater than the marginal cost of funds, it is not worth raising the taxes to finance it. The problem is that it is practically impossible to calculate the social rate of return of all projects that are to be financed, and estimating the marginal cost of funds is fraught with problems. Nevertheless, evidence from various studies suggests that the marginal cost of funds tends to be greater, the higher the existing level of tax revenue, the greater the use of corporate and personal taxes to raise revenue, the higher the administrative costs of raising the revenue, the higher the taxes on exports, and the larger the informal sector in the economy (Walters and Auriol 2005). With tax revenues in Bulgaria at levels about 40 percent of GDP, and given its relatively small formal private sector and a large informal sector, marginal cost of funds is likely to be high and above social returns for a whole range of programs or projects that are currently funded by the public sector. Tanzi and Schuknecht (2000), in fact, argue that state spending need be no more than one-third of nominal GDP in order to achieve the most important social and political objectives.⁹

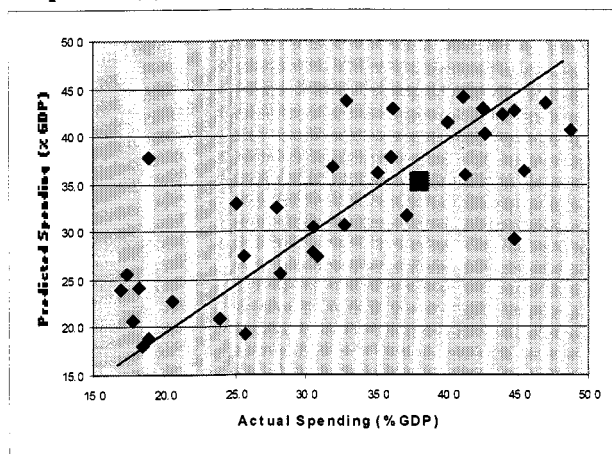
2.11 To get some idea of whether the size of government in Bulgaria is too large, its size is compared with other countries. The hypothesis tested here is whether one can find countries with consistently high rates of growth that, other things equal, have public sectors that are smaller than that of Bulgaria. Graph 2.1(a) presents the results of a statistical regression for 38 countries with the fastest GDP per capita growth rates in the world during 1994-2003.¹⁰ The graph plots the level of public spending predicted by the estimated regression against the actual level of public spending. The diagonal line through the origin gives the set of points for which the two levels are the same. Points to the right of the line represent the set of countries that have actual expenditures that are higher than predicted.

⁹ Tanzi V. and L. Schuknecht, *Public Spending in the 20th Century: A Global Perspective*, Cambridge: Cambridge University Press, 2000. For a more recent discussion of the role and size of government see Vito Tanzi, "The Economic Role of the State in the 21st Century", mimeo, July 29, 2005.

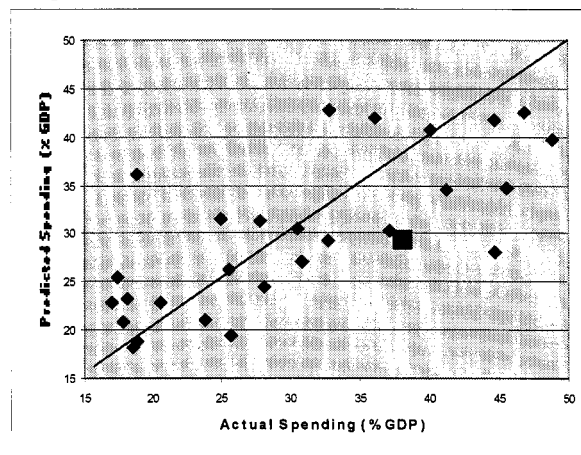
¹⁰ The observations were selected by putting together a dataset containing only countries whose per capita GDP growth rates (on a purchasing power basis) were at least 2.5 percent (or one-quarter standard deviation more than the average growth rate in about 130 countries). Average expenditure ratios in 2000 to 2004 were regressed on real per capita GDP in 2000 (in logs) and on regional dummy variables. The predicted expenditure ratios are calculated from the coefficients of this regression.

Graph 2.1: Actual vs. Predicted Levels of Public Expenditure

Graph 2.1 (a)



Graph 2.1 (b)



2.12 Graph 2.1 (a) suggests that, given its economic and social circumstances, the overall spending level in Bulgaria could be lower (square dot). Since Bulgaria was not included among the 38 fastest growing countries during the period in question, its position on the graph was determined by applying the coefficient estimates to Bulgaria's data. At an average of about 39 percent of GDP (excluding social contributions of the government) in 2000-2004, the country's level of spending was about 3.0 percentage points of GDP higher than predicted by the model.¹¹

2.13 Repeating the regression without European and Central Asian countries indicates that the level of expenditures predicted for Bulgaria is fully eight percentage points lower than the actual. Graph 2.1(b) presents the results for a subset of this sample (28 out of the 38 countries). Removing Europe and Central Asia from the sample helps to control for the effect on growth rates and expenditure associated with massive economic restructuring and other processes. It also controls for a possible European bias towards larger government. As before, the predicted value of spending for Bulgaria is derived by applying the coefficient estimates for the twenty-eight-country sample to Bulgaria's indicators. The two sets of regressions thus suggest that there is room for Bulgaria to reduce its public expenditures relative to GDP without jeopardizing its prospects for rapid growth.

2.14 The question now becomes which public expenditures can be reduced without reducing growth. There are the classic core functions of the public sector, such as law and order, protection of property, sound money, defense, or public health. These services have the common property that they would be provided in less than desired quantities for the country as a whole, if they were left to the individuals to provide them. There are a variety of reasons why such market failures would occur, and it is important that the reasons for failure of under-provision of any one good or service be understood before action is taken. The country might, for example, believe that it needs many university graduates, and that the poorer but talented students may not be able to attend the university or some technical school. This, however, does not imply that such training should be free for all, or even for a substantial portion of the relevant population, or that it has to be provided by a public institution. Since much of the benefit of higher education is typically appropriated by the students in the form of higher salaries, part of the

¹¹ The results are confirmed with a larger sample of about 130 countries including Bulgaria, which controls for the effect of living standards, demographic factors, external openness, and region-specific conditions.

solution to financing university education is to correct the imperfections in the capital market so that the students in question can be lent money to pay for the studies with an assurance that the loans will be repaid. Whether this education should be provided by public or privately owned institutions is probably a matter of a mix of both to provide competition and variety (see Canning et al 2005).

2.15 Every program in the public budget should be subjected to a scrutiny with the intent that what cannot assuredly be done best, in the sense of “in sufficient amount and efficiently”, through the public sector, should be left to the private sector. This is much easier said than done in part because publicly provided or financed goods and services have a range of outcomes that are hard to weigh against each other, and hence making choices among them becomes an intensely political matter. Yet public funds are limited and trade-offs must be made. It is, however, unclear how one can sensibly trade off improvements that promote different sorts of well being (mortality, morbidity, clean air, clean water, or security) with those that contribute more directly to growth (roads or marketable skills). Nothing definitive is known in this area, other than that a whole array of goods and services will be needed and/or desired by the population.

2.16 One way of tackling the problem is to examine public expenditures by sector and look for inefficiencies. Some countries have organized their public expenditures into programs aimed at achieving certain measurable results. Those that have done it for some years have a track record of being able to relate the input costs of these programs (expenditures) with the outputs and outcomes, and they can then compare these results with other countries that have attempted to do the same so as to see whether they are spending their money effectively. Bulgaria is just moving in the direction of program-based and results-oriented budget management, and thus it will be some time before the above approach can be applied.

2.17 Another way of tackling the problem is to look at the share of GDP that is being spent in a certain sector, and compare it to other countries. If similar amounts of money are achieving very different results or similar outcomes are the result of rather different levels of expenditures, then there are presumably grounds for an examination of the causes of this discrepancy, and thus a basis for a correction.

2.18 At the overall budget level it seems that some of the most rapidly growing countries that have had extended periods of rapid growth, have been able to achieve that with expenditure to GDP ratios below (and some well below) 40 percent. Table 2.3 presents' budget data, by sector, for a number of countries—mostly the cohesion countries and new EU candidates in Europe, but also some Asian countries that have a record of two or more successive decades of rapid GDP growth. The countries with extended periods of growth were not all in Asia, witness Ireland and Chile.

Table 2.3: Functional Allocation of Expenditure in Selected Countries (in percent of GDP, 2003 or 2004)

	P. C. GDP Growth (PPP)	Total Expend. 1/	General Public Services	Defense & Public Order	o/w Defense	Education	Healthcare	Social Protection	o/w Pensions	Economic Affairs
Bulgaria 2/	2.4	40.0	2.9	5.1	2.3	4.3	4.7	13.8	9.3	5.0
Croatia	5.2	51.6			2.3	4.3	6.6			
Czech Rep.	2.9	42.1	4.4	4.3	1.6	4.4	6.4	13.7	9.0	7.1
Estonia	6.6	38.9			2.0	5.7		14.3	6.1	3.8
Hungary	3.9	48.2	8.9	3.5	1.4	5.5	4.2	20.9	9.7	
Latvia	6.2	33.2	4.6	3.7	1.7	5.8	3.4	14.3	8.2	3.5
Lithuania	4.7	32.9	4.6	3.6	1.6	5.9	3.8	15.2	6.2	4.4
Poland	4.5	45.7	7.0	2.2	1.8	5.6	4.1	22.1	11.6	3.2
Romania	2.4	30.9	1.6	3.2	1.4	3.0	3.4	9.9		5.5
Slovakia	4.2	38.4	5.0	3.8	1.8	4.4	8.4		7.6	4.7
Slovenia	3.7	48.2	8.5	3.4	1.4	5.9	6.8	18.3	11.6	3.5
Ireland	8.2	34.3	2.3	2.2	0.5	4.4	6.4	9.5	3.1	5.1
Portugal	2.3	47.9	4.5	3.3	2.1	7.0	7.1	15.7	11.9	4.4
Greece	3.2	48.9	3.3	4.6		3.6	2.9	20.9	12.8	5.6
Spain	2.7	40.5	2.5	3.3	0.9	4.4	5.4	13.5	8.9	4.2
Chile	3.7	22.1	0.3	2.7		3.8	3.0	7.5	2.9	2.7
Korea	4.5	24.8								
Malaysia	3.1	28.8	1.42	2.9		7.6	2.0		1.5	5.3
Singapore	2.9	15.2								
Thailand	2.9	16.8	0.9	2.4	1.3	4.0	1.3	2.0		3.36

Source: Bulgaria MoF; IMF Government Finance Statistics; Eurostat; and OECD. Data for Bulgaria are for 2004. All other countries are for 2003 and 2004.

1/ Expenditure components in the table may not add to the total due to the use of different sources and because some expenditure items may have been excluded (e.g., net lending and unallocated outlays).

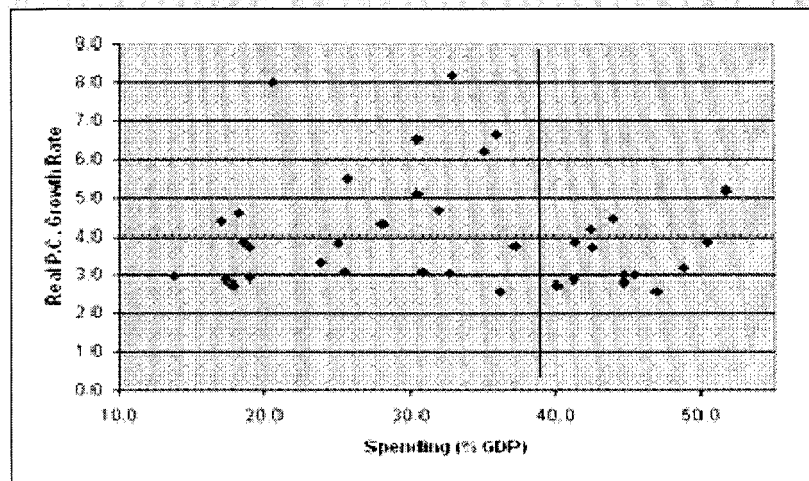
2/ Includes government employer contributions to the social insurance funds.

2.19 The same relationship—that fast growing countries have had expenditure to GDP ratios below 40 percent—can be gleaned from the scatter of such ratios for the sample of rapidly growing countries used in the regression analysis noted above (see Graph 2.2). What is interesting in Graph 2.2 is that while per capita GDP growth of 3-5 percent per year was achieved by countries with expenditure to GDP ratios ranging anywhere from 15 to 50 some percent, countries growing faster than that all had a ratio of less than 40 percent.

III. ROOM FOR EXPENDITURE REDUCTIONS

2.20 One can also identify some areas where savings of public monies are possible by comparing sectors across countries (see Table 2.3). For example in the defense and public order category, Bulgaria is an outlier, although most of the difference seems to come from the public order (security) side. However, even in defense it seems that savings are possible, if one takes new EU member states as comparators. In contrast to Bulgaria, the other new NATO member states spent on average only 1.7 percent of GDP on defense, while facing the same demands for upgrading their military systems.

Graph 2.2: Per Capita Income Growth and Government Spending



Source: World Bank, World Development Indicators and IMF World Economic Outlook Database

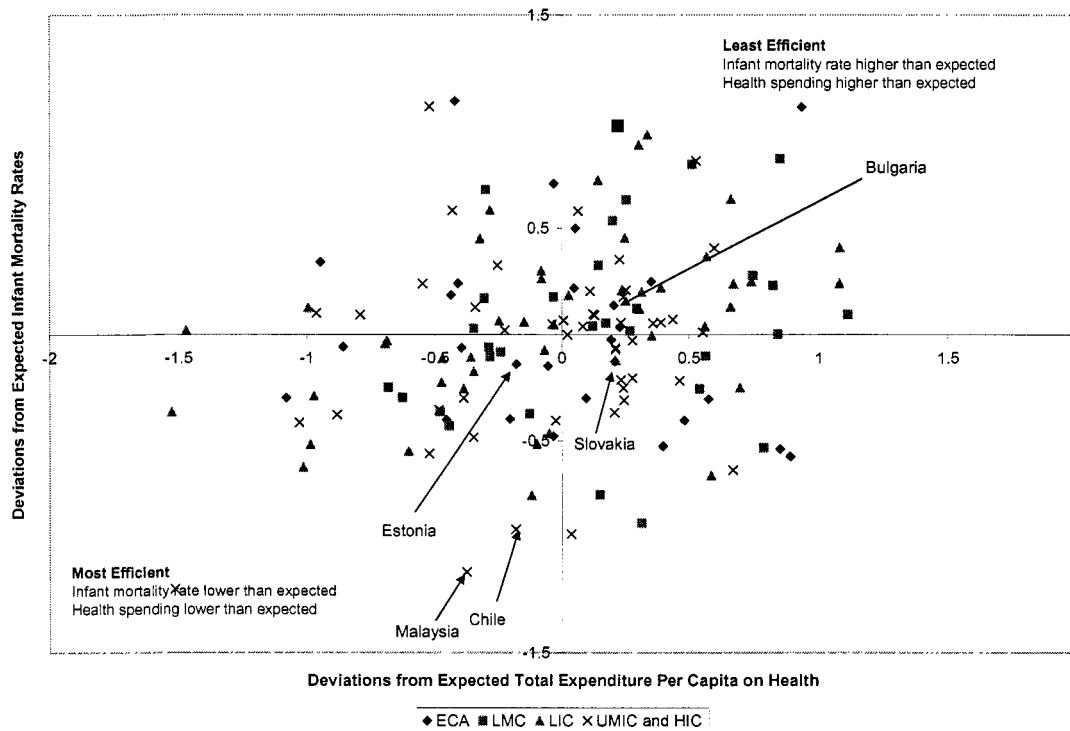
2.21 In education, Bulgaria is not an outlier in terms of the level of public spending, but a series of reviews of Bulgarian education have found considerable room for improvement in terms of efficiency of public spending and quality of education at all levels. One of them is possible overstaffing. The number of workers in education represents 2.4 percent of Bulgaria's population (42 percent of general government employment). The comparable ratio for Ireland, which is known to have a modern and effective education system, is 1.7 percent. While salaries in education in Bulgaria consumed 42 percent of the total government wage bill, this ratio was only 22 percent in Ireland in 2003. Other problems identified in Bulgaria were sub-par teaching materials, inadequate training of teachers, and unsatisfactory supervision of the entire system. Also, the rapidly declining population implies the need for consolidation of school and university facilities in the years to come. In sum, there is a need to reallocate the existing level of expenditures in ways that would improve overall education system performance.

2.22 In health care, Bulgaria spends roughly in line with other European countries. However, why can Chile and Malaysia get equal or better outcomes in terms of indicators such as infant mortality or life expectancy, but with a smaller resource envelope? For example, Graph 2.3 shows that it should be possible for Bulgaria to achieve considerably lower rates of infant mortality with the same (or lower) per capita expenditure on health. The deviations of actual from expected values for infant mortality, plotted on the vertical axis, are based on a regression of infant mortality on expenditures per capita devoted to health (controlling for per capita GDP and regional factors), for a large world-wide sample of countries. The deviations plotted on the horizontal axis are based on a regression of total per capita expenditures on health as a function of per capita GDP and regional factors.¹² Countries that are spending less per capita than would be expected, yet achieving higher than expected infant mortality reductions find themselves in the south-west quadrant of the graph ("most efficient"). Those in the north-east quadrant, among them

¹² There is currently no consensus regarding the relationship between aggregate public spending on health and health outcomes, though microeconomic evidence provides strong evidence in favor of medical interventions to avert the largest causes of child mortality in developing countries. The impact of public spending has been shown to be quite complicated and can depend on cultural, demographic and other characteristics, including public sector efficiency (see D. Filmer and L. Pritchett, "The Impact of Public Spending on Health: Does Money Matter?", *Social Science and Medicine*, No. 49, 1999, pp. 1309-1323). However, real GDP per capita has been shown to be an important control variable and regional dummy variables may capture much of the other key social determinants.

Bulgaria, are “least efficient”. Comparators such as Slovakia and Estonia are achieving better results with the same or fewer resources. However, countries such as Chile or Malaysia not only perform better in reducing infant mortality, but are also more efficient in achieving rapid growth (see their expenditure to GDP ratios in Table 2.3) than any other country in the sample with the same or lower per capita spending.

Graph 2.3: Infant Mortality and Public Health Expenditure



2.23 In the social protection area, despite a series of reforms in the pension system and social assistance programs to achieve fiscal sustainability, the social security system in Bulgaria remains largely unsustainable, and will remain so without a long term plan to limit the growth of both average benefits and the number of beneficiaries. While aggregate social spending is not high by European standards, it is higher than in several high-performing economies, and has been financed by some of the highest social contribution rates in the region. The public pension system is close to one-quarter of general government spending, and is running a deficit of about 1.2 percent of GDP, which is financed mainly with transfers from the central government. In comparison with Estonia and Slovakia, for example, pensions account for 2 to 3 percentage points of GDP less than in Bulgaria. In Ireland, pensions cost only slightly more than 3 percent of GDP, and total social protection is over 4 percentage points of GDP less than in Bulgaria, although admittedly Ireland has a different demographic situation.

2.24 At under US\$ 50 per month in 2004, the average pension is low, but the number of pensioners is quite large, covering 57 percent of households. Although the total number of pensioners has been coming down since 2001, the number of beneficiaries accounts for over 30 percent of the population.¹³ This makes total pension outlays sensitive to changes in benefits, and therefore pension increases should be

¹³ This excludes the number of self-employed pensions. The decline in the total number of pensions is mainly due to the increase in the minimum retirement age to 62.5 for men and 57.5 for women.

limited as much as possible.¹⁴ Furthermore, the decline in the number of pensioners could have been larger were it not for the very fast increase in disability pensioners. While the number of regular and farm pensioners has declined about 11 percent between 1998 and 2004, the number of disability pensioners increased nearly 50 percent to 455,000 during this period. This calls for a closer examination.

2.25 Review of public expenditures from the economic classification point of view as in Table 2.4 (for the group of countries in Table 2.3), yields some further insights about possible areas for consolidation. The general government wage bill in Bulgaria is not high by European standards, since even though the government labor force is large, wages are low. The civilian work force in general government of 450,000 is 5.7 percent of Bulgaria's population. The comparable ratio for Chile, which is reputed to have an efficient public sector, is 2.9 percent.

Table 2.4: Economic Classification of Expenditure in Selected Countries (% of GDP)

	Wages & Salaries	Operations & Maintenance	Subsidies	Interest Payments	Social Security & Health	Capital Expenditure
Bulgaria 1/	5.1	6.3	2.3	1.8	14.6	4.1
Croatia	10.8			2.1		
Czech Rep.	3.2	4.8	2.6	1.1	17.7	6.3
Estonia			1.1	0.3	12.1	2.2
Hungary	7.2	6.7	1.7	4.2	17.1	4.7
Latvia	6.9	6.0	0.7	0.9	9.5	4.4
Lithuania	7.1	5.7	0.8	1.1	30.5	3.1
Poland		9.6	0.6	2.6	20.1	
Romania	4.9	4.9	2.1	1.3	12.0	3.3
Slovakia	6.5	6.4	2.0	2.3	13.6	3.9
Slovenia	11.1	4.9	1.4	1.5	19.0	4.7
Ireland	9.1		0.7	1.2	10.6	3.9
Portugal	14.9		1.4	2.9	17.2	3.3
Greece	11.8		0.1	5.3	17.3	3.9
Spain	10.3		1.1	2.2	14.9	3.5
Chile	4.3	1.5		1.0	6.6	3.4
Korea	6.5				4.1	5.3
Malaysia	5.5	10.3		2.7	1.5	6.5
Singapore				0.8		5.6
Thailand	5.8	3.0				2.8

Source: Data were compiled from a variety of sources (see Table 2). For Bulgaria and Romania they are for 2004. For all other countries the data are for the latest available year, which in most cases is 2003.

1/ Excludes government employer contributions to the social insurance funds.

2.26 Bulgarian budget subsidies are high by international standards. They accounted for 2.3 percent of GDP in 2004, or 5.8 percent of the general government budget. The averages for the EU15 and the new EU member states were about half that amount. Almost half of the subsidies in Bulgaria went to the health sector, principally to loss-making hospitals and clinics. Much of the rest of the subsidies went to agriculture, most of them to support the production of tobacco.

¹⁴ For example, the increase in the maximum pension in 2003 changed the projected date of achieving financial balance from 2007 to 2013.

2.27 Pushing the points made on public expenditure consolidation a bit further, one could construct the public finances of a hypothetical composite country, with the idea of replacing a less efficient sector in Bulgaria with a more efficient one from another country. Table 2.5 below presents two illustrative allocations in which selected economic components of expenditure in Bulgaria are replaced with the values of relatively more efficient countries. They are called illustrative because expenditures among countries for categories that are seemingly the same are often not entirely comparable (not least because the outcomes stemming from the expenditures can be different), and the choice of countries is somewhat arbitrary. However, it is another way of confirming that there is probably room for reduction of public expenditures in Bulgaria without reducing, and possibly increasing, the rate of growth of GDP.

2.28 Keeping in mind the caveats, the first illustrative case suggests that Bulgaria could save about 5.4 percentage points of GDP by adjusting its spending to the levels in comparable sectors of a combination of better performing countries. In the first row of the table, the expenditure ratios of wages and salaries, operations and maintenance, subsidies and capital expenditure in Bulgaria are replaced with the corresponding values for Chile, Slovenia, Ireland and Malaysia, respectively. In each case, none of the values chosen are the lowest among the set of high-performing economies, and in the case of capital expenditures, the amount assumed is about 2.5 percentage points of GDP higher than is currently spent in Bulgaria, on the assumption that Bulgaria will need to increase its investment in physical infrastructure for some years to come. The composite is thus somewhat conservative.

Table 2.5: An Illustrative Allocation of Economic Components of Expenditure

Wages and Salaries	Operations and Maintenance.	Subsidies	Capital Expenditure.	Total	Bulgaria 2004*	Potential Savings
(Chile)	(Slovakia)	(Ireland)	(Malaysia)			
4.3	5.0	0.7	6.5	16.4	21.8	5.4
(Chile)	(EU15)	(EU25)	(Malaysia)			
4.3	5.2	1.0	6.5	17.0	21.8	4.8

*The allocation of the economic components of defense and public order is based on the average distribution in the NMS10 in 2004. Social contributions are not included in the wage bill.

2.29 The second illustrative case (row 2 of Table 2.5), shows that savings of about 4.8 percentage points of GDP may be possible. This is done by making further adjustments in the direction of being even more conservative than in the first case. Since Ireland's level of subsidization is relatively low, it is replaced with the higher average level of spending for this category in the EU15. Similarly, Slovakia's expenditure on operations and maintenance is replaced by the higher level of spending in the EU15.

2.30 The same exercise can be replicated using functional categories of expenditure, which would suggest that fiscal savings between 4.6 and 5.5 percent of GDP could be achieved. The first row of Table 2.6 shows the expenditure ratios of general public services, defense and public order, recreation, culture and religion, and economic affairs being replaced with the respective values of Ireland, Latvia, Ireland, and Chile. Using these values, the fiscal savings come to about 4.6 percent of GDP. Another, less conservative scenario could involve lower values of the latter three expenditure components, implying savings of 5.5 percent.

Table 2.6: An Illustrative Allocation of Functional Components of Expenditure

General Public Services	Defense and Public Order	Recreation, Culture and Religion	Economic Affairs	Total	Bulgaria 2004	Savings
(Ireland) 2.3	(Latvia) 3.7	(Ireland) 0.54	(Chile) 2.7	9.2	13.8	4.6
(Ireland) 2.3	(NMS10) 2.7	(France) 0.67	(Chile) 2.7	8.4	13.8	5.5

2.31 Whether one looks through the lens of economic components of expenditure or through the prism of expenditures by function, there seems to be ample room, on the order of five percent of GDP, for Bulgaria to reduce its public expenditures relative to GDP without a negative effect on the rate of growth and with improved outcomes in areas such as education and health. Although difficult, this is far from impossible. It has been done before in other countries. Between 1984 and 1990, Chile reduced its public expenditures from 34.3 percent of GDP to 21.9 percent of GDP, with two percentage points coming from real expenditure reductions and 10.4 percentage points from rapid growth in GDP. Since 1990 the average GDP growth has continued to be strong, and the ratio today is 22.1 percent of GDP. In a more celebrated case, and under a democratic government, Ireland cut its public spending from 53.2 percent of GDP in 1987 to 40.8 percent of GDP in 1990, two thirds of which came from real cuts in expenditures in what was then labeled as “expansionary contraction.” The ratio today is 34.3 percent (see Table 2.2), and Ireland today has an income per capita that is above the EU15 average. Other countries that have gone through conscious and successful reductions in government relative to GDP in the 1990s include Canada, and New Zealand (see Lewis 2004).

2.32 Further, Table 2.7 shows expenditure reductions of a selection of EU countries that have gone through adjustments of 7.2 percent of GDP on average at some point during the past 15 years, while keeping up or improving the performance of their economies. The most significant cuts were made in the following areas:

- **Social security and welfare payments:** In most Nordic countries eligibility criteria for unemployment benefits were tightened, and the pension system was thoroughly reformed in the 1990s. Over the long term, Bulgaria faces significant fiscal pressures from aging population that might render public finances unsustainable in the absence of a continuing effort to carry out the reform of the social insurance system.
- **Wages and salaries:** Cuts in the government employment accompanied by moderate increases in public sector wages in an environment of high economic growth were central to fiscal consolidation in Ireland in 1988-89. Irish government adopted schemes for early retirement of public employees, while the rapidly expanding economy absorbed the released employees in the private sector. The wage moderation was largely accepted after the government carried out broad tax cuts and thus boosted after-tax incomes.
- **Health and education:** Ireland and Sweden successfully reformed their healthcare and education systems, respectively.
- **Subsidies and transfers:** Subsidies were cut in most countries in the 1990s. Hence, there is a reason to believe that Bulgaria should be able to find room for further judicious cuts in these areas.

- **Defense:** As hard as it seems, all countries managed to achieve some cost cuts in this area in the 1990s.

Table 2.7: Expenditure Adjustments in Select EU Countries in 1990s

(Changes in percent of GDP)	Ireland		Denmark		Finland	Spain	Sweden	Netherlands	United Kingdom	Average
	1988-89	1995-97	1984-85	1996-97	1994-97	1994-97	1994-97	1994-96	1994-97	
Total expenditures	-11.4	-6.2	-5.2	-3.8	-10.9	-6.6	-8.5	-5.2	-5.9	-7.2
Defense	-0.1	-0.2	-0.5	-0.1	-0.2	-0.3	-0.4	-0.4	-0.9	-0.3
Education	-1.3	-0.7	-0.8	0.3	-1.5	-0.5	-0.9	-0.6	-0.1	-0.7
Health	-1.3	-0.5	0.0	-0.1	-0.2	-0.5	0.6	-0.2	-0.2	-0.3
Social security and welfare	-2.5	-2.1	-1.7	-2.5	-8.0	-2.6	-5.8	-2.2	-2.3	-3.4
Housing and other amenities	-1.6	-0.3	-0.6	0.0	-0.3	-0.1	-1.9	-1.1	-0.3	-0.7
Economic services	-2.9	-0.3	-0.8	-0.1	-1.9	-1.6	-3.7	-0.1	-0.9	-1.4
Interest payments	-1.1	-1.8	1.3	-0.8	0.9	-0.3	1.0	-0.5	0.4	-0.2
Current expenditures	-10.8	-5.7	-4.7	-3.5	-10.8	-5.5	-7.6	-4.5	-4.4	-6.6
Expenditures on goods and services	-1.8	-1.3	-1.1	-0.2	-1.1	-1.8	0.0	-0.2	-1.2	-0.9
Wages and salaries	-1.0	-1.1	-0.8	-0.2	-1.0	-1.1	-0.3	0.0	-2.0	-0.7
Other purchases of goods and services	-0.8	-0.2	-0.3	0.0	-0.1	-0.6	-0.6	-0.2	1.0	-0.4
Interest payments	-1.1	-1.8	1.3	-0.8	0.9	-0.3	-0.6	-0.5	0.4	-0.2
Subsidies and transfers	-7.9	-2.7	-4.8	-2.5	-10.6	-3.4	-8.6	-3.8	-3.6	-5.5
Subsidies	n.a.	n.a.	n.a.	-0.1	-6.0	-0.4	-3.4	0.5	-0.3	-1.9
Transfers to other levels of government	n.a.	n.a.	-3.5	-1.6	-4.4	-0.6	-1.7	-1.4	-1.7	-2.2
Capital expenditure	-0.6	-0.4	-0.5	-0.3	-0.2	-1.2	-0.8	-0.6	-1.5	-0.6

Source: IMF(2003), Hungary: Selected Issues, IMF Country Report No. 03/125.

IV. CONCLUSION

2.33 This section has tried to establish that there is ample room for Bulgaria to reduce the size of its government, and thus accommodate the likely level of EU transfers that will come after the expected accession without having to reduce growth. Bulgaria's public expenditures in the aggregate seem to be between 3 and 8 percent of GDP higher than would be necessary to keep pace with the growth of a sample of rapidly growing countries, while improving many of the outcomes. The fact that the private sector is still weak and is carrying a large tax burden also argues for a smaller state. Finally, an examination of public expenditures in terms of functional and economic classifications suggests that there should be room for expenditure reductions relative to GDP in health, social benefits, and size of the civil service, defense, and security. The envelope for education may be about right at the moment, but like in other sectors, there is a need for pressing on with the reforms to improve outcomes. As will be argued in the next chapter, reforms in education, because of its relevance to long-term rapid growth, are crucial to the successful (growth-promoting) absorption of the EU funds.

2.34 Reductions will take political resolve. Change means losses for some of the vested interests, so there is bound to be resistance. The challenge is to be able to convince people that a large majority will gain from well designed changes. Bulgaria's government has gone through difficult changes in the past 15 years and in the past few years has finally seen some fruits of these efforts. It would be important not to lose the unique opportunity of substantial grants from the EU to help it weather further reforms. This chapter suggested some areas where greater efficiency in the use of funds could help Bulgaria maintain its momentum. It is crucial that the same discipline that has been suggested for the various categories of expenditure also be applied to the funds that will be coming from the EU. The areas that are likely to receive the most funds are agriculture, transport, human resource development, and environment. For this reason the remaining chapters (Chapters 3-6) of this report focus on how the expenditures in these areas could best be targeted on programs that would help raise productivity and sustain the convergence of Bulgaria's GDP per capita to that of EU25.

3. PREPARING THE LABOR FORCE FOR GLOBAL COMPETITION

3.1 The continued rapid decline in population, due to low birth rate and emigration, presents both a challenge and an opportunity for Bulgarian education and labor policies.¹⁵ The challenge makes it imperative that every effort is made to optimize the human resource development of the reduced population ensuring that the quality of the existing pool of talent is promoted. The opportunity exists in improving the efficiency and application of the existing educational spending so that it is targeted at improving the quality of education and training, and increasing the coverage of the dwindling cohorts of students (including from socially disadvantaged families) expected in the future.

3.2 Over recent years Bulgarian education has benefited from a range of analyses, and a review of these indicates a large degree of consensus on the problems and issues which need to be tackled. Also, action has been taken on a number of recommended reforms. Among these are developments on standards and quality of vocational education, quality assurance in higher education, curriculum reform initiatives, plans for reform of teacher education, and plans for lifelong learning through the Strategy for Continuing Vocational Education. New agencies such as the National Agency for Vocational Education and Training (NAVET), the National Evaluation and Accreditation Agency (NEAA), the Curriculum Council, the National Pedagogical Centre are being established, and could have much to contribute to reforms in the period ahead.

3.3 However, much more remains to be done and there is a large unfinished reform agenda that would need to be implemented to make the education system cost-efficient and responsive to labor market needs. Bulgaria's education system needs urgently modernization to ensure it equips students with skills required in a competitive market economy. This chapter briefly reviews some of the key challenges facing the country in the field of education, but above all highlights, based on the experience of Ireland, specific proposals to promote human resource development, which the Bulgarian government may want to develop for funding from the EU Structural Funds.¹⁶

I. EDUCATION IN BULGARIA

3.4 While Bulgaria has a highly educated population given its level of income, it needs to improve the efficiency of public expenditures on education in order to further upgrade its human capital. The efficiency and effectiveness of public expenditures on education are particularly important, given the objectives of improving labor productivity. Better labor productivity is needed to raise wage income and catch up with the EU standards of living.

¹⁵ The fact that primarily working age and highly educated people emigrate magnifies the challenge.

¹⁶ A more thorough assessment of the education sector is included in the education sector policy note, which was prepared in parallel by the World Bank. It was agreed at the outset that this report will focus on proposals to meaningfully program the forthcoming EU structural funds for human resource development based on the experience of Ireland.

3.5 According to the Bulgarian administrative data, participation rates in basic and secondary education have increased in the past ten years, but declined at the tertiary level. Most children of the basic education age (7-14) participate in some level of education, but participation rates in secondary and tertiary education drop significantly (see Table 3.1). There is also significant difference in school participation rates in urban and rural areas, and between income and ethnic groups, with non-poor Bulgarian urban dwellers having highest participation.

Table 3.1: Participation Rates By Level of Schooling, Gender, Location and Poverty Status

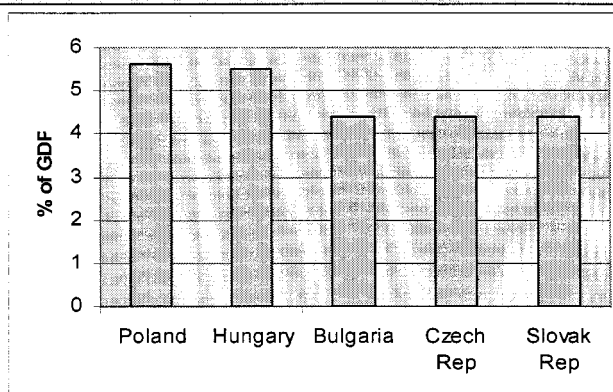
	Basic	Secondary	Tertiary
Total Population	95.5	78.8	29.8
Males	94.1	81.6	27.1
Females	96.8	76.4	32.4
Rural	96.5	49.5	6.3
Urban	93.2	87.5	39.1
Poor	89.9	50.0	6.2
Non-Poor	97.9	88.7	38.1
Bulgarians	97.6	89.7	37.5
Roma	81.0	37.0	2.1
Turks	94.1	39.5	1.6

Source: World Bank Bulgaria Education Policy Note 2005.

3.6 Drop out rates are relatively high, and a large share of the young population is leaving school early, and entering the workforce with only basic education qualifications. In Bulgaria, about 20 percent of the 18-24 year olds have only lower secondary education¹⁷, and are not participating in further (tertiary) education and training, while tertiary enrollment rates are steadily increasing in EU countries. Further, participation in lifelong learning is low in Bulgaria compared to EU countries: while 9.4 percent of adults aged 25-64 in the EU participated in education and training activities in 2004, in Bulgaria the fraction was only 1.4 percent.

3.7 Consolidated general government expenditures on education in Bulgaria, have been rising gradually over the last few years, to reach a level of 4.3-4.4 percent of GDP during the period of 2003-05. This is comparable to the average for select CEE countries (see Figure 3.1), though below the EU15 average of slightly above 5 percent of GDP. At the same time that outlays for education gradually increased, the size of each age cohort has continued to decrease, implying rising public spending per student. The age cohorts in Bulgaria are projected to continue to decline rapidly: the number of 5-9 year olds is expected to fall by 26.3 percent, the number of 10-14 year olds by 39.2 percent, and the number of 15-24 year olds by 35.6 percent in 2000-2010. This compares with the average projected falls for the EU countries of 7.9 percent, 10.9 percent, and 17.4 percent, respectively.¹⁸

Figure 3.1: Public Expenditures on Education, 2003



3.8 Despite increased public spending per student in Bulgaria, the quality of education has declined. Figure 3.2 shows that the performance of Bulgarian students in the Third International Study of Mathematics and Science (TIMSS) has steadily declined between 1995 and 2003, while public spending

¹⁷ This is significantly above the Lisbon target of ten percent by 2010.

¹⁸ European Commission (2005d), Key Data on Education in Europe.

per student has significantly risen. In 1995, the performance of Bulgarian students in TIMSS was above the average of EU8 countries, but by 2003 Bulgaria's average scores had dropped below the average of EU8 (see Figure 3.3). Furthermore, among the European countries participating in the Program for International Student Assessment (PISA) in 2000, Bulgaria ranked 33rd among 41 participating countries. Even though Bulgaria's performance at that time was close to the predicted level for its per capita income, about 40 percent of Bulgarian 15 year olds who took the test performed at a level which indicates serious difficulties in using reading literacy as an effective tool to advance and extend their knowledge and skills in other areas. Moreover, there is evidence that the education system is not producing the skills needed in the labor market, and thus contributing to high unemployment rate among recent graduates. Bulgaria should be able to deliver better outcomes for the same cost.

Figure 3.2: Real Expenditure per Student and TIMSS math Scores

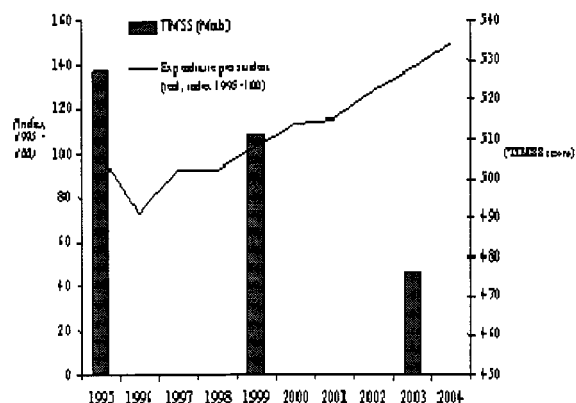
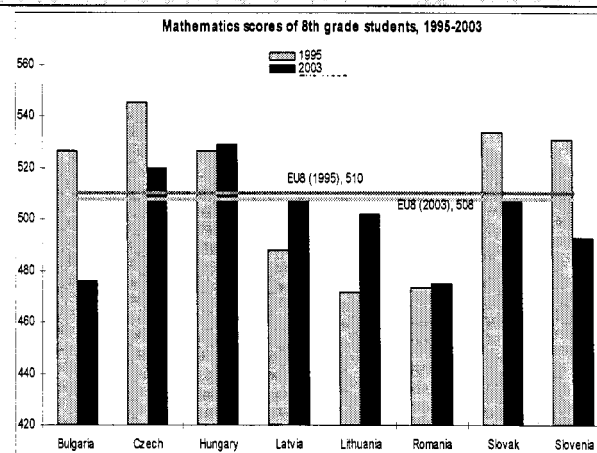


Figure 3.3: Mathematics Scores of 8th Grade Students in 1995 and 2003 on TIMSS



Source: World Bank Bulgaria Education Policy Note 2005.

Source: World Bank Bulgaria Education Policy Note 2005.

3.9 To put matters into further perspective, some successful countries like Chile and Japan have managed to do well with spending on education at levels that were more modest than Bulgaria's is today. Chile spent 2.5 percent of GDP on education in 1990, and is still spending only about 3.8 percent of GDP. Japan spent four percent of its GDP on education in 1970.

3.10 The education system in Bulgaria has overcapacity in staffing and facilities in particular at the tertiary level, as is illustrated by the low student to teacher ratios (see Table 3.2), and the excess capacity will only increase in light of declining age cohorts, unless action is taken. In 2003, in primary education the student-teacher ratio was comparable to the average OECD levels; but in lower and upper secondary education Bulgaria's ratio was about 12 and eight percent, respectively, below the OECD average; and in higher education Bulgaria's ratio was about 30 percent below the OECD average. By 2005, some of the overcapacity at the secondary level had been pruned, but the ratios at the lower and upper levels were still about 10 and seven percent above the OECD average, respectively. Further, these aggregate ratios mask significant differences in ratios across regions, and urban and rural areas. While some schools are congested, there are some schools with small class sizes. Owing to low student to teacher ratios, about two-thirds of total public expenditures on education go to wages and salaries for staff. Because of this, limited resources are available to keep curricula and textbooks updated, and to provide adequate teaching materials and maintain facilities.

Table 3.2: Ratio of Students to Teaching Staff in Educational Institutions (2003)

	Primary	Lower Secondary	Upper Secondary	Tertiary
Bulgaria (2004/05)	16.3	12.6	11.9	10.5
Czech Republic	18.3	14.3	12.6	17.3
Germany	18.7	15.6	13.7	14.9
Ireland	18.7		13.7*	15
United Kingdom	20	17.4	12.6	18.2
United States	15.5	15.5	15.6	15.2
OECD Average	16.5	14.3	13.0	14.9

Source: World Bank Bulgaria Education Policy Note 2005 based on OECD Education at a Glance, 2005; National Statistics Institute, Education in the Republic of Bulgaria (2005).

* all secondary

3.11 There is a need to eliminate inefficiencies in the education system, and raise the quality of education and school participation rates. Measures needed include the consolidation of schools and higher education institutions¹⁹ in light of rapidly declining student cohorts to increase the student-teacher ratios and to release funds for curriculum and textbook reform, and upgrading of consolidated facilities. There is also a need to improve the training of teachers, quality controls on education delivery, and to introduce regular national student assessments, in particular, the external Matura exams, to evaluate the impact of public spending on education on learning outcomes. Lifelong learning would need to be encouraged, and, above all, linkages forged between the education system and labor markets to ensure graduates with needed skills are produced. Unwieldy bureaucracy continues to be a feature of the system and administrative reforms have been difficult to achieve. Also, there is need to consider of further engaging the private sector in the delivery of education at different levels.

3.12 Unless determined moves are undertaken to address such inefficiencies in the outdated educational framework, it will be difficult to persuade external donors to invest significant resources in the system. The Bulgarian system has been through a great deal of travail over the past fifteen years. It is a relatively short time-span to absorb so much change and meet so many new demands. It may well be that the stage is now being reached when reform measures can be undertaken with greater confidence for success than was hitherto the case.

II. GOVERNMENT EDUCATION POLICY DIRECTIONS

3.13 The Bulgarian government has set out its policy objectives on human resource development in the draft *National Development Plan (NDP)*. These include: (i) improved and enhanced competitiveness of labor force and the economy through the development of appropriate educational and training institutions adequately responding to the needs of the economy; (ii) life-long learning; (iii) improved socio-economic cohesion and social inclusion of disadvantaged groups (young people, long-term unemployed, disabled persons, ethnic minorities); (iv) development of entrepreneurial culture through training and advancement of skills; (v) gender equality in employment; and (vi) improved capacity and infrastructure of vocational training and human resource development systems. Also, an operational program on development of human resources is being developed.

3.14 *The Employment Strategy 2004-2010*, adopted by the Council of Ministers, spells out in further detail the policy intentions to improve the quality of human capital and activate the lifelong learning

¹⁹ Bulgaria has a large number of tertiary education institutions relative to the size of the country, and the bulk of them are small. In 2003/04, there were 51 higher education institutions, of which 42 were universities or equivalent institutions. Of the 51 institutions, 17 had fewer than 1,000 students; 32 fewer than 5,000; and only 7 had more than 10,000 students.

policy. These cover a wide range of issues, which are pertinent to the plans for using the EU funds and have ambitious timescales for implementation. The education policy priorities include guaranteeing equal access to quality education with reformed curricula, and extension of mandatory education until 16 years of age. The education is proposed to be made results-oriented; by linking funding to the quality of the education product. Education delivery is to be decentralized and autonomy given to schools, while the school network is proposed to be optimized. Above all, conditions for real and profitable student learning as well as for lifelong learning are to be created by establishing linkages between the educational establishment and business.

3.15 These policy objectives and priorities are in line with EU policy concerns. Traditionally, the EU policy on education has been tentative and cautious, the prerogatives on education being left very much to the member states. In recent years, and particularly since the Lisbon Objectives of 2000, while still respecting the rights of member states, the EU has adopted a strong stance on education policy. It regards education and training as central to making Europe “the most competitive and dynamic knowledge-based economy in the world capable of sustainable economic growth with more and better jobs and greater social cohesion.” In 2004 the European Commission stated: “The European Social Fund will also support actions to expand and improve investment in human capital, in particular by improving education and training systems, and action aimed at developing institutional capacity and the efficiency of public administrations at national, regional and local level” (European Commission, 2004).

3.16 However, as highlighted above, the record of education policy implementation in Bulgaria is not an impressive one, and hence it would be important to ensure that the current plans are effectively implemented. Earlier policy plans set out in such documents as the *Strategy for Higher Education* (1999) and the *National Programme for Modernising Education* (2001) highlighted many issues on which reform was needed, but implementation of these plans failed and the set outcomes were not achieved. It may be that they were over-ambitious, but they may also have erred in the lack of in-built strategies for implementation and lack of political will to reform. Legislative changes are not a sufficient condition for results on the ground. Systematic implementation and enforcement of policies are critical for impact. It would be important that current plans do not suffer from the same defect. The achievement of large scale educational reform is difficult in all countries. Issues of timing, communication, preparedness, resources and sophisticated planning are integral to the achievement of educational reform. The current education system in Bulgaria has contrasting policy priorities and practices, and the successful transformation of such a system does not come quickly or easily, but it can be achieved.

III. THE OPPORTUNITY PROVIDED BY EU ACCESSION.

3.17 The forthcoming EU accession provides an unprecedented and historical opportunity to move the planned reforms forward, by providing incentives and leverage for change as well as external funding. The government would need to take the advantage of the period prior to accession to prepare for and launch the planned major educational reforms.

3.18 In preparation for reform, five steps would need to be taken:

3.19 *First*, a communications and public relations campaign would need to be initiated to inform the government staff and the public of the purposes, and advantages to be gained from the reforms. The advantages and changes to teachers, school leaders, VET personnel, and higher education personnel would need to be clearly spelled out. The role and benefits of EU Structural and Regional Funds would need to be explained to people who are often removed from policy planning and negotiations, and the historical window of opportunity would need to be highlighted.

3.20 *Second*, the core principle which ought to drive the process of change is the centrality of human resource development—education and training—for a modern knowledge-based economy. It is essential that education and training reforms have a high priority on the government agenda. The responsibilities of the Ministry of Education and Science (MES) and the Ministry of Labor and Social Policy (MLSP) converge on a number of areas of human resource development. Therefore, since the successful implementation of reforms requires an inter-ministerial effort, it would be of utmost importance that these ministries collaborate and closely coordinate the reforms for the common good.

3.21 *Third*, MES should also consider establishing a Strategic Policy Unit to focus on policy and strategy implementation, and devolving routine activities to other units. MES, which together with MLSP has a crucial role to play in driving the reforms, is compartmentalized and fragmented, and there is inadequate communication between sectors within the ministry. As a result of this, the big picture, into which the components fit, tends to get lost. The top management team of MES should also meet on a regular basis on policy matters to ensure consistency, cohesiveness and lack of contradiction in what emerges.

3.22 *Fourth*, to ensure all staff is aware of ongoing activities and reforms, MES would need to prepare and regularly update a position paper summarizing all the education and training initiatives underway, indicating the timetable for their development and implementation. By highlighting the potential interconnectedness of the initiatives including across various ministries (for example, with MLSP and MES), the paper would help the staff to reach possible synergies between tasks. Mention should also be made of the role of social partners within the operations, giving them a sense of ownership of the work.

3.23 *Fifth*, there is a need to build the capacity and provide re-training to MES, desirably with international help based on best practice. It would also be advisable to have an MES representative in new Regional Development Councils, which will have some responsibility for the disbursement of EU Regional Funds, to ensure capital investments in the education and training infrastructure are appropriately considered.

IV. STRATEGY TO LEVERAGE EU FUNDS TO SUPPORT EDUCATION REFORM

3.24 The Bulgarian government will need to make the strategic decision of the proportion of the European Social Fund (ESF) and European Regional Development Fund (ERDF) to be applied to human resource development in 2007-2013 through the education and training systems. In the case of Ireland, education and training was given a high priority with 39 percent of ESF resources allocated to that sector. Ireland submitted thirty-four specific education and training related proposals for ESF support for the 2000-2006 periods, in addition to proposals to ERDF.

3.25 The specific project proposals presented below for Bulgarian government consideration are based on an analysis of Bulgaria's needs and policies regarding the development of its labor force, and are influenced by the Irish approach to drawing down EU support to assist the achievement of policy objectives in this area. It is important to recognize that in Ireland the projects supported by EU funds were built on an already strong general education system. Since that is not the case in Bulgaria, to repeat the Irish success, Bulgaria would have to accelerate its education reforms. Assuming general education reforms are deepened and accelerated, there are twenty four proposals for ESF and four for ERDF. The government may select just a few proposals from the list, or present them all together as a coherent package to the EU with a clear focus on human resource development.

3.26 The proposals do not deal with all general issues affecting contemporary Bulgarian education, but focus on how the EU structural funds can be beneficially drawn upon and applied to human resource

development needs within the education and training system. In setting out the proposals, it was taken into account that precedents have been established for assistance in the areas proposed, which, by reflecting government's commitment for reform, should make negotiations over EU funds easier. When preparing a submission for EU funds, reference would need to be made to the relevant Bulgarian policy and practice in relation to the issues, supplemented by reference to recommendations from reports of external analysts, and authoritative national reports. The government would need to maximize the synergies of proposed projects, emphasizing their inter-connectedness so that the whole education system is demonstrably moving to the planned direction in a coherent way.

3.27 While ERDF funds could be used to support four proposals for rationalization of school, vocational colleges, and universities (see sub-section A), the other twenty-four specific proposals for ESF support fall under four broad areas.

- Improving the quality of the education and training system (see sub-sections B and C);
- Promoting social inclusion and lifelong learning (see section D);
- Addressing skill needs and vocational training (see section E); and
- Investing in research, technology and innovation (see section F).

3.28 Individual proposals would need to include targets, rationale, costing, and modes of implementation. It is particularly important that evaluation processes form an integral part of the submission. The National Evaluation Unit, being planned by MLSP for 2007, may turn out to be useful for that purpose. While co-funding from national resources is required for activities, it is worth remembering that most of the suggested areas for action already form part of government policy, and would need to be pursued regardless of EU entry. Since these activities focus on human resource development of the Bulgarian population, if implemented, there will be many winners from these reforms.

A. School and University Rationalization and Infrastructure Upgrading

3.29 The funds from ERDF could help Bulgaria achieve some of its educational policy objectives, in particular rationalization of schools, vocational colleges, and universities that is urgently needed and a pre-requisite for any other reform. ERDF has been designed to assist capital infrastructure initiatives, which are in line with the policy objectives of the EU. The potential proposals would include the following:

3.30 ***School Rationalization:*** There is a need to consolidate small schools in Bulgaria, some of which are outdated, with a view of creating bigger and better equipped units, and hence having a greater chance of achieving the curricula and pedagogic reforms. Based on experience from other countries, there is typically local opposition to consolidation, and good communication that explains the benefits of school mergers become vital, as well as establishment of satisfactory transport arrangements to bring children from outlying districts to the schools.

3.31 ***Reorganization of Vocational Colleges:*** Bulgaria has a large number of post-school, non-university institutions. With a declining population and new policy concerns, the future viability of some of these single purpose institutions has become an issue as they are not cost-efficient. Merging some of these institutions and giving them a new role might be worth exploring as a pilot. If the restructured institutions were designed to be multi-purpose with a curriculum designed to serve the regional needs of the labor market, involving a variety of courses with an applied emphasis, their might become viable and be catalysts for regional development. In Ireland, the scheme of Regional Technical Colleges proved to be a great success in early 1970s, and the colleges are now called Institutes of Technology. They have

benefited from ERDF and from the European Investment Bank, and regarded by the EU as one of the ERDF success stories in education.

3.32 *Rationalization of Universities:* Bulgaria has a large number of universities set up during the Communist era, which are now incommensurate with the needs of a declining population and with the type of university education required. Though rationalization of universities to enhance quality and efficiency has been planned for years, little progress has been made. Yet it is imperative that this issue is tackled before any further investments are made in the university system. Rationalization is likely to be costly in the short run, but the long-term benefits can be expected to outweigh the costs. It is important that Bulgaria engages in serious planning for university rationalization so that ERDF could be enlisted to support this move. Such funds could provide the needed “lubrication” for the change.

3.33 *Improving University Infrastructure:* In the context of university rationalization, ERDF support could be solicited for infrastructure improvements in universities. There is a great need to build up library holdings, upgrade science laboratories, and improve ICT equipment. However, a proposal for infrastructure investment would only have credibility if it supplemented rationalization of universities.

B. Improving the Quality of Basic and Secondary Education

3.34 There are seven ideas for projects to improve quality of the education system under ESF:

3.35 *Curriculum Reform:* Curriculum Reform continues to be a major policy concern. While progress has been made and reforms are in place up to the fourth school year, there is a need to speed up the reform and sequence the vocational elements better with general education. There is also a need to strengthen the staff expertise involved in the design of new syllabi. The new Curriculum Council, which is a representative advisory body, plays a useful role in fostering a sense of ownership of curricula changes. Proposals in this area would need to spell out the objectives, degree of progress, agencies involved, significance for reform, and costing.

3.36 *Assessment Reform:* There would be a need to reform the education assessment and evaluation system. Lack of external assessment of the school system is a long standing problem, and little progress has been made owing to tradition, old practices, and some social opposition. The National Assessment Centre was established in July 2005 as a service centre for MES, but would benefit from further expertise to achieve its aims. An external exam may now be taken at the end of the fourth year (Stage of Primary), but the exam is not mandatory. There is also discussion about an external assessment at the end of the next stage of schooling, either at the end of the seventh or eight year, but it is unclear whether this exam would be made mandatory. The assessment situation at the end of secondary schooling is most problematic. An external Matura examination was to be introduced in 2003, but that has not happened yet. According to the law, the exam needs to be operative from the school year 2006 onwards, but also that seems uncertain. Introduction of the Matura exam would, however, be of pivotal importance for the quality of the education system, and it should cover generic pupil competencies as well as specific subject competencies. Depending on the quality of national policy on the matter, EU support for assessment reform, and particularly for the implementation of the Matura test would be well targeted.

3.37 *Support for Modern Language Teaching in Schools:* Precedents exist for seeking EU funding for the extension of modern language teaching at various levels of the education system. With the brain drain of young language teachers out of the school system, there may also be justification to temporarily provide modern language courses for in-career teachers.

3.38 **Training of Trainers (T.O.T.): Initial Teacher Education:** Weaknesses in the existing initial and in service teacher education were identified by OECD as a major constraint to the transformation of the Bulgarian school system (OECD 2004). Since then a law on the reform of initial teacher education has been prepared, and is due for implementation early in 2006. The content of the law is consistent with the OECD recommendations, and emphasizes practical experience of student teachers and the use of internships. Resources to finance the reform would be needed.

3.39 **T.O.T. In-Career Teacher Development** Bulgaria established a National Pedagogic Centre in 2005, which is to have a major role in in-career teacher development. The center will have a network of regional branches, and there are precedents for EU Structural Fund support for such an entity. The integration of ICT within teaching and learning in the schools is at the moment the priority. The in-career work should also be seen to be in support of the curriculum and pedagogic reforms. Also, school based in-service education for school staff, in particular in vocational education schools, would be needed.

3.40 **Staff Development for the Inspectorate:** There is also a need to reform the regional school inspectorate, and shift its role to promoting quality teaching and curriculum reform. It would also be desirable to equip the inspectorate so that it can engage in whole school evaluations based on best international practice.

3.41 **Retraining of School Directors:** In line with the policy decision to devolve more responsibility and discretion to school directors, it is important that directors get re-training. School directors are key for improving teaching, learning, and assessment in schools. In addition to extended professional responsibilities, 660 school directors already manage a devolved school budget, and the plan is to extend this to further schools in the years ahead. Budget management calls for financial management skills and improved administration, though. Training courses in these areas have been supported by EU funds in other countries.

C. Improving Quality in Higher Education

3.42 Six proposals to improve quality of higher education under ESF would be as follows:

3.43 **Career Guidance Services:** *The Employment Strategy 2004-2010* emphasizes the importance of career guidance in secondary education, but EU support could also be elicited for career guidance provision at higher education. At present, there is a great imbalance in the subjects taken by students at university. For example, of the students engaged in Bachelor and Masters Courses in 2002, 57.2 percent were studying the social sciences, economic management and administration, and technical sciences, but only 5.9 percent were studying the natural, physical, or mathematical sciences and informatics. There is a need to develop good information on employment/career prospects for students, and guidance on where their aptitudes might best be suited. This would need to be linked to data on labor market trends and needs. A career guidance service could be of major benefit to students, but also to the labor market in the longer term. Career guidance services would be most relevant in multi-faculty universities and, over time, guidance personnel in a number of such institutions could build up a valuable data base. Such services are likely to be viewed favorably by ESF.

3.44 **Graduate Conversion Courses:** Owing to mismatch between the courses offered and the needs of a changing labor market, Ireland pioneered a scheme of conversion courses for graduates, which obtained ESF support. It permitted graduates in fields such as the social sciences and humanities to engage in one or two-year courses called Advanced Technological Skills, which allowed them qualify in

subject areas of technological competence (for example, I.T. qualifications). Such an approach could be beneficial also to Bulgaria.

3.45 ***Quality Assurance:*** The National Evaluation and Accreditation Agency (NEAA) issued in 2005 a new policy document on quality assurance, which provides a framework for action in this area. The framework is supported by the university rectors, and NEAA has established links with the European Universities Association (EUA). The NEAA policy and procedures are in line with good international practice. NEAA has made good progress on Institutional Accreditation and Program Evaluation, and procedures exist for the publication of NEAA reviews. However, the agency would need further financial and staffing support to engage international experts in its review panels to promote transparency.

3.46 ***Staff Development for Teaching Staff in Higher Education:*** Universities in most countries have recognized the importance of staff development programs, and built these into their quality improvement policies. Staff development is particularly important when past practices are to be changed. When a system is also suffering from a brain drain of bright young academic staff, staff development programs gain further importance to help staff to cope with new curricula, new styles of teaching and learning, and new forms of assessment. Without such programs, the quality assessment process would unlikely yield the desired outcomes, and hence using structural funds for this purpose would be valuable.

3.47 ***Staff Development for University Leadership:*** Improving the quality of university leadership, both for academic and administrative purposes, would also be needed. At present, Rectors are chosen through elections by their academic colleagues, and may have limited leadership and administrative experience. They would benefit from being exposed to appropriate courses and experiences among peers, such as those organized by EUA and OECD's Institute for the Management of Higher Education. Since universities enjoy considerable autonomy, unless their leaders are exposed to new management styles and best practice in modern universities, reforms in higher education will be further delayed. External stakeholder representation in university governing bodies would be needed, and university leaders would need to develop the skills to liaise with external bodies, including the solicitation of external funding.

3.48 ***A Distance Education Service:*** Distance education is still not well developed in Bulgaria. Working in conjunction with the EU-PHARE program for distance education (1995-1999), the National Centre for Distance Education, involving an association of 20 Bulgarian universities, was set up within the University of Sofia. With the termination of the PHARE project, funding for the center also disappeared, though. In November 2004 the Council of Ministries issued a decree on the Organization of Distance Learning in Higher Schools, by which distance education can deliver all the traditional degrees of a university to distance education students. Up to 30 percent of the courses will involve personal attendance for tutorial purposes. The standards of provision are set by NEAA. Even though the EU is less enthusiastic about distance education than before, the government may want to seek EU support for it.

D. Social Inclusion and Equal Opportunity

3.49 To promote social inclusion and equal opportunity, the following five proposals could be considered for ESF support:

3.50 ***Program for School Drop-outs:*** In 2005, MLSP instituted a new program for unemployed youth who had dropped out of school. Also in 2005, a survey was undertaken on the reasons for drop-out. Currently, less than 70 percent of age cohort completes secondary education, but detailed analysis of attendance figures suggests that the share may be lower than the official statistics indicate. The problem

is most acute in areas of significant socio-economic disadvantage, and there is a risk that the drop-outs, by falling on social safety nets, will become a drain on the society. The *Employment Strategy 2004-2010* expresses a serious concern, and suggests a range of interventions to ameliorate it. Social inclusion is also an EU policy concern, and ESF would be geared to assist national initiatives of this type. The government would need to prepare a program building on the initiatives underway.

3.51 *School Completion Program:* A school completion program is designed to prevent drop-outs before they happen, focusing on pupils at risk of dropping out. A number of actions can be taken to prevent dropping out, including counseling, provision of extra resources and facilities such as breakfast, free books, organized study time, and extra curricular activities, which help the vulnerable pupil view the school in a more positive light. MLPS is already implementing programs that provide a glass of milk and breakfast at school, and textbooks, school aids, and accessories for specific categories of children. In Ireland, ESF supported a School Completion Program.

3.52 *Career Counseling and Guidance for Youth:* The *Employment Strategy* indicates that “Centers for Information and Career Development” for youth are planned, which is supported by the PHARE program, “Clearing the Pathway to Employment for Youth”. A provision for such career guidance is essential to reach youth and long-term unemployment. If based on existing initiatives in this area, the ESF support could legitimately be sought.

3.53 *Second Chance and Back-to-Education Initiatives:* Second chance and back to education initiatives are directed to the long-term unemployed, whose education levels are often low. MLSP has initiated schemes to encourage employers to engage unemployed, and to provide and/or facilitate training for them. In Ireland, the Vocational Training Opportunity Scheme was designed to attract unemployed people back to education and training. It began as a pilot scheme in 1989, and continues to operate with the support of ESF. The scheme provides courses up to two years, focusing on vocational skills along with enterprise training, personal development, job search, skills and general studies. Course delivery is integrative and flexible, with a strong emphasis on core skills—literacy, numeracy, communication and new technology. Experience built up over sixteen years of its existence could be helpful for Bulgaria. Funds could also be available to assist general adult literacy schemes.

3.54 *Assistance Schemes for Certain Categories of Students:* ESF support could be obtained to assist a range of student schemes, including: (i) schemes for students with disabilities to participate in higher education; (ii) assistance to disadvantaged students to gain access to higher education courses through special provision made between schools in severely disadvantaged areas and universities; (iii) student hardship funds which are used when students encounter unexpected hardship through bereavement, accidents, or loss of employment by parents; (iv) education and training programs for groups such as Roma children. Bulgarian government is planning to launch a new national training program for Roma on literacy and vocational training in 2006. In Ireland, a roughly equivalent social group is known as Travelers, and ESF has supported programs designed with their needs in mind and which, hopefully, lead to engagement in mainstream education.

E. Addressing Skills Needs and Vocational Training

3.55 In order to promote lifelong learning and strengthen vocational training, the following ideas might be considered for support under ESF:

3.56 *Training of Trainers:* Vocational education and training (VET) in Bulgaria has benefited from a variety of EU schemes. Yet, there is still a need for the in-career training of teaching staff, linked to the

continuing reform of VET curricula and pedagogic practice. The inclusion of practical, apprenticeship style experience should form part of such reform. A structured three-year program might be appropriate.

3.57 *Lifelong Learning (Continuing Vocational Training):* An inter-ministerial group developed in 2005 a “National Strategy for Continuing Vocational Training, 2005-2010”. This is a major step to promote lifelong learning, and consistent with the EU policy, though successful implementation of the strategy will take time and effort. Elements of the program could be packaged in line with common priorities of Bulgarian and EU policy to be eligible for ESF support.

3.58 *A National Qualification Framework:* According to the EU policy, in the wake of the Copenhagen objectives, a framework of national qualifications is to be developed to set out a level of award, in relation to other qualifications, for all accredited courses available to the population. This framework would facilitate recognition for various forms of work and academic experience. Such a framework would bring clarity and coordination to diverse features of an education system. It is a particularly useful mechanism for the promotion of a lifelong learning policy. The National Qualifications Framework of Ireland, whose establishment benefited from EU Structural Funds could be a useful model to follow.

3.59 *Quality Assurance in VET:* The National Agency for Vocational Education and Training (NAVET) has made good progress in establishing standards and setting up licensing procedures for adult vocational training centers. NAVET operates on a tripartite partnership between MLSP, MES, and the employers. It has successfully liaised with relevant international agencies such as the European Training Foundation, and played a crucial role in ensuring quality and standards in VET. Inclusion of further partners, and development of a monitoring system of vocational education centers would be welcome, though. The injection of EU funding to an Agency which is already progressing well would likely yield returns.

3.60 *Improving Data on Labor Market Needs:* Data on labor market needs is inadequate in Bulgaria. In 2005, a new program was planned, with bilateral support from Sweden, to establish a comprehensive survey of employers about their labor market needs, which hopefully will provide information to the National Statistics Office. The Bulgarian government could seek support for the establishment of a standing Expert Skills Group to continuously monitor the changing employment needs of the labor market and feed these into policy agencies. Such an Expert Skills Group, working in association with the National Competitiveness Council, which reports on competitiveness issues for the Irish economy, has proved to be highly beneficial in the case of Ireland.

F. Research, Technology, Innovation

3.61 MES and the Ministry of Economy prepared a Position Paper entitled “Towards a Science, Technology and Innovation Policy for Bulgaria” in 2002. While the paper contained valuable recommendations, it was never implemented. On research funding, the paper proposed a peer review structure by mixed teams of Bulgarian and foreign researchers, with a shift from input to output financing. The European Commission has agreed that funding for basic research will be provided, while maintaining a balance with other priorities and activities in the research area. The Bulgarian Government may want to seek EU support for the setting up of a competitive funding scheme for promotion of research and innovation, perhaps along the lines of the Irish Program for Research in Third Level Institutions. This would be consistent with the 2002 Position Paper, and lay the basis for a new tradition of research funding, in line with aims for the European Research Area.

V. CONCLUSIONS

3.62 There is an urgent need to modernize the Bulgarian education system to ensure the country has a highly skilled and productive labor force that can make the catching up with the EU standards of living a reality. This is critical since while the level of education is steadily increasing in EU countries, participation in higher education and education outcomes have been declining Bulgaria.

3.63 The forthcoming EU accession provides an unprecedented opportunity for the Bulgarian government to move the planned education reforms forward, by providing incentives and leverage for change as well as external funding. At the same time, the demographic change and declining age cohorts provide an opportunity to improve the efficiency of public education spending, and ensure it is spent effectively on improving the quality of education and training of reduced numbers of students. The government would need to take advantage of the period prior to accession to make this happen.

3.64 The government would be well-advised not to increase public spending on education as a share of GDP, but rather to seek efficiency gains within the existing resource envelope. This is important, since public spending on education is already on a level comparable to many CEE countries, and spending per student has been steadily increasing, yet education outcomes remain well below CEE levels. Provided credible reform is taken, maintaining the existing level of public spending on education as a share of GDP, including possible EU funds, would be warranted.²⁰ While consolidation of spending is recommended in many other sectors (see chapter 2), given the importance of education for promotion of labor productivity a differential treatment is recommended on education, assuming reforms proceed.

3.65 The reform measures needed would include gradual consolidation of schools, universities, and vocational colleges in light of declining age cohorts to increase student-teacher ratios and release funds for upgrading of curriculum, textbooks, and facilities. The savings from school consolidation should be retained within the sector, and used to finance quality improvements. To raise the quality of education, there is a need to undertake a curriculum reform, develop better teaching materials, improve teacher training, open University administration to outside representatives, and establish periodic national student assessments to evaluate the impact of education. At the upper secondary and university levels, it is critical to establish links with the labor markets to ensure students are equipped with skills required in the market. In a global economy, continuous upgrading of labor skills is increasingly important to stay competitive, and hence would need to be encouraged. Further engaging the private sector in education delivery could also be very important.

3.66 The proposals for projects presented in this chapter for possible EU funding, even though, tilted towards the interface between the education system and the market, form a broad education and human development reform package consistent with the EU education policy. The concerns underpinning the proposals relate to key policy concerns of the Bulgarian authorities, such as: (i) improving the quality of education; (ii) modernizing the educational infrastructure; (iii) improving access for all; (iv) reducing inequalities; (v) promoting social inclusion; (vi) building the capacity of personnel; and (vii) promoting lifelong learning.

3.67 With careful planning and strong leadership Bulgaria can achieve its human resource development goals. The potential is there, but it will not be realized without a great deal of effort. There is a lot at stake for current and future generations of Bulgarians.

²⁰ It is important to note that above considerations mainly apply to education, and not the wider field of human resources development. In light of the present labor market situation with low activity and employment rates particularly for low or unskilled labor, and excess demand for skilled labor, resolute action would need to be taken to raise the low level of participation in life long learning, which may require additional funding.

4. TOWARDS IMPLEMENTATION OF THE EU COMMON AGRICULTURAL POLICY

4.1 Agriculture is still a relatively important economic sector in Bulgaria, accounting for about 25 percent of employment and 11 percent of GDP in 2004. Its productivity, as measured by value added per worker in agriculture, is low and has actually gotten worse in the past few years (Table 2.2 in WB 2005c). This makes it imperative to raise efficiency in agriculture, thereby increasing the competitiveness of the sector in a national and European perspective, and raise the standard of living of people in the rural sector.

4.2 The importance of agriculture is even more pronounced when viewed from the perspective of EU grant transfers. With the expected accession to the EU, Bulgaria will become eligible for large financial support and agriculture is likely to be the largest recipient of transfers from the EU. These funds will come within the EU's Common Agricultural Policy (CAP) support framework, particularly under the two *Pillars* of the CAP. *CAP Pillar 1* will fund direct income support payments to farmers and market measures (buying of products into public storage, surplus disposal schemes, and export subsidies). *CAP Pillar 2* will support rural development measures and, with the evolving shift of CAP priorities, will come to dominate policies in the rural sector over time.

4.3 With the signing of the *Act of Accession* to the EU, Bulgaria has been granted an overall ceiling of CAP-related support payments of about EUR 2 billion for the first three years of EU membership. Of this, the national ceiling for direct income support via the CAP Pillar 1 *Single Area Payment Scheme* (SAPS) amounts to EUR 722 million²¹, EUR 388 million would support CAP-related market measures, and EUR 940 million²² would support rural development under CAP Pillar 2²³. While EU allocations on CAP Pillar 1 funds are not subject to counterpart-financing from the national budget, Pillar 2 funds require a 20 percent national counterpart-financing. From a cash-flow perspective, CAP Pillar 1 income support payments need – although not subject to counterpart-financing requirements – to be pre-financed by the Bulgarian Government, because actual transfer from the EU to the national budget of Bulgaria, covering the financial resources spent in the form of direct payments, will only be made retroactively. As is discussed below, a potential topping-up of EU income support by additional *Complementary National Direct Payments* (CNDPs) would further increase the potential need for fiscal space in the state budget.

4.4 The implementation of CAP raises a chain of strategic decisions to be evaluated and taken into account by the Government. These decisions involve:

- selection of a payment scheme for direct income support measures;
- a potential topping up of area based payments by CNDPs;

²¹ See Act of Accession (AA2005/P/Annex III, point 8, Art. (ab) replacing Annex VIIIA regarding the national ceilings referred to in Article 71c of 32003 R 1782: Council Regulation (EC) 1782/2003 as of 29 September 2003, amended several times afterwards the latest of which is 32004 R 0864: Council Regulation (EC) No. 864/2004 of 29.04.2004).

²² The significant allocation to measures under Pillar 2 of the CAP underlines the increasing importance of rural development and structural aid measures in the reformed CAP; it also reflects the need for such assistance in Bulgaria. Among the candidate countries and new member states, Bulgaria has one of the highest relative ceilings/allocation on such measures.

²³ This includes EUR 733 million under the EAGGF guarantee section (EUR 183 million for 2007, EUR 244 million for 2008 and EUR 306 million for 2009) and EUR 207 million under the guidance section (EUR 47.6 million for 2007, EUR 68.3 million for 2008 and EUR 91.1 million for 2009), now combined under the *European Agricultural Fund for Rural Development* (EAFRD) recently established under Council Regulation (EC) No.1698/2005. The amounts under the EAGGF guidance section added to the EAFRD would lead to equivalent reductions in the Structural Fund allocation for the respective years.

- payment modalities;
- eligibility criteria; and
- transfers among CAP-related EU budget allocations to Bulgaria.

4.5 It is important to understand the implications of these strategic decisions, as they will affect the incomes of farmers and the balance of state budget in Bulgaria. To explore these issues, a partial equilibrium model of the Bulgarian agricultural sector was used to simulate a series of scenarios on the adoption of the EU agricultural policies, and to provide information about the likely impact on producers' value added (farmers' income) and on the state budget.²⁴ Eleven major agricultural products were considered in the model, namely wheat, barley, grain maize, sunflower, tobacco, potatoes, milk, beef, pork, poultry, and sheep and goat. These products account for 68.4 percent of the combined gross agricultural production.

I. PRINCIPLES OF CAP INTRODUCTION IN BULGARIA

4.6 Key aspects of implementing CAP in Bulgaria will include details on the phasing-in of CAP payments, application of the Single Area Payment Scheme (SAPS), and the possibility for Bulgaria to "top-up" EU direct payments to farmers by so-called *Complementary National Direct Payments* (CNDP). CNDPs would have to be primarily financed from the national budget; for the first years of membership they could, however, be partially funded by reallocating up to 20 percent of the EU allocation on rural development measures under CAP Pillar 2. The transitional provisions for introducing CAP in Bulgaria are aligned with the latest overall reform of the CAP currently being implemented in the EU's current member states²⁵. These reforms include the introduction of a Single Farm Payment Scheme (SFPS), which ties direct payment support to farms rather than to production; cross-compliance of farmers with EU environmental, sanitary, and veterinary provisions; and a redirection of overall financial support towards Pillar 2. In order to understand the impact of joining CAP on Bulgarian farmers' incentives, it is necessary to understand the possible modalities of the new payments and the likely amounts involved. It is also important to understand that the negotiations with the EU only set the framework for the overall budgetary ceilings and envelopes, leaving the specification of eligibility criteria for direct payments, the issue of top-ups and the possible re-allocation of budgetary resources among CAP-Pillars to the Bulgarian government.

Phasing-in of Direct Payments

4.7 As is the case in the new EU member states, Bulgaria is scheduled to gradually phase-in CAP direct payments over a period of ten years. During this period, farmers in Bulgaria will receive only a percentage of the direct payments received by farmers in

Table 4.1: Phasing-In Percentages for CAP Direct Payments in Bulgaria, Romania and the 10 New EU Member States

Year		10 new member states	Bulgaria and Romania
2004		25	—
2005		30	—
2006		35	—
2007		40	25
2008	percent of	50	30
2009	EU direct	60	35
2010	payment	70	40
2011	level	80	50
2012		90	60
2013		100	70
2014		100	80
2015		100	90
2016+		100	100

Source: Ministry of Agriculture and Forestry

²⁴ Specifications of the model, its results and detailed analysis underlying this chapter are available in a paper prepared for this report (Kray *et al.*, 2006).

²⁵ Implementation for all current member states is underway. For the NMS-10 that joined the Union in 2004, target date for SFPS implementation is January 01, 2009.

EU15. Assuming that Bulgaria accedes to the EU in 2007, direct payments will start at 25 percent of the EU level in 2007, increase yearly by five percentage points until 2009, and then increase by ten percentage points per year to reach 100 percent of the then applicable EU level in 2016 (Table 4.1).

Single Area Payment Scheme

4.8 The 2003 CAP reform contains a *Single Farm Payment Scheme* (SFPS) to be introduced in the EU starting in 2005. SFPS is based on direct payments that EU15 farmers had been receiving under certain support programs during a reference period (2000 to 2002). As Bulgaria (like most new member states) did not operate CAP-conforming direct payments during the reference period, and due to the high administrative requirements for implementing direct payments on an SFPS basis, Bulgaria, like all new EU member states, has been given the option to choose the *Single Area Payment Scheme* (SAPS) instead²⁶. However, using the SAPS rather than the new SFPS approach is limited to the first three years of membership, with the possibility of two one-year extensions. Before the new member states can change over to SFPS, they must demonstrate that they have the management and control systems in place to do so. If they do not have these systems in place by 2008 (2011 for Bulgaria), they will continue under SAPS, but their aid percentage will be frozen at 50 percent of the EU15 level.

4.9 SAPS provides for a flat-rate, per-hectare payment to farmers to be made once a year, irrespective of the crops produced, or even whether any crops are produced at all, as long the land is maintained in good agricultural and environmental condition. The amount is calculated by the total amount of direct payment funds²⁷ available in the calendar year, divided by the eligible utilized agricultural area²⁸.

4.10 To be eligible for direct payments, agricultural holdings (that is, farms) and parcels used by them have to satisfy a set of criteria regarding their size.²⁹ The minimum size for an agricultural holding to be eligible for support under SAPS is 0.3 hectares. Bulgaria can decide to set it at a higher level, but not higher than 1.0 hectare. In practice, all new member states applying SAPS except Cyprus have chosen to set one hectare as the minimum holding size (Hungary is applying a system with two minimum sizes). Under SAPS, this criterion is complemented with a minimum parcel size eligible for the area payment (for example, Poland has opted for 0.1 ha; Slovakia has opted for 0.3 ha). The application of minimum holding of, say, one hectare would exclude three-quarters of farmers in Bulgaria. However, it would significantly reduce the budgetary and administrative costs associated with providing direct payments to small-scale production units (most of which are of limited viability), and enhance structural change towards a less fragmented farm and land use structure in Bulgaria.

4.11 The simulations are based on an initial SAPS payment level of 68.96 EUR/ha for 2007 (and respective increases in the following years) estimated on the basis of the national ceiling for direct payments and the maximum *utilized agricultural area* (UAA) that could be eligible for support (2,904 million hectares)³⁰. This level of SAPS payment would be among the highest within the new EU member

²⁶ Among the ten new EU member states, all except Slovenia and Malta opted for the SAPS. Source: USAID FAS, GAIN Report E34004.

²⁷ "Market support" payments in certain sectors (that is, payments mainly granted for processing of certain outputs and producer organizations) are not included in the SAPS payment. These schemes will operate in addition to SAPS.

²⁸ The definition of "utilized agricultural area" is total area of arable land, permanent grassland, permanent crops, and kitchen gardens that have been maintained in "good agricultural and environmental condition".

²⁹ Art 143b of Council Decision 2004/281/EC and Art 134 & Annex XX of Commission Regulation (EC) No 1973/2004 of 29 October 2004 laying down detailed rules for the application of Council Regulation (EC) No 1782/2003 as regards the support schemes provided for in Titles IV and IVa of that Regulation and the use of land set aside for the production of raw materials.

³⁰ As of the time of our assessment, the Bulgarian government had not taken a final decision regarding the size of the holdings and parcels eligible for support under the SAPS, though the Ministry of Agriculture has signalled that it

states, where initial (i.e., 2004) SAPS levels amounted to 70.2 (HU), 57.3 (CZ), 44.5 (PL), 43.8 (SR), 35.9 (LT), 26.8 (EST), and 20.7 (LV) euro per hectare of agricultural land. For Romania, the initial (2007) SAPS level is estimated to amount to about EUR 62.7 per hectare³¹. A decrease in the agricultural area eligible for support might even increase the per-hectare payment in Bulgaria to the highest per-hectare support level in the region.³²

Topping-up of Direct Payments

4.12 Within carefully defined limits, Bulgaria (as all new member states) will have the option to supplement the EU direct payments with so-called *Compensatory National Direct Payments* (CNDP). These are also commonly referred to as "top-up" payments. CNDPs are part of the compromise reached with new EU member states to offset the impact of the 10-year phase-in period for direct payments. Under the principle of equal treatment, CNDP will also be applicable in Bulgaria and Romania.

4.13 As a new member state, Bulgaria will have the possibility of topping-up EU direct payments.

Under the so-called *30 percent formula*, Bulgaria could top-up EU direct payments (for all or a specified list of agricultural products) by up to 30 percentage points of EU15 average above the applicable SAPS phasing-in level in the relevant year³³. This would allow Bulgaria to reach 100 percent of the EU15 payment by 2013 instead of 2016³⁴ (Table 4.2). Through 2009, Bulgaria will be allowed to fund the top-up payment in part by diverting up to 20 percent of the rural development funds that the EU will provide after accession (Pillar 2 funds). Whereas the EU direct payments are not subject to national co-financing requirements, all top-up payments that divert funds from Pillar 2 to Pillar 1 must be matched by funds from the national budget. Starting in 2010, Bulgaria would have to finance the topping-up entirely from national funds.

Table 4.2: Maximum SAPS and CNDP Levels in Bulgaria Relative to the EU15, 2007–16

Year		SAPS	Max. CNDP	Total
2007		25	30	55
2008		30	30	60
2009		35	30	65
2010	Percent of EU direct payment level	40	30	70
2011		50	30	80
2012		60	30	90
2013		70	30	100
2014		80	20	100
2015		90	10	100
2016+		100	0	100

Source: Ministry of Agriculture and Forestry

might favor the "Hungarian model". Because of this, the estimate in this report is based on the highest possible threshold (respectively lowest possible SAPS payment per hectare), that is, on the assumption that the total UAA in Bulgaria would become eligible for support. Factors such as minimum parcels' size, minimum holdings' size, etc. would reduce the eligible area. However, most importantly, any decrease in the eligible area would result in an increase in the SAPS payment as well as an increase of the CNDPs that may be granted to farmers, and hence to higher positive impact on farmer incomes than those presented in this report.

³¹ Sources: AGRA FACTS calculations (AF25-04) for EU-8; World Bank estimate for Romania.

³² In principle, Bulgaria would have the option to include public (commonly used) pastures and meadows (about 1.6 million ha) for direct support. This would lead to increase in the supported area and consequently decrease the per-hectare SAPS payment (in which case it would be above 44 EUR/ha). Direct support to public lands in the EU is a rare case (e.g. Ireland, parts of Spain) and is a very difficult to implement process. It requires the establishment of a complex system of contracts and procedures between the parties involved and/or benefiting from the direct support.

³³ Under the alternative *pre-accession-formula*, Bulgaria could top-up EU direct payments to the total level of direct support the farmer would have been entitled to receive, on a product-by product basis, prior to accession under a CAP-type national scheme. Due to the characteristics of Bulgaria's agricultural support policies prior to accession and their limited financial budgets, this option is of rather hypothetical nature (see Kray et al., 2006).

³⁴ In no case at any time should the payment be higher than 100 percent of the EU-15 level of direct payments.

II. ANALYSIS OF THE IMPACTS OF INTRODUCING THE CAP IN BULGARIA

4.14 As outlined in the previous section, CAP implementation in Bulgaria raises a chain of important strategic decisions to be evaluated and taken by the Government. A simple but powerful model of the Bulgarian agricultural sector has been used to simulate the impact of these potential decisions on the financial situation of producers and on the state budget.

4.15 The scenarios evaluated reflect the strategic choices to be made by the government, based on the options negotiated and confirmed in the Act of Accession. The simulations assess the impact of all relevant combinations of the listed options (see Table 4.3)³⁵. The impact of the various scenarios is measured against a baseline scenario represented by the pre-accession support environment. The analysis is performed in both a short-term and a medium-term perspective. Eleven major agricultural products are considered in the model, namely wheat, barley, grain maize, sunflower, tobacco, potatoes, milk, beef, pork, poultry, and sheep and goat. These products account for about 70 percent of the value of gross agricultural production.

4.16 In the model, farmers' income is determined by the difference between revenues from the sale of agricultural produce and tradable input costs and by any non-price-related direct monetary transfers provided to farmers (for example, per-hectare payments to crops or per-head payments to livestock). Any change in agricultural policies that affects input prices, output prices, or direct monetary transfers (such as CAP direct payments) translates into changes in the value-added of agricultural production (that is, farmers' income). In the medium to long term, the level of production of the various commodities is responsive to the level of profitability of each commodity. Therefore, changes in value-added will also lead to changes in total output, further compounding the initial changes in farmers' income. For each commodity the extent of the medium- to long-term supply response is reflected.³⁶

³⁵ Since the number of permutations of all variables (options) yields a large number of scenarios, the discussion of results concentrates on the most significant aspects.

³⁶ The elasticities of supply used in this study are consistent with the elasticities used in similar World Bank studies for other countries in the regions. See for instance, C. Csaki, J. Nash, V. Matusevich, and H.A. Kray, 2002, *Food and Agricultural Policy in Russia: Progress to Date and the Road Forward*, World Bank Technical Paper No. 523; C. Csaki, J. Nash, A. Fock, and H.A. Kray, 2000, *Food and Agriculture in Bulgaria: The Challenge of Preparing for EU Accession*, World Bank Technical Paper No. 481; and A. Valdes and H.A. Kray, 1999, *Lithuania: Adjustments of Agricultural and Trade Policies*, World Bank ECSSD Working Paper No. 16.

Table 4.3: Current Options for Implementing the CAP in Bulgaria

GENERAL OPTION	OPTIONS REFLECTED IN SCENARIOS
<i>Background</i> Introduction of Single Farm Payment (SFP) scheme vs. Single Area Payment Scheme (SAPS)	The MAF has confirmed that Bulgaria will opt for SAPS. The scenarios therefore reflect this scheme only (reduction of complexity of simulation). The simulation is based on the 2007 25 percent phasing-in level.
<i>The current EU15 CAP reform introduces a decoupled SFP. This payment is based on direct payments farmers received under certain support programs during a reference period from 2000 to 2002. As Bulgarian farmers did not receive these direct payments during the reference period and due to the high administrative burden of implementing direct payments, the country will be able to opt for SAPS instead</i> CNDPs: Topping-up EU payments from national budget <i>Subject to approval by the European Commission, Bulgaria may use its own funds (national envelope) to increase payments by a maximum of 30 percent (of the EU payment).</i> CNDPs: Payment modalities for topping-up EU payments	The scenarios reflect four levels of topping-up: TU00*: 0% TU10*: 10% TU20*: 20% TU30*: 30% No decision has been made yet, but a strong preference exists in the MAF for CNDPs to the livestock sector (especially dairy farming) and certain crop sectors (tobacco). The scenarios therefore reflect several types of payment modalities: PM1*: top-up as flat SAPS payment PM2*: top-up as product-specific payment (crop vs. livestock; various CNDPs within livestock sector).
<i>In case of topping-up the EU payment, Bulgaria can decide to grant the top-up as a 'flat' per-area payment or as a payment to specific production activities (e.g. livestock sector activities). From the distributional point of view it is important to note that 'flat' payments to areas producing fodder crops can be considered support to livestock producers.</i> CNDPs: Transfer of budgetary resources from Pillar2 to Pillar1 support measures <i>Bulgaria will be allowed (for a limited time) to fund the topping-up in part by diverting up to 20 percent of the rural development funds that the EU will provide after accession. All such funds must be matched by funds from the national budget (co-financing requirement).</i> Eligibility criteria for holdings	The scenarios reflect three levels of Pillar 2 to Pillar 1 transfers: TP00*: 0% TP10*: 10% TP20*: 20% The MAF considers setting the minimum holding size at the 1 ha level for cereals and protein crops and at the 0.3 ha for intensive crops (vegetables, tobacco, etc.) Minimum holding size HS03*: 0.3 ha (for tobacco and potatoes) HS05*: 0.5 ha (for all crops or only for tobacco and potatoes) HS10*: 1.0 ha (for grains and oilseeds or for all crops)
<i>The minimum size for an agricultural holding to be eligible for support under the SAPS is 0.3 hectare, but Bulgaria can decide to set it at a higher level, but not higher than 1 hectare. In practice, all NMS applying SAPS have chosen to set 1 hectare as the minimum holding size.</i> <i>Under SAPS, this criterion can be complemented with a minimum parcel size eligible for the area payment (Poland has opted for 0.1 ha).</i>	

* name of variable in simulation model.

III. EFFECTS OF INTRODUCING A SINGLE AREA PAYMENT SCHEME ON FARMERS' INCOMES AND STATE BUDGET

A. Effects on Farmers' Incomes

Introduction of the CAP Price Regimes

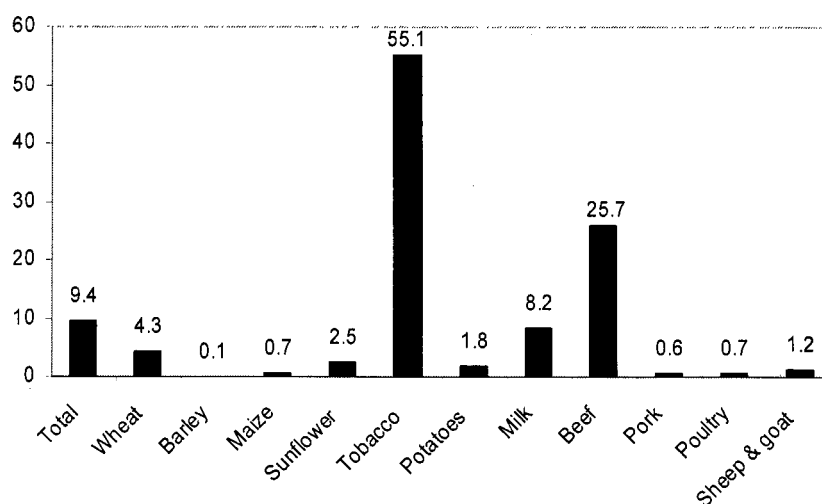
4.17 The introduction of CAP will replace currently applied direct income support schemes. In 2004 these schemes accounted for 9.4 percent of farm income on average (see Box 4.1).³⁷ Support to tobacco farmers accounted for 55.1 percent of the value added, by tobacco farmers, with support to all other sectors except beef paling in comparison.

4.18 Bulgarian farmers have relatively low production costs in crops such as cereals and oilseeds, and high production costs in livestock and intensive crops, but still obtain positive incomes from all of these activities. As the first step in approaching the potential importance of CAP direct payments, current income levels are compared with those that would occur under a pure CAP pricing regime (without direct payments). Although, in reality, the mere introduction of CAP prices without accompanying CAP direct payments is not possible, the results of this (hypothetical) scenario provide important insights into the potential future role of direct payments.

Box 4.1: Increased Importance of the Direct Payment Scheme

Introduction of the CAP in Bulgaria will significantly increase the importance of direct payment schemes granted to Bulgarian farmers. Bulgaria already operates a number of direct payment schemes, but they are not aligned with CAP provisions. Figure 3.1 presents the situation as it was in 2004.

Figure 3.1. Share of direct subsidies and premia payments in farming income from selected crop and livestock production activities, 2004



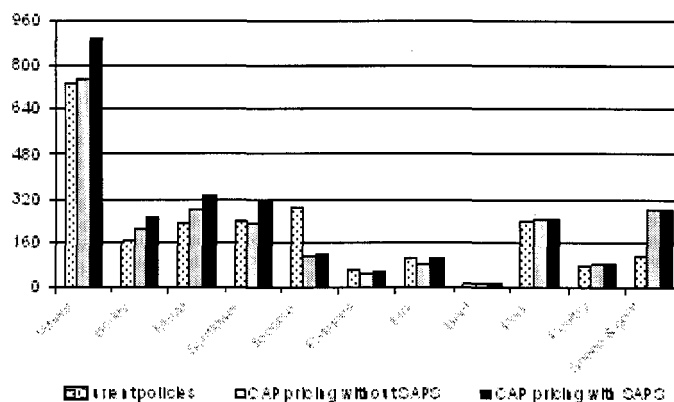
Source: World Bank calculations

On average, current direct payment schemes account for 9.4 percent of agricultural income generated from the activities included in our analysis, with substantial variations across production activities and farm types. Variations among farm types occur due to differences in the product mix of farms and eligibility for direct payments.

³⁷ Estimated as share of total state support for all sectors in the model in total value added (for all sectors in the model).

4.19 The introduction of a hypothetical CAP pricing regime without granting direct payments would increase *average* producer incomes by 3.3 percent (light grey area vs. the solid straight line in Figure 4.2). The impact, however, would be uneven across the various sub-sectors/activities (see Figure 4.1). CAP pricing levels (at current productivity, efficiency, and quality levels) without direct payments would induce income increases for crop sub-sectors with relatively consolidated production structures (wheat, barley, etc.), whereas incomes in sub-sectors with strongly fragmented production structures would decline—some of them substantially (tobacco) (Figure 4.1).

Figure 4.1: Activity Specific Income Effects of Introducing the CAP in 2007, million BGN

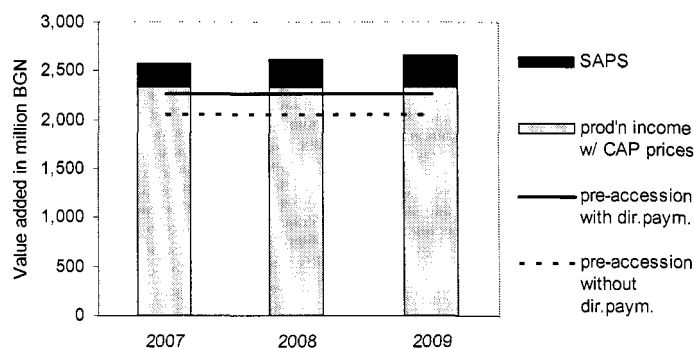


Source: World Bank calculations

4.20 Introducing CAP with SAPS (but in this scenario still without top-ups) further improves the income situation of the farmers. However, the improvement is clearly greater for wheat, barley, maize and sunflower than for tobacco, potatoes, or livestock.

4.21 The basic SAPS (2007 level, without CNDP) will result in an immediate and significant (19.1 percent) increase in primary producers' average income from the analyzed activities, more than compensating for some of the aforementioned negative income effects of an introduction of CAP pricing. The increase in average income for farmers will continue to grow in subsequent years as SAPS payments are stepped up. Although no SAPS payment will be granted to livestock activities as such, livestock producers would enjoy income increases as a result of price increases for outputs, price decreases for some inputs (forages other than concentrated ones) and SAPS payments to their forage area.

Figure 4.2: Sectoral Income Effects of CAP-Introduction, 2007-09



Source: World Bank calculations

Topping up SAPS Payments with CNDP

Box 4.2: Topping Up EU SAPS Payments

The Government faces several options in setting CNDPs:

- CNDPs can be granted as a flat (decoupled) per hectare payment to agricultural land or as production-specific (coupled but in conformity with the CAP) premia granted to specified crop or livestock production activities (subject to the premium quotas negotiated). If implemented as coupled payments, CNDPs contribute to the establishment of payment claims and titles to some producers in the sector. In most cases, allocation of these payment titles increases distributional inequality in the medium to long run and is almost impossible to withdraw once granted.
- CNDPs could have significant implications for the public purse since they must be entirely or partially financed from the state budget. In contrast, SAPS payments are covered entirely by the EU budget.
- CNDPs do not have direct income effects on consumers through the price mechanism, but they nevertheless represent an income transfer from taxpayers to farmers through the fiscal mechanism.
- The impacts on the incomes of the regional or individual agricultural producers vary with the regional or individual product mix and the size of the operations or holdings that are targeted. Simulations show that product mix is more important than the size of the operation/holding in determining the magnitude of these impacts.

4.22 Table 4.4 shows the estimated payments under a SAPS regime used in the model, including the magnitude of a variety of hypothetical CNDP top-ups potentially granted as a flat per hectare payment. CNDPs intended to top up SAPS can be granted to agricultural producers up to a maximum level of 30 percent of direct payments granted to EU15 farmers. This implies that CNDPs could, if granted by the government, be larger than the SAPS payments during the first year of accession. Their relative importance will decrease over time, as SAPS levels increase on a yearly basis.

For the period 2007–09, the phasing in schedule (Table 4.2) suggests that SAPS payments (not including CNDPs) will increase from EUR 68.96 per hectare in 2007 to EUR 96.55 per hectare in 2009.

Table 4.4: Alternative SAPS/CNDP Payment Scenarios for First Year of EU Membership

Item	Value
Base area (hectares) ^a	2,904,479
Allocation on direct payments, 2007–09 (EUR million) ^a	721.7
SAPS support level, 2007 (percent of EU-15) ^a	25
CNDP support level, 2007(percent of EU-15) ^a	max. 30
SAPS payment, 2007 (EUR/ha) ^b	68.96
SAPS + 10% CNDP, 2007(EUR/ha) ^c	96.55
SAPS + 20% CNDP, 2007(EUR/ha) ^c	124.13
SAPS + 30% CNDP, 2007(EUR/ha) ^c	151.72

a. Results of negotiations on Chapter 7 (Agriculture).

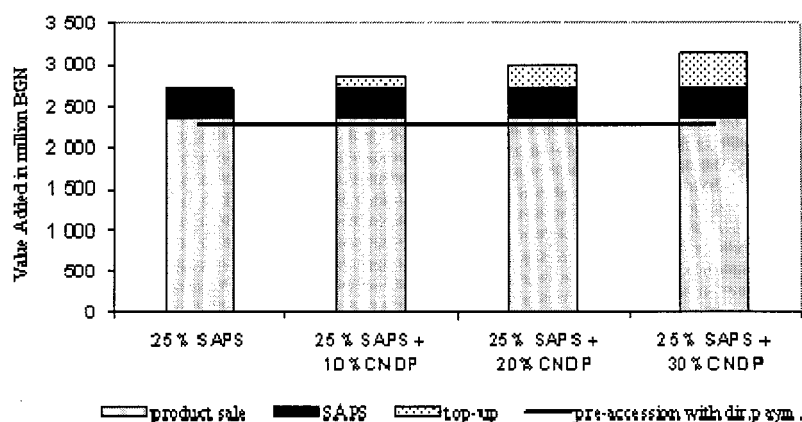
b. Based on World Bank calculations.

c. Assumed for flat disbursement as per-hectare payment.

4.23 **Average income effects for the sector:** As shown in Figure 4.3, the implementation of CAP SAPS will lead to substantial average income increases already in the first year of EU membership. These increases will become even more pronounced as payment levels increase along the phasing-in schedule (Table 4.2). Due to the relatively high level of basic SAPS payment levels in Bulgaria and their positive income effects, the average income of farmers will be more than maintained at pre-accession levels even without the introduction of CNDPs. Introduction of CNDPs would simply add to the already positive effects of the introduction of CAP and the basic SAPS (compare the total of lightly and darkly shaded areas in Figure 4.3 with the solid line). The figure also shows the increase in farmers' income for

hypothetical top-ups of 10, 20 and 30 percent as compared to “basic” SAPS (i.e. without top-ups). The increases in the total farmers’ incomes for year 2007 would amount to 25.5, 31.9 and 38.3 percent for the three levels of top-ups, respectively.

Figure 4.3: Sectoral Income Effects of SAPS and CNDP at 2007 Levels



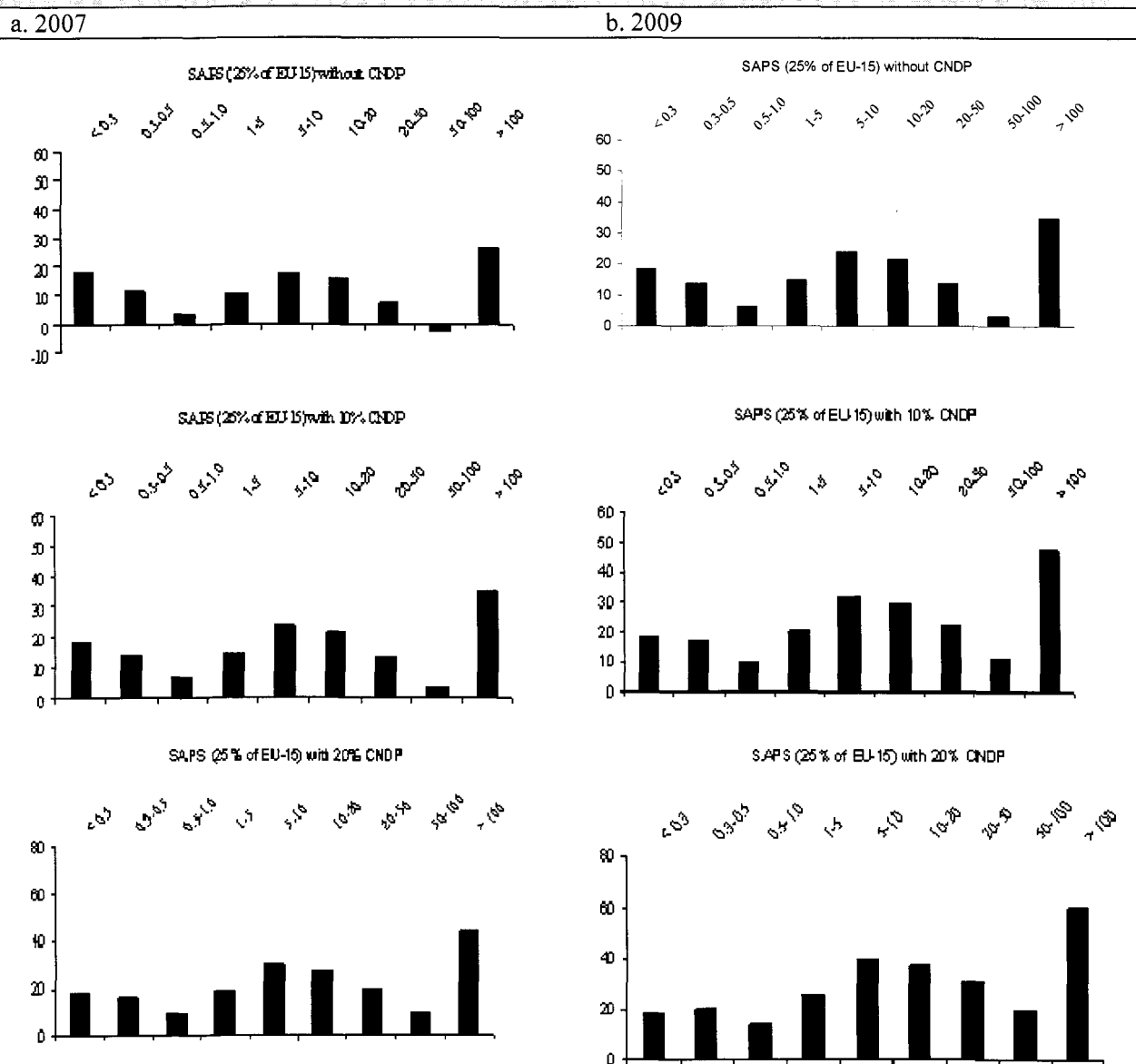
Source: World Bank calculations

4.24 Income effects by farm size: Figure 4.4 shows that the introduction of SAPS payments, even without CNDPs, will, by 2009, have a positive impact on the incomes of all farm size clusters. These income effects are, however, not equally distributed among farm sizes. While almost all farm size clusters would realize substantial income increases, the cluster of farms between 50 and 100 hectares would see their income remaining at pre-accession levels (the displayed 0.9 percent income decrease for 2007 is not analytically significant, potentially originates from the model’s specification, and is considered a crop-effect rather than a size effect³⁸). With the continued phasing-in of SAPS in 2008 and 2009, even this size class would see their incomes increase³⁹ over pre-accession levels.

³⁸ The model does not cover fruit and vegetable farms as well as wine-grape farms. The main reasons for the slight negative impact (-0.9 percent) in the group of farms in the category of 50-100 ha for 2007 are (a) the substantial difference in the number of farms in certain size class, (b) the area cultivated by the size class, and (c) the share of the size class in sectoral value added. Tobacco, for instance, covers only 5.1 percent of the total crop area in the size class, but accounts for 35.4 percent of the total value added of the size class. The decrease in the profitability of tobacco production when its current subsidies are replaced by lower SAPS direct payments is therefore the main factor in the overall income development of farms in this farm size category.

³⁹ The effects for 2008 and 2009 are non-cumulative, i.e. are based only on the SAPS payment for 2008 and 2009 and the respective CNDPs options. No price changes that could further increase the positive impact have been taken into consideration.

Figure 4.4: Projected Change in Producer Income by Farm Size, 2007-2009 Relative to 2004 Levels, in Percent.



Source: World Bank calculations

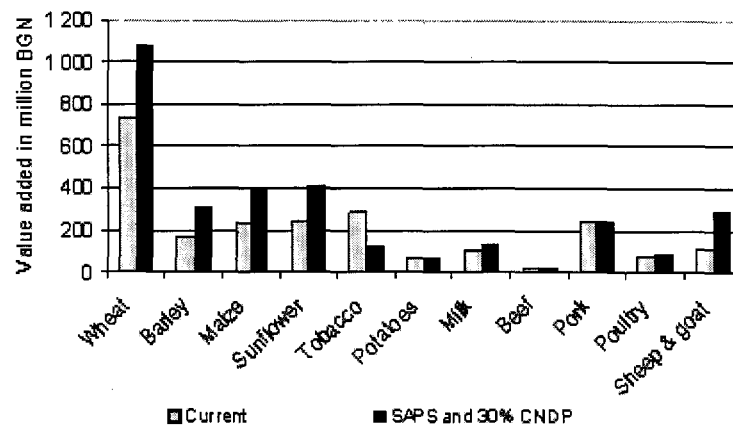
Note: Income effects for farms below 0.3 ha are due only to price effects. Prices in 2009 assumed equal to 2007 price levels

4.25 Income effects by activities: The impact of SAPS also varies between farmers producing different commodities. Figure 4.5, for example, displays the impact on producer incomes in a SAPS scenario with a 30 percent top-up. Such a top-up, while yielding a substantial boost to the incomes of farmers in most activities, would still leave the tobacco and potato growers with income decreases. The farmers in these sub-sectors will no doubt call for targeted (specific, coupled CNDP) support.⁴⁰ The

⁴⁰ Of all reformed agricultural sub-sectors, tobacco is the only one that will be partially de-coupled and partially coupled over the period 2006-2009. By 2010, however, the tobacco sector in the EU should be entirely de-coupled.

tobacco sub-sector is, however, likely to receive support outside the SAPS payment given the combination of coupled and decoupled payments in the EU⁴¹. However, this should be weighed carefully against the fact that commodity-specific CNDPs distort the production patterns, and thus impose efficiency losses on agriculture. From the competitiveness point of view, it would be better for such farmers to be encouraged to improve production efficiency and diversify their income generating activities, using the funds available under Pillar 2.

Figure 4.5: Income Effects on Producers of Selected Commodities of Introducing SAPS and CNDP at 2007 Levels



Source: World Bank calculations

4.26 Income effects by regions: Introduction of CAP

will also have a differentiated regional impact. The regional absorption of the SAPS/CNDP funds will strongly depend on the importance of the supported production activities in total regional agricultural output. Bulgaria has six NUTS2 regions: North-West, North-Central, North-East, South-East, South-Central and South-West. All regions are characterized by a specific agricultural product mix due to the specificity of the climate conditions, soil characteristics, and pre- and post-transition development.⁴²

⁴¹ The amount of additional support to the tobacco sub-sector could not be quantified, since no detail information on the (dis-)continuation of tobacco support schemes was available.

⁴² Bulgaria's NUTS-2 regions differ in product specialization and farm concentration (by type and number). The South-Central is the most advanced in milk, beef, vegetable and intensive crops production (including perennials like vineyards, orchards, etc.). It concentrates about 61 percent of the total tobacco area, half of the area sown with potatoes (49 percent) and more than one-third of the cattle (38 percent). The North-East region specializes in cereal production (wheat, barley and maize) and has the highest share in pigs, poultry and sheep and goat heads. It is the region with the highest concentration of extremely large cereal producing farms (some of them reaching up to 2,000-6,000 ha) and the highest concentration of land cultivated in a consolidated way (despite its ownership fragmentation). The North-Central region is the most balanced in terms of crop and livestock production, with tobacco and potatoes keeping insignificant shares. The South-West region is relatively unimportant from agricultural perspective, while the North-West and the South-East regions are relatively well developed, but not to the extent of the agriculturally three most developed regions.

4.27 The results from the simulations show that already in 2007 significant income gains (of more than 25 percent) will characterize all Northern regions, as well as the South-East, if only the basic SAPS is applied. As SAPS phasing-in levels increase, these gains will rise to more than 35 percent by 2009. The South-Central region, however, will not see an immediate first-year income gain from the introduction of the basic SAPS (Figure 4.6). In general, income gains for the South-Central and South-West regions will lag well behind the gains of the other regions, even after the first year.

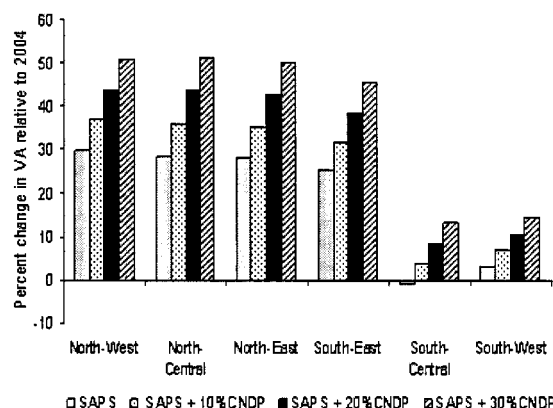
4.28 These results show that whether the introduction of the SAPS is viewed from the angle of the overall sector, the production of specific commodities, farm size, or by region, its impact will be positive (and over time increasingly so), except in a few isolated cases. These results, combined with the negative implications for the budget (discussed later in this chapter) of the introduction of CNDPs and the general reorientation of the CAP toward increasing the competitiveness of the rural economy, strongly suggest that top ups should be resisted

Setting Eligibility Criteria

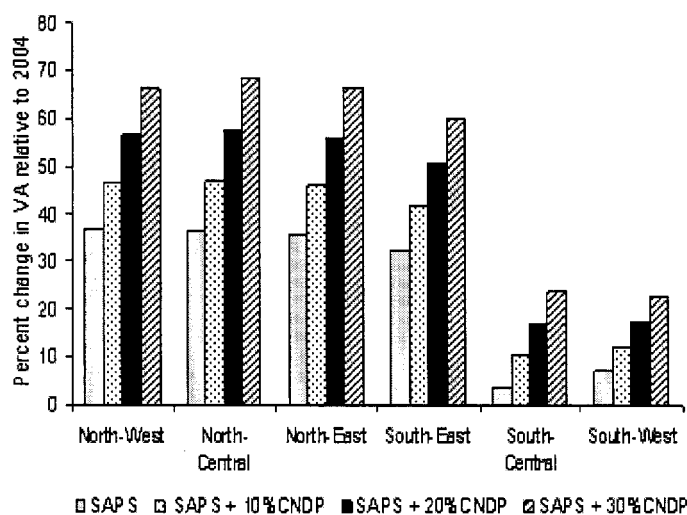
4.29 As indicated before, the government has to set up a minimum holding size eligible for support under CAP. Although thresholds need to be set in order to facilitate structural change and avoid the prohibitive administrative costs of operating SAPS for extremely small farms, they are likely to be controversial. According to the 2003 Agricultural Census conducted by the Ministry of Agriculture, half of the 0.65 million Bulgarian farms are on less than 0.5 hectares of agricultural land and 76.6 percent of them are on less than one hectare of agricultural land⁴³. Setting the eligibility threshold for agricultural holdings at 1.0 ha would thus exclude three-quarters of the farms from support under any SAPS/CNDP direct support schemes, but would only exclude 6.6 percent of agricultural land from SAPS eligibility. Cereal and oil seeds producers would hardly be affected by the highest threshold level of 1.0 hectare. The highest relative exclusion of farms from the analyzed sectors would occur for farmers from the tobacco and potato sub-sector.

Figure 4.6: Producer Income Effects of CNDP Levels by NUTS-2 Regions, 2007-2009

a. 2007



b. 2009



Source: World Bank calculations

⁴³ Strikingly, 96.7 percent of all agricultural farms are on land smaller than 5 hectares.

4.30 If the farm size threshold were set at the EU minimum eligibility threshold of 0.3 hectares, it would exclude 31.6 percent of agricultural holdings, but only one percent of agricultural land from SAPS/CNDP payments. Small farms also have a relatively low share in many of the crop production activities analyzed in this report. If the threshold level is set at 0.5 ha, only 13 and 16 percent of the land used for production of tobacco and potatoes, respectively, will not qualify for SAPS/CNDP payments. However, this would affect about 22 percent of all producers in each sub-sector.

4.31 The potential exclusion of large shares of farmers from direct payment schemes is understandably a controversial issue. This discussion, however, should take into account that the projected introductory level of SAPS payment amounts to EUR 68.96 per hectare a year—an amount that is not likely to substantially increase the incomes of farmers operating less than one hectare of land, yet would greatly reduce the administrative burden. Instead of focusing on the exclusion criteria, attention should rather focus on enabling small and medium-size farmers to effectively absorb the funds allocated to CAP Pillar 2 to raise their competitiveness and/or ease their exit from farming by providing them with adequate retirement schemes. CAP direct support schemes cannot substitute for the need to raise the productivity and improve the competitiveness of the Bulgarian agriculture.

4.32 Based on the simulation results, a one-hectare threshold will correspond to a sectoral income increase of 18.0 percent (for SAPS without CNDPs)⁴⁴. For a 0.5 hectare threshold, this value goes up to 18.6 percent. If a mixed approach is considered (1.0 hectare for cereals and oil seeds, and 0.3 or 0.5 hectares for tobacco and potatoes), the value would amount to 18.2 percent (or 18.1 percent)⁴⁵. These sectoral averages do not provide full insight into the distributional inequalities between farmers who are eligible and those who are not. The average income from the production of the 11 analyzed commodities reaches 95 percent of that of farmers eligible for support under these schemes. These results are based on an assumption that farmers not eligible for direct payments still benefit from the anticipated price increases in the livestock production sector⁴⁶.

Efficiency Considerations of SAPS Payments: Important Caveats

4.33 Without underestimating the social dimension of exclusion from direct income support payments, the role of direct payments in solving agricultural income problems must not be overestimated. Direct payment schemes such as the SAPS can provide some level of income stability and help mitigate farmers' exposure to income risk by guaranteeing a minimum income from farming. They cannot, however, substitute for the need to raise the productivity and improve the competitiveness of Bulgarian agriculture. Increasing productivity and competitiveness remains the only long-term and sustainable solution to raising agricultural income. Agricultural support schemes targeting specific sectoral adjustment needs, such as farm consolidation, productivity, or competitiveness-enhancing farm-level investments, would be better addressed under the CAP Pillar 2 umbrella. Similarly, the application of a direct income support scheme cannot substitute for the implementation of Pillar 2 measures, such as an (early) retirement scheme for the agricultural sector.

4.34 In addition, direct income support payments may reduce the incentives for farmers to restructure, consolidate, and modernize. They may lower incentives for consolidation, because they guarantee a minimum income from farming, reducing the incentive for unprofitable or elderly farmers to sell or lease

⁴⁴ Respective increase of the SAPS payment per hectare due to a decrease in the supported/eligible area is not taken into consideration. If taken into account, the estimated impacts would be higher mentioned above.

⁴⁵ Dairy and beef producers are also taken into account when setting up the threshold levels based on the used area by the milk and beef producers.

⁴⁶ Eventual inclusion of the commonly used (public) pastures and meadows in the area eligible for support would result in a positive sectoral income effect of about 13 percent on average.

their assets to more productive entrepreneurs. SAPS payments and their effects on land prices or land rental rates may also make land consolidation more difficult, in particular in a farming environment in which agricultural credit is not well developed. Evidence from EU member states indicates that support payments granted on a per hectare basis lead to significant increases in prices for buying, and especially leasing agricultural land. This will significantly disfavor tenants, slow any consolidation toward more viable farm sizes, and in effect support to landowners rather than farmers. These tradeoffs between direct income support payments, and the need for higher productivity and competitiveness point to the need to clearly separate rural social measures (such as measures to reduce social tensions and provide social protection in the rural areas) from the major instruments of agricultural income support.

4.35 Direct payments may boost the financial sector, and increase the attractiveness of the agriculture to the banking sector. Experience from the new EU member states shows that banks are assisting farmers in applying for direct payments and crediting their activities against the guarantee of the secured direct payment. Though such assistance may lead to a small outflow of funds out of agriculture, it will strengthen the relationships between the banking sector and agricultural producers, and create necessary conditions for increased investments in agricultural assets and efficient use of the direct support.

Targeting Income Disparities through CAP Pillar 2 Funds

4.36 Bulgaria's rural sector requires restructuring to make it competitive and efficient. Many of the new EU member states required such restructuring, but the need is even more pronounced in Bulgaria, where half of all holdings are less than 0.5 hectare, and another 26 percent are between 0.5 and 1.0 hectares. A recent poverty study found that rural poverty in Bulgaria had declined. One of the hypotheses for this somewhat surprising result was that many rural farming households supplement their fixed pension income with transfers from relatives (mostly their own children) working in towns or abroad, from household plots, and from renting out part of their land. However, rural/agricultural credit remains a major problem in Bulgaria--with important implications for co-financing of investment proposals. Other issues implied in the rules for state support or access to financial sources (for example, to document several years of experience in the business, the need to have been active in the specific activity, the need of some residency status, etc.) have complicated applications of small and starting companies for investments and state support, and have been sometimes prohibitive for proposals of new enterprises. However, measures need to be adopted that allow inefficient farms to close down (through effective early retirement schemes, for example), and obstacles to the expansion of new and more efficient farming units to be removed.

4.37 Although direct income support measures can maintain or increase sectoral income at pre-accession levels in the short term, they cannot be relied on as a means to safeguard agricultural income in the medium to long term. The most powerful and only sustainable way to sustain agricultural incomes is to make agriculture more productive and competitive. Any attempts to solve structural problems through the SAPS/CNDP regime will not improve the sector's productivity and efficiency, but only weaken its structural adjustment. It is therefore recommended that rural development measures and rural social measures be clearly separated from the major instruments of agricultural income support.⁴⁷ The targeted

⁴⁷ If, for instance, coupled CNDPs were implemented to safeguard the incomes of small producers in less favored areas (with the aim of maintaining extensive grazing systems in mountainous areas, for example), further analysis on regional holding structures and the differences in their profitability would be needed. Experience from the Slovak Republic has shown that a coupled CNDP scheme, initially intended to support suckler cow production in less favored areas (LFA), has mainly favored larger scale herds in the most fertile regions of the country. Decisions on specific cases that might be under consideration by the Ministry of Agriculture should always be evaluated against the opportunity to achieve a similar sectoral income effect with a more flexible and less specific type of instrument (specific payments coupled to dairy cattle versus payments based on forage area, for example), and against the opportunity to achieve a similar (or even higher) sectoral income effect by effectively targeting the

and judicious application of rural development measures under CAP Pillar 2 that meet the structural needs of the agricultural sector has a key role to play in providing agriculture with the ability to sustainably take advantage of its integration to the highly competitive EU agricultural market.

4.38 Bulgaria's accession to the EU is expected to take place the same year that a reformed policy framework to CAP Pillar 2 measures will be introduced. The new framework will seek to reinforce the EU's rural development policy and simplify its implementation. According to the current proposal, the European Commission expects this reformed rural development policy to play a more important role in the new, reformed CAP (the 2003 reform of CAP Pillar 1). The proposal recommends introducing a single funding and programming instrument, in an effort to simplify management and control of the program. Under this concept, the member states and regions are expected to have more freedom in implementing the program. This freedom may be advantageous in appropriately targeting country-specific rural development needs; the country-specific strategic framework for Bulgaria, however, is still in the process of being prepared and thus could be included in the simulation model.

4.39 The main features of the new EU rural development policy include the following:

- One funding and programming instrument, the European Agriculture Rural Development Fund (EARDF).
- A genuine EU strategy for rural development, with sharper focus on EU priorities.

Box 4.3: Potential Measures under Axis 1 of CAP Pillar 2

Measures aimed at improving human potential:

- vocational training and information actions for people engaged in the agricultural and forestry sectors
- setting up of young farmers
- early retirement of farmers and farm workers
- use of advisory services by farmers and forest holders
- setting up of farm management, farm relief, and farm and forestry advisory services.

Measures aimed at restructuring physical potential through:

- modernizing farms
- improving the economic value of forests
- adding value to primary agricultural and forestry production
- improving and developing infrastructure related to developing and adapting agriculture and forestry
- restoring agricultural production potential damaged by natural disasters and introducing appropriate prevention actions.

Measures aimed at improving the quality of agricultural production and products:

- helping farmers adapt to demanding standards based on European Community legislation
- supporting farmers who participate in food quality schemes
- supporting producer groups through information and promotion activities for products under food quality schemes.

Transitional measures for new member states:

- supporting semi-subsistence farms undergoing restructuring
- supporting the establishment of producer groups.

funds available under Pillar 2 of CAP (in the case of milk producers in less favored areas, the LFA schemes under Pillar 2 would represent a more targeted approach). Direct payment schemes granted only to specific production activities require a more complex administrative framework and give rise to distributional inequalities in the medium to long run.

- Reinforced control, evaluation, and reporting.
- A strengthened bottom-up approach. Member states, regions, and local action groups will have more influence on tailoring programs to local needs. The scale of the CAP Pillar 2 measures expected for Bulgaria (EUR 940.0 million for 2007–09) is larger than direct income support for the farming community (EUR 721.7 million for 2007–09). In light of the lessons learned from SAPARD implementation, a key concern for the first years of Bulgaria's EU membership is whether both administrative and sectoral absorption capacity will be adequate effectively absorb the CAP Pillar 2 allocation.
- In the light of the challenges faced, it is advisable to address the structural adjustment needs of Bulgarian agricultural by concentrating on the measures foreseen under Priority Axis 1, (measures improving the competitiveness of farming and forestry) of the new EU rural development policy (Box 4.3). According to the tentative allocations of the recent European Commission proposal on the reformed EU rural development policy, the Bulgarian government will have to allocate at least 10 percent of the national envelope on Axis 1; the maximum allocation determined by the minimum allocations for the other two axis amounts to 62.5 percent of the national envelope.

4.40 If not addressed in a timely and appropriate manner, the implementation of Pillar 2 measures may create a situation in which already well-performing larger farms (100 hectares and above) are the main beneficiaries of most of the support schemes. Half of all farmers (99 percent if all farms less than 100 hectares are considered) may not benefit from or take advantage of Pillar 2 allocations in adjusting to the future policy framework of Bulgaria's agri-food sector. The economic and social implications of their failure to benefit from Pillar 2 allocations could be more pronounced than their lack of eligibility for Pillar 1 support measures.

Table 4.5: Axes of Reformed EU Rural Development Policy (CAP Pillar 2)

<i>Axis</i>	<i>Goal</i>	<i>Examples</i>	<i>Minimum allocation</i>	<i>Max. EU co-financing</i>
1	Improve competitiveness of farming and forestry	Improving and developing infrastructure related to the development and adaptation of agriculture and forestry, supporting farmers who participate in food quality schemes, setting up of young farmers, and helping semi-subsistence farmers in new member states become competitive	10%	50% (75%)*
2	Improve the environment and land management	Natural handicap payments to farmers in mountain areas, NATURA 2000 payments, agri-environment measures, and animal welfare payments.	25%	55% (80%)*
3	Improve quality of life and diversification	Diversification to nonagricultural activities, support for the creation of micro enterprises, encouragement of tourism, village renewal	10%	50% (75%)*
4	LEADER+		2.5%	50% (75%)*

Source: Council Regulation (EC) No 1698/2005

^a To apply only from 2010-2013

* in convergence regions

4.41 In addition to rapidly scaling up its efforts to upgrade the required administrative infrastructure and designing a more targeted programming strategy, the government must focus on facilitating the uptake of the program by small and mid-size farms. This adjustment should be carried out in the light of the findings on potential effects of CAP Pillar 1 measures. The measures aiming at structural improvements need to be appropriately complemented by measures aiming at improving the environment and land management (Axis 2), and improving the quality of life and diversifying rural areas (Axis 3) (Table 4.5).

B. Implications of CAP Implementation Options for the State Budget

4.42 In choosing the final SAPS implementation package, the Government will have to base its decision not only on its likely impact on the incomes of producers but also on the impact on the national budget (which affects all taxpayers). While the overall support ceilings and commitment envelopes are set for CAP Pillars 1 and 2, the ability for Bulgaria to top up the direct payments (either fully from its own budget or by diverting up to 20 percent of the funds available under Pillar 2) confronts the government with a number of financing options.

4.43 The ceiling for EU support to Bulgarian agriculture for its first year of EU membership amounts to about EUR 560 million. Of this, about EUR 200 million will be available for direct income (SAPS) support measures under Pillar 1 of CAP in the first year of accession, which need to be pre-financed from the national budget (but are expected to be reimbursed in early 2008). A further EUR 231 million will be available for agriculture-related rural development measures from the EU budget in 2007, out of a total of EUR940 programmed for this purpose for the period 2007-09. Since Pillar 2 support is subject to counterpart-financing requirement of 20 percent, an additional EUR 58 million would have to be raised from the national budget in 2007. Total CAP-related support to Bulgaria's rural sector could therefore amount to about EUR 618 million in the first year of accession (Table 4.6).

Table 4.6: CAP-Related Budget Allocations, 2007-09

	<i>Programming period</i>			<i>First year of membership</i>		
	<i>2007-09</i>			<i>2007</i>		
	<i>Ceilings for EU transfers*</i>	<i>National counterpart-financing</i>	<i>Total public spending</i>	<i>Ceilings for EU transfers*</i>	<i>National counterpart-financing</i>	<i>Total public spending</i>
<i>Measure</i>	<i>(EUR million)</i>			<i>(EUR million)</i>		
Direct payments	722	0	722	200	0	200
Market measures	388	0	388	129	0	129
Rural development	940	235	1,175	231	58	289
Total CAP Bulgaria	2,050	235	2,285	560	58	618

* Actual transfer from EU to Bulgarian budget carried out retroactively in t+1

4.44 The disbursement of funds under all three categories (direct payments, market measures and rural development) will be subject to the receipt of applications. Actual disbursement will depend on Bulgaria's absorption capacity for these funds. The figures in Table 4.6 should therefore be viewed as commitments only, and they are given in constant 2004 prices.

4.45 In raising the budgetary resources that would be required should the government decide to top up the SAPS payments, the government faces two options. The *first option* would be to finance topping up entirely from the national budget. The *second option* would be (for a limited time) to fund CNDPs in part by diverting up to 20 percent of rural development funds that the European Union will provide for that

purpose. However, all such re-allocated funds would be subject to the counterpart-financing requirement for CAP Pillar 2 funds. Simulations were conducted to examine the implications for the national budget of the introduction of SAPS without top-ups, and SAPS with various levels of top ups under the two financing options.

4.46 The simulations reveal that the national budgetary allocation required to co-finance CAP in Bulgaria for the period 2007-09 range from EUR 235 million (SAPS without top-up, and with own budget resources used only as co-financing for Pillar 2 activities) to EUR 957 million (30 percent top-up with entirely national sourcing, plus funds needed for co-financing of Pillar 2 activities) (see Table 4.7). As expected, the potential draw on the national budget is larger, the greater the hypothetical CNDP payments, and the less the Pillar 2 funds are diverted to Pillar 1 activities.

Table 4.7: Budgetary Implications of Sourcing Alternatives for Topping Up SAPS, 2007–09

<i>Required national contribution to:</i>					<i>Funds available for:</i>			
<i>Top-up (% of EU-15)</i>	<i>Sourcing of top-up*</i>	<i>DP</i>	<i>Pillar 2 RD measures</i>	<i>Total</i>	<i>DP</i>	<i>Pillar 2 RD measures</i>	<i>Total CAP (Pillars 1 and 2)</i>	<i>Share of national contribution in total available public funds (%)</i>
		<i>(million EUR)</i>			<i>(million EUR)</i>			
0	n.a.	0	235	235	722	1,175	2,285	10.3
10	National budget only	241	235	476	962	1,175	2,525	18.8
10	10 % of EU P2 allocation plus national budget	165	216	382	962	1,081	2,431	15.7
10	20 % of EU P2 allocation plus national budget	90	197	288	962	987	2,337	12.3
20	National budget only	481	235	716	1,203	1,175	2,766	25.9
20	10 % of EU P2 allocation plus national budget	406	216	622	1,203	1,081	2,672	23.3
20	20 % of EU P2 allocation plus national budget	331	197	528	1,203	987	2,578	20.5
30	National budget only	722	235	957	1,443	1,175	3,006	31.8
30	10 % of EU P2 allocation plus national budget	646	216	863	1,443	1,081	2,912	29.6
30	20 % of EU P2 allocation plus national budget	571	197	769	1,443	987	2,818	27.3

Source: World Bank calculations

Note: EUR 388 million for market measures included in Total CAP column.

4.47 Bulgaria's entire budget for agriculture and forestry is currently running at about 1.6 percent of GDP (EUR 334 million). This includes EUR 122 million under the State Fund Agriculture and EUR 82 million under the Tobacco Fund. These are the two principal sources of current agricultural subsidies. Since these subsidies, in terms of an aggregate amount, will be more than replaced by funds from the EU, there should be plenty of room for about EUR 80 million (EUR 235 million over the period 2007-09) needed to co-finance rural development activities under Pillar 2, either from the two Funds or from other parts of the national budget for agriculture and forestry. However, using the national resources for CNDP payments could quickly put a strain on the budget, and call into question the opportunity cost of these funds. In an economic environment in which the Government is trying to keep in check the size of the state in the longer run, and run a budget surplus to put a lid on the burgeoning current account deficit, directing any national resources toward CNDP payments would be poor economic policy.

IV. SUMMARY: IMPLICATIONS OF THE GOVERNMENT'S STRATEGIC DECISIONS

4.48 **Selection of a Payment Scheme:** Owing to the complexity of implementing a Single Farm Payment Scheme (SFPS), Bulgaria should opt for a Single Area Payment Scheme (SAPS) as was done by almost all new EU member states. It will then have a window of three to five years to prepare for a conversion to SFPS, before penalties for non-conversion kick in.

4.49 **Topping-Up (Complementary National Direct Payments):** Calculations show that the introduction of SAPS payments to agricultural producers, starting at a level of 25 percent of the EU15 average in 2007, would increase average income of agricultural producers by 19.1 percent already in the first year of accession. This would suggest that no top-ups are required to maintain and increase average producer incomes. However, the impact would be uneven across the producers of various agricultural products. Income from tobacco production would even decline as current direct payments are replaced with those under CAP. Producers of dairy would also suffer an initial small loss of income, as would some medium to large-size farmers (50-100 hectares) due to crop-specific effects rather than size effects. The latter, however, would only be a short-term impact, and turn into income increases in the first year after accession. It, therefore, does not justify top-ups. Topping up is expensive, competes with other more productive programs, and establishes a sense of entitlement which makes it politically difficult to remove it later. Given the arguments in this report on keeping the size of the state in check and on focusing all new expenditures on improving productivity, topping-up is not the way forward. A different mechanism of support should be sought if it is necessary for political reasons, perhaps best done through the financing potentially available under Pillar 2.

4.50 **Payment Modalities:** If, nevertheless, topping-up is chosen, simulations carried out for this report suggest that the application of a specific, coupled CNDPs to limited sub-sectors (tobacco) should be favored over the implementation of broad, sector-spanning top-ups. Moreover, such support should have a sunset clause. Also, since support to specific products introduces distortions into production patterns, the form of CNDPs chosen should be as simple as possible. Support to particular regions or sub-sectors would best be addressed under Pillar 2 as part of general rural development of a specific region. This would also fit better into strategic considerations of internal convergence among Bulgaria's regions, and with the expected strategy for rural development in the EU in general.

4.51 **Eligibility Criteria:** According to EU rules, the minimum parcel size for which payments may be requested under SAPS is 0.3 ha, and the minimum holding size eligible for support under SAPS is 1.0 ha. Bulgaria can decide to set the minimum size of eligible area per holding at a level higher than 0.3 ha, as long as it does not exceed one hectare. Fixing the eligibility threshold at the 1.0 hectare level would exclude about 76 percent of all farms, but only 6.6 percent of agricultural land in Bulgaria. Some preference was noted in the Ministry of Agriculture for a mixed threshold model of 0.3 ha for intensive farming and one ha for all others. This is possible, although this report has argued for a single eligibility criterion of one hectare. In making this decisions the authorities will have to weigh carefully the benefits of structural change that would be promoted through consolidation of farms into larger more efficient units, and the reduction of costs stemming from not having to administer the SAPS for very small farms versus the adjustment costs for the farmers that would retire.

4.52 **Transfers among Budgetary Allocations:** If Bulgaria decides to top up, there are two ways to do it. It can top up the direct payments coming from the EU from its own resources, or it can top them up by using up to 20 percent of the monies available under Pillar 2, but it must provide counterpart-financing of these diversions from Pillar 2 funding from its own resources. The Pillar 2 route is obviously fiscally less demanding, but it would take funds away from activities that potentially have a much larger impact on the competitiveness of Bulgaria's agriculture than direct payments to farmers. Detailed calculations of the various options are provided in the previous section.

4.53 Efficiency Considerations: Direct payment schemes such as SAPS can provide some level of income stability, and help mitigate farmers' exposure to income risk by guaranteeing a minimum income from farming. Direct support schemes, however, cannot substitute for the need to raise the productivity and improve the competitiveness of Bulgarian agriculture. Productivity growth and improved competitiveness remain the only sustainable solution to agricultural income problems. Agricultural support schemes targeting specific sectoral adjustment needs, such as farm consolidation, productivity or competitiveness-enhancing farm-level investments, and diversification to off-farm activities is the way forward. These can be tailored to the specific needs of the different regions, and it can all be done under Pillar 2.

4.54 In addition, direct income support payments may reduce the incentives for farmers to restructure, consolidate, and modernize. Direct income payments may reduce incentives for consolidation, because they guarantee a minimum income from farming, and therefore reduce the incentive for unprofitable or older farmers to sell or lease their assets to more productive entrepreneurs. SAPS payments and their effects on land prices or land rental rates may also make land consolidation more difficult, in particular in a farming environment in which agricultural credit is not well developed. Evidence from EU member states indicates that support payments granted on a per hectare basis lead to significant increases in prices for buying and especially leasing agricultural land. This will significantly disfavor tenants, slow the consolidation toward more viable farm sizes, and would in effect support land owners rather than farmers. These trade-offs between direct income support payments, and the need for higher productivity and competitiveness point to the need to clearly separate rural social measures (such as measures to reduce social tensions and provide social protection in the rural areas) from the major instruments of agricultural income support.

5. ROAD INFRASTRUCTURE FOR GROWTH

5.1 Bulgaria is in line to receive significant resources through the EU Cohesion Fund that is targeted for the transport sector. While the government needs to explore options for the optimal use of these EU grants and Bulgaria's own funds to improve roads, railways, ports, airports, and other forms of transport, this report only examines expenditures and plans for the road network. A similar analysis will be subsequently required for the other transport sub-sectors, particularly rail, inland waterways, ports, aviation, and urban transport.

5.2 In light of the overall condition of the existing road infrastructure, and Bulgaria's commitments in the Accession Treaty to improve these, this chapter presents an assessment of road investments that could be financed within the government's medium term financial framework that includes the projected EU grants for the transport sector.⁴⁸ The aim is to identify investments that would be important for promoting Bulgaria's growth potential, in part by removing the major bottlenecks on the road network. Based on projected economic rates of return, traffic densities, and the eligibility requirements for EU grants, a prioritized list of road segments for construction and rehabilitation will be proposed, and suggestions made for utilization of the EU funds.⁴⁹ Since the final amounts of the EU Cohesion Funds to be allocated to roads will be confirmed later, the government will need to adjust the proposed road investment strategy to fit within the agreed medium term expenditure framework and the available EU grants.

5.3 Roads provide the dominant mode of transport in Bulgaria. In a survey of road freight transport costs among EU accession countries carried out in 2001 by the International Road Union (IRU), Bulgaria was recorded to have the lowest transport cost at about 75 percent of the mean value for all EU accession countries. Roads carry about two thirds of the combined road plus rail freight market (measured in ton-km including international traffic), and effectively dominate the short distance, high value, and time sensitive cargo. In the passenger segment, road transport competes aggressively with rail and has gained a share of over 70 percent of the intercity passenger transport.

5.4 The road network in Bulgaria is well developed totaling over 37,000 km of which 19,000 km are national roads managed by the Road Executive Agency (REA). While the road infrastructure has coped reasonably well during the significant transformation of the economy over the past 10 to 15 years, it is now feared that the quality of the infrastructure as well as its capacity on certain sections could become bottlenecks hindering the efficient movement of goods and people. This is becoming critical as Bulgaria aspires to join the EU, and the economy is growing at a relatively fast pace. Recent surveys have shown that only one-third of the national road network is in good condition, and road capacity around the larger urban centers is not sufficient to satisfy the growing traffic demand. Furthermore, a considerable portion of the core national road network still needs to be upgraded to European technical and safety standards despite the ongoing Government upgrading program implemented under Transit Roads projects. Bulgaria has committed to completing the upgrade of its main road network to EU standards by 2014, with a number of roads upgraded by 2008. The road infrastructure problems are largely due to insufficient

⁴⁸ The analysis was also carried out to assist the Government in developing its own Sectoral Operational Program for Transport (SOPT) as required by the EU. Hence, the format of this chapter follows as much as possible the standard format of EU SOPTs.

⁴⁹ The analysis presented draws on earlier work by the World Bank, namely: (i) Bulgaria - Policy Note on Road Investment Financing Options, Bank report, June 2005; (ii) Bulgaria - Review of Motorway Projects and Potential for Public Private Partnerships, Bank report No. 27917, March 2004; and (iii) Bulgaria - Preparation of a 15-Year Road Maintenance and Rehabilitation Strategy, World Bank Consultant report, June 2004.

allocation of funds for road maintenance and rehabilitation. In addition, there is a need to strengthen the road agency's capacity to assess priorities and to efficiently implement improvement programs.

I. BULGARIAN ROAD NETWORK AND ITS STRENGTHS, WEAKNESSES, OPPORTUNITIES, AND THREATS

Strengths

5.5 Aligning Bulgaria's transport policy and regulations with those of the EU has put the transport sector in a strong position to support the country's integration with the EU. This has established the necessary pre-requisites for Bulgaria's full integration within the EU transport market. In addition to transitional arrangements regarding road freight, which will have an impact on road transport demand, Bulgaria has until December 31, 2013 to gradually increase axle-load limits on the national road network. Implementation of this policy will require significant investments and financial resources both from the state and the private sector.

5.6 In general, access is not an issue in Bulgaria as the road network is well developed and reaches all communities. The road network extends over 19,000 kilometers of national (Republican) roads managed by the REA, in addition to 18,000 kilometers of other roads managed by local jurisdictions. More than 98 percent of the national road network is paved. There are also about 3,310 bridges that span 5 m or more, and about 31 major tunnels with an average length of about 340 m. The core network consists of 7,000 km of national roads that includes the motorways, and Class I and II roads. Of these about 3,000 km are classified as E roads of international importance, mainly aligned in the East-West direction reflecting the transport flow patterns. International traffic is carried primarily by motorways and Class I roads all of which begin or end at an international border crossing. Class II roads connect major population centers, and are often used by international traffic as they provide good level of service and direct travel. However, the road network connectivity with neighboring countries is relatively limited, given the length of the Bulgarian borders. There are some 21 border control check points in operation, but only about six of these carry the bulk of the traffic.

5.7 Only a few roads experience capacity constraints at present. This is confirmed by the average speeds currently reported on the network—about 70 km/h for cars on Class I roads and 110 km/h on motorways. Trucks achieve overall speeds of 40-50 km/h on Class I roads and 90-100 km/h on motorways. However, road traffic flow is constrained through the major urban centers because of the absence of bypasses. For international truck traffic, delays at border crossings and through large urban centers are far more significant than delays along the roads.

Weaknesses

5.8 While road infrastructure supply seems adequate for the present demand, the quality of the road infrastructure is an issue. The main problem for road users is the poor condition of roads. In addition, the road network was originally designed with low bearing capacities (below EU requirements) resulting in restrictions on axle loads along some roads, which is a concern for truck operators. The road network also has a major

Table 5.1: Road Network and Condition

Road Class	Length (km)	Good (percent)	Fair (percent)	Poor (percent)
Motorways	328	70	27	3
Class I	2,961	51	19	30
Class II	4,012	37	29	34
Class III	11,964	28	33	39
Total (Average)	19,265	34	30	36

Note: Around 18,000 km of Class IV local roads are not included.

backlog of road rehabilitation owing to lack of maintenance and poor quality construction in the past. Only one-third of Bulgaria's national roads are in good condition (Table 5.1). About 900 km of Class I roads are in poor condition requiring immediate rehabilitation.

5.9 The major reason for the poor road condition is inadequate financing for road maintenance. The annual financing required for routine maintenance and repair of the national road network (excluding Class III roads) has been estimated by REA to be about Euro 61 million⁵⁰. If allowance was made for periodic maintenance (including the need for pavement rehabilitation), the total annual financing required would amount to at least Euro 200 million. However, total maintenance and rehabilitation expenditures have amounted to only Euro 90-130 million in recent years, with typically Euro 50 million spent on routine maintenance and repair, and Euro 70 million spent on rehabilitation.

5.10 The total spending on roads in Bulgaria over the period 1999-2003 was on average about 1.21 percent of GDP, of which maintenance represented 0.39 percent, rehabilitation 0.48 percent, new construction 0.28 percent, and debt service 0.12 percent. However, total spending on roads as a percent of GDP has been decreasing and reached 1.05 and 1.03 percent in 2002 and 2003, respectively (see Table 5.2). Although maintenance spending has been increasing during the past five years from 0.21 percent to 0.38 percent, it is well below the levels in EU15 countries which range between 0.5 and 1.0 percent of GDP⁵¹. The high share of rehabilitation spending in the past five years has been due to the Transit road projects, which aimed at reducing the large backlog in maintenance and rehabilitation. This will need to continue in the medium term until the backlog of rehabilitation has been cleared. However, Bulgaria's planned accession to the EU in 2007, combined with the emphasis on developing the major transport corridors in order to expand infrastructure capacity, has led to a renewed focus on motorway development. As a result, there is a risk that expenditure on maintenance and rehabilitation might be displaced in the coming years by the need to finance construction of the motorway network.

Table 5.2: Annual Road Expenditure (million Euro), 1999-2005

	Actual						Planned
	1999	2000	2001	2002	2003	2004*	2005
Current Expenditures Total	45.0	60.8	58.5	80.3	87.0	116.7	98.7
Routine Maintenance and Repair	25.9	30.9	40.0	61.4	67.9	96.7	66.3
Interest payment (EIB, EBRD)	5.8	7.3	9.4	8.6	8.0	8.0	9.8
Others	13.3	22.6	9.1	10.3	11.2	12.1	22.5
Capital Expenditures Total	96.7	132.4	119.6	82.6	81.9	79.8	121.7
Rehabilitation	62.2	70.8	87.6	52.4	32.9	27.9	44.9
New construction	34.5	61.6	31.9	30.3	49.0	51.9	76.8
Repayment of Principal (EIB, EBRD)	5.6	9.7	12.6	12.1	11.4	14.3	16.1
Others	0.1	0.1	1.0	0.1	1.4	3.0	1.2
Total road expenditure	147.4	203.0	191.6	175.1	181.6	213.8	237.6
GDP at current price	12,200	13,700	15,200	16,600	17,700	19,500	22,300
Total road expenditure percent of GDP	1.21	1.48	1.26	1.05	1.03	1.09	1.06

Source: REA

* Estimate

5.11 In order to secure a sustainable source of funding for road infrastructure, Bulgaria introduced a vignette system in 2004 to collect charges for using the main road network. It is expected that this source

⁵⁰ Over Euro 86 million, if all Class III roads are included.

⁵¹ International Road Federation (IRF), 2000. World Road Statistics, Geneva.

of funds would generate about Euro 80 million annually. The level of fees for Bulgarian-registered vehicles are significantly lower than those for foreign-registered vehicles at present, but the discrimination between domestic and foreign users will be gradually phased out. It is also important to note that EU regulations forbid the charging of vignettes in parallel with direct tolling of roads. This is particularly relevant for the proposed new motorways that would be financed through PPP schemes with revenues generated from direct toll collections.

5.12 The Government has recognized the problem posed by core roads not meeting European standards and has embarked on a series of rehabilitation programs on transit roads. About 2,000 km of roads were upgraded or rehabilitated under the EIB financed Transit Roads Projects I, II, and III. An additional 340 km of mainly Class I roads that are in poor condition or do not meet European standards, will be rehabilitated under the forthcoming Transit Roads IV Project. This will result in most of the Class I roads, the backbone of the road network in Bulgaria, being restored to good or fair condition, allowing modern trucks to operate without undue axle load restrictions.

5.13 In addition to road rehabilitation, REA's capacity to program, prioritize, and implement investments would need to be further strengthened. The standard of project preparation is relatively weak, and economic analysis of projects is often lacking. Particular attention should be paid to strengthening administrative capacity and ensuring sound financial planning for infrastructure projects. In addition, there is a need to introduce modern road management tools, and provide training for REA staff to adopt management methods and standards commonly found in other EU countries.

Opportunities

5.14 Bulgaria's geographic location gives special importance to road transport. Over 70 percent of the population of 7.7 million lives in urban areas. The capital Sofia has about 15 percent of the total population. Other important towns (Plovdiv, Varna, Burgas, Ruse, Stara Zagora and Pleven) have another 15 percent. The rest of the urban population lives mainly in small and medium sized towns. The density of the population is low compared to other European countries about 71.1 people per square km. This spatial distribution of the population tends to increase the demand for transport, especially roads. Over half of Bulgaria's territory is also sparsely populated hilly or mountainous terrain, which makes the cost of constructing transport infrastructure higher than in countries with mostly flat terrain.

5.15 Bulgaria's increasing links with the EU will have an impact on the directions of trade flows and create new opportunities for road transport. Bulgaria's trade flows were previously dominated by the Eastern Corridor because of its links with the CIS, and these flows were served by maritime transport in the Black Sea and by rail. Today, the EU is the main trade partner, which is expected to further intensify after Bulgaria joins the EU. The EU membership combined with Bulgaria's geographic position makes transport infrastructure potentially attractive for transit transport to and from other countries. However, Bulgaria's transport infrastructure, particularly roads, was not planned to satisfy such demand.

5.16 The growth of Bulgaria's economy will continue to shape transport demand and favor road transport. Foreign trade has been expanding at a rapid rate and is now growing at close to 15 percent per year in dollar terms. As a result, freight transport measured in terms of ton-kms increased between 2000 and 2003 by an average 5.5 percent per year for national transport, and by 10.4 percent for international transport. Most of this increase was carried by road transport, which grew by 14.4 percent per year, while rail freight decreased by an average 1.6 percent per year. The market share for road transport has therefore increased to 55 percent of national transport, and to about two thirds for all freight transported in Bulgaria, including international transport (Table 5.3). The dominance of road transport is still below the EU15 average where modal shares are 85 percent for road transport and 15 percent for rail. The

proportion of imports and exports handled by road has also increased in the past five years and is currently at about 15 percent for exports and 22 percent for exports measured in terms of tonnage.

5.17 In countries with levels of vehicle ownership similar to that of Bulgaria (about 270 cars per 1,000 people), the rate of growth of road traffic can be much higher than GDP (the elasticity of demand with respect to GDP is greater than one—sometimes as high as about 2.0 for car traffic⁵²). Recent trends in Bulgaria would suggest an elasticity of about 1.0 because traffic and GDP have grown at a similar pace. Nevertheless, in the future, international experience would suggest a higher elasticity, particularly as disposable income increases.

5.18 The deregulation of road freight together with the transformation of the Bulgarian economy has had an impact on the modal split in the country. The economy is becoming less dependent on heavy industries and agriculture. Consequently, the movements of heavy and low value cargo, and of bulk raw materials that were mostly carried by rail and water transport, are being reduced and substituted by the transport of compact high value goods, carried mainly by road transport. The effect of this trend is important because it has many impacts on the modal split in a competitive market where rail and water transport mainly carry long-haul heavy “cheap” freight or multi-modal container traffic. As an illustration, international freight transported by road almost doubled in 2004, although it is not certain if this trend will continue (see Table 5.3).

Table 5.3: Goods Carried (million ton-kms)

					percent	percent
		Rail	Road	Total	Rail	Road
National	2000	4,504	3,061	7,565	60%	40%
	2001	4,139	3,310	7,449	56%	44%
	2002	3,838	3,931	7,769	49%	51%
	2003	4,292	4,586	8,878	48%	52%
	2004	4,117	5,296	9,413	44%	56%
International	2000	1,034	3,343	4,377	24%	76%
	2001	765	4,737	5,502	14%	86%
	2002	789	4,873	5,662	14%	86%
	2003	982	4,911	5,893	17%	83%
	2004	1,094	7,569	8,663	13%	87%
Total	2000	5,538	6,404	11,942	46%	54%
	2001	4,904	8,047	12,951	38%	62%
	2002	4,627	8,804	13,431	34%	66%
	2003	5,274	9,497	14,771	36%	64%
	2004	5,211	12,865	18,076	29%	71%

Source: ECMT

(*) The sharp increase recorded in 2004 in international road traffic merits further analysis

Threats

5.19 The declining and ageing population may, however, limit the pace of transport demand growth. Hence there is a risk of over-investing in the road network. Bulgaria’s population of 7.7 million has decreased by about 7 percent since 1992, and it is estimated that by 2010, the population might further decrease by another 7 percent. At the same time, the population is ageing, with an increase in the group

⁵² Based on evidence from the UK and other European countries in the 1950’s and 1960’s as car ownership was fairly low.

of the population over 65 (that is, the group with the least mobility), combined with a decrease in the share of the young population (that is, under 15). The workforce is also shrinking in absolute terms, although its share of the population is increasing.

5.20 In the medium term, passenger transport demand growth will be constrained by the per capita income. Vehicle ownership is increasing, albeit from a low base with car ownership at 0.27 per capita, which is lower than in other Central and Eastern European countries. The annual increase in vehicle ownership was 12.6 percent between 1996 and 2000, significantly higher than during the 1990's. However, by 2001 the rate had fallen to 3.7 percent. Despite this growth of vehicle ownership, recent trends in road traffic volumes are mixed. The ongoing 2005 traffic counts in Bulgaria should capture the effects of economic growth over the past five years. Intermediate traffic counts carried out since the last national survey in 2000 show that some main road sections have experienced a decline in traffic, while others have an increase. There is no obvious overall increase despite the increase in vehicle ownership--possibly because many cars are being used mainly for travel within cities, and therefore contribute little to inter-urban traffic volumes. Regardless of the actual growth of the economy and income in the past five years, the standard of living in Bulgaria has remained low. The income per capita of the population (in terms of purchasing power parity) in Bulgaria is currently roughly 30 percent of the EU average.

5.21 Bulgaria has made significant achievements in the past ten years to attain fiscal discipline, and to keep the size of the state in check. It is unlikely that under these circumstances national resources committed to public investment in infrastructure can increase by much as a share of GDP. Bulgaria will need to increase the efficiency and effectiveness of public expenditures, particularly in road management, where investments tend to be large. In a situation where the country is faced with significant need for basic public infrastructure combined with the challenge of improving competitiveness, the Government will need to be highly selective in choosing road investments with high economic impacts.

II. STRATEGIC PRIORITIES FOR THE ROAD NETWORK

Identification of Road Sector Investment Priorities

5.22 The government's stated priorities for investments in the road sector include:

- (i) Expansion and upgrading of motorways and class I roads along the European corridors;
- (ii) Expansion and upgrading of complementary national priority corridors that connect to the European corridors;
- (iii) Improved cross-border links with neighboring countries;
- (iv) Improvement of transit through cities and towns (bypasses); and
- (v) Maintenance and rehabilitation of the existing national and regional roads.

5.23 Whether investments in road assets are financed from public funds or by the private sector, road users and other beneficiaries ultimately pay for the investment through user fees or taxes. Consequently, investments should only be made in assets with high economic return and high strategic value for providing efficient road transport. It is therefore necessary to rigorously set priorities on the basis of economic efficiency and strategic goals within an integrated framework for planning and budgeting. Investment programs will need to be constrained by realistic estimates of available financial resources, taking into account existing commitments for ongoing projects and for debt servicing. The recurrent expenditure implications of the public investment program will need to be systematically considered and

closely integrated within the budgeting process in order not to reduce the productivity of earlier road infrastructure investments.

Table 5.4: Summary of SWOT Analysis

Strengths	Weaknesses
• Harmonization of legislation with the EU, concluding the negotiation of Chapter 9—"Transport Policy" • Sufficient level of access provided by the road network over most of Bulgaria's territory • As a whole, sufficient capacity to satisfy future demand in the short run • Adequate institutional arrangements with a separate road agency • Progress in road rehabilitation and upgrading program • New sources of funding (vignette) • Increasingly competitive road transport services • Sustained growth of road transport in the past few years	• Only one third of all national roads is in good condition • Inadequate maintenance of existing road assets • Lack of bypasses around major agglomerations for high-intensity traffic roads • Relatively under-developed road connections with neighboring countries. • Weak capacity of the road agency • Lack of program budgeting and prioritization • Only a small share of budget revenues from road sector is channeled to roads • Poor record of road safety
Opportunities	Threats
• Geographic features and population density make road transport very important • Strategic geographical position with major international transport corridors passing through the territory of the Bulgaria • Economic integration with EU should translate into stronger trade flows with EU partners and requires efficient road infrastructure integration into EU transport structures • Economic growth translating into increased mobility and demand for road infrastructure • Increasing modal competition favoring road transport • Availability of new EU funds	• Continued low maintenance levels • Over-investment without sound justification • Over investment in light of demographic trends and per-capita income • Delay in implementation of priority road programs

5.24 In view of the above, it is proposed that priorities for investments in the road transport sub-sector in Bulgaria be based on the following criteria:

- (i) Projects should satisfy the minimum accepted economic criteria (one being a rate of return of five percent for EU funded projects or higher for projects financed by other IFIs) and be ranked on the basis of the estimated benefit/cost ratio in order to maximize the economic return per dollar of investment.
- (ii) Projects to be largely financed from EU grants must meet the eligibility criteria, in particular that Cohesion funds should primarily be focused on the TEN-T network, and may be used to support environmentally sustainable transport.

- (iii) All projects should be accompanied by estimates of the required annual recurrent spending during their operating life.
- (iv) All projects proposed for financing through Public-Private Partnerships (PPP) must already be included in the public investment program, and should be competitively and transparently bid when selecting the private sector partner⁵³.
- (v) Projects should satisfy projected demands and not create excess capacity, particularly with regard to the timing for construction of motorways.
- (vi) Projects should be prioritized on the basis of reliable information, particularly on the projected future traffic flow, and the estimated costs per km of construction.

Proposed Strategy for Bulgaria's Road Infrastructure Investments

5.25 The Government plans to give high priority to the rehabilitation and maintenance of the national road network, while at the same time continuing to construct selected motorways and other high capacity roads in response to the demand for high quality roads along the strategic TEN-T corridors. All proposed investments will be subject to the minimum economic return criterion.

5.26 The maintenance of the existing road assets, including clearing the large backlog of road rehabilitation and repair of critical bridges that require urgent attention, will be largely financed from Bulgaria's own resources supplemented by loan financing if necessary. These investments generally have high economic returns.⁵⁴ The low spending on road maintenance in the past has led to the poor state of the road network in Bulgaria thereby affecting the competitiveness of rural areas and of the national economy as a whole. Rehabilitation of the core network will also encompass improving the bearing capacity of road pavements together with improvements to road geometry and riding quality in order to align these with EU standards. The government expects that these measures will lead to lower transport costs and higher road safety standards in Bulgaria.

5.27 From the EU integration perspective, investments of national importance will also include the improvement of road and rail links with neighboring countries, incorporating provision of cross-border infrastructure and implementation of trade facilitation measures.

5.28 The Government has proposed a program of capital-intensive projects, including the construction of motorways along most of the country's main transport corridors, with particular attention to the four major Pan European TEN-T Corridors; IV, VIII, IX, and X that pass through Bulgaria. In light of the significant total investment required to implement these projects, a key issue for the Government is to ensure that the limited financial resources available for road investments are carefully planned and implemented in order to maximize economic returns, while at the same time effectively utilizing the EU grants (ISPA up to 2006, and Cohesion funds thereafter) that are available for the transport sector. Private sector financing may also be selectively utilized for the construction of motorways through PPP schemes where feasible.

5.29 Based on this analysis, the following investment prioritization strategy is proposed for the main road network:

- Completion of the planned road rehabilitation under the Transit programs.

⁵³ See World Bank Policy Note on Road Investment Financing Options in Bulgaria (June 2005) for further details.

⁵⁴ Based on World Bank analysis of economic rates of return using data supplied by REA in 2005.

- Increased financing for road maintenance towards achieving the EU15 average as a proportion of GDP in the long run.
- Rehabilitation and repair of bridges on critical road links identified by recent bridge study.
- Removal of capacity bottlenecks particularly through urban centers with the construction of bypasses in conjunction with traffic management improvements.
- Completion of the Trakia motorway and construction of the Struma motorway.
- Improving the capacity of the main roads that carry high traffic flows, such as the Sofia Ring road, and bypasses to other large towns.
- Construction of the Vidin-Sofia motorway and the Maritsa motorway, subject to meeting the minimum economic criterion.

Selection of Measures for the Sector Operational Program for Transport SOPT for Roads

5.30 In keeping with the generic criteria outlined in the previous section, the following criteria were applied for the selection of projects to be included in the EU Sector Operational Program for Transport (SOPT) for roads.

- (i) *Alignment with EU strategic priorities:* To be eligible for financing from the EU Cohesion Fund, proposed road projects must be on the TEN-T corridors. Proposed projects outside the TEN-T may be eligible for financing from the Cohesion Fund, if they support EU priorities such as environmentally sustainable transport.
- (ii) *Economic feasibility:* Proposals for EU grant financing require a minimum economic rate of return of five percent.
- (iii) *Project readiness:* As Cohesion Funds will be available from 2007 onwards, project preparations should be well advanced, with technical design and pre-feasibility studies already undertaken. In addition, they should not face any major statutory issues, such as land acquisition or compliance with safeguards.

5.31 In addition to the above criteria, it should be noted that other investments in Class I, II and III roads that lie outside the European corridors may be eligible for financing under the European Regional Development Fund (ERDF), although it is stated policy of the EU that preference should be given to the TEN-T road network.

Projects Eligible for Financing under the Cohesion Fund

5.32 Preliminary estimates of the economic return on proposed road investments were made using data provided by REA, with the exception of construction costs and updated traffic data. Construction costs were estimated from data provided by the Road Construction Chamber, and the 2000 traffic count data were projected to 2005 using past GDP growth rates. The results of the economic analysis should therefore be treated as indicative only. REA should update the economic and financial analysis using the latest information on road condition and investment costs as well as the 2005 traffic counts once they are available. This would be particularly important in the case of motorways, since by using the 2000 traffic levels, projected traffic may be underestimated. This will enable REA to fine-tune its investment program to satisfy current and future transport demand.

5.33 Expansion and upgrade of motorways: Bulgaria has an ambitious motorway construction program comprising of nine motorway projects, two of which (Trakia and Liulin) are already funded (see the Road Network Map in Annex I). Of the remaining seven, one (Hemus) does not lie on a European corridor, and no detailed studies have been conducted for three others (Vidin-Sofia, Ruse-Veliko Tarnovo, and Kalotina-Sofia). In addition, the currently low traffic levels on the Hemus, and Ruse-Veliko Tarnovo proposed motorways imply that these would not be economically justified in the short to medium term. Therefore, only the following motorways would remain as candidates for potential financing from the Cohesion Fund during the period 2007-2013:

- (i) *Struma motorway:* This proposed project would link Sofia with the Greek border (Kulata) along Trans-European corridor IV, and is the only road project in Bulgaria included in the list of EU priority projects as defined in the Van Miert report. The planned motorway is 116 km long, of which 18 km are currently under construction. Total construction costs are estimated at €600 million or €5.2 million per km. The relatively high construction costs are partly due to the mountainous sections along the road. A PHARE-funded feasibility study was conducted in 2000. Historical and projected traffic levels are relatively high, resulting in economic rates of return above the EU threshold of five percent. Traffic levels indicate the potential for private sector participation. However since the geographic characteristics of the surrounding area do not allow for an alternative free public road (which is required under EU regulation in case of a toll road), the PPP modality will have to be some sort of an availability payment. The government does not propose to submit projects with potential for PPP financing as candidates for co-financing from the Cohesion Fund. Project preparation for this proposed motorway is well advanced and construction could start as soon as 2007.
- (ii) *Maritsa motorway:* This proposed project would go from Chirpan to Kapitan Andreevo, connecting the Trakia motorway (Sofia–Burgas) with the Turkish border, along the Trans-European corridors IV and IX. This is 114 km long, of which 3 km are operational and one carriageway has been fully constructed on a further 17 km. An additional 3 km are required for the border crossing point. Total construction costs are estimated at €385 million, or €3.4 million per km. This includes 30 major structures along the motorway, with a bridge across the Maritsa at an estimated cost of €40.3 million. Historical and projected traffic levels are significantly lower than those on the proposed Struma motorway and below the levels that would support a private concession. Estimated economic returns are below EU requirements based on available traffic data. For this reason, a lower priority has been assigned to the construction of this motorway compared to Struma. While detailed designs have been carried out for the proposed motorway (excluding the border crossing section) and the majority of the land has been acquired, no feasibility study has been conducted to date. Construction of the proposed Maritsa motorway should therefore be delayed until such time when the projected traffic would justify construction.
- (iii) *Vidin-Sofia motorway:* This 190 km planned motorway connects Sofia with the Romanian border along Corridor IV. Construction would take place in two sections; (i) 84 km from Botevgrad to Montana at an estimated cost of €236 million (€2.8 million per km); and (ii) 106 km from Montana to Vidin at an estimated cost of €298 million (€2.8 million per km). While current traffic volumes are low, they can be expected to increase following the completion of the Danube Bridge at Vidin (funded by ISPA). Construction of this motorway should only be considered after the second Danube bridge crossing has been completed. However, no feasibility studies have been carried out to date and a detailed economic analysis is required taking into account the impact of the Danube

bridge. If traffic volumes are sufficient, private sector participation could be contemplated. As a parallel class I road exists (I-1/E79), tolling could be considered, subject to economic and financial viability.

- (iv) *Trakia Motorway remaining sections.* An alternative to public financing for the remaining section of the Trakia motorway has also been considered. Trakia is a four lane motorway that serves the main east-west corridor between Sofia and Burgas and has a planned length of 368 km, of which 179 km is in operation (mainly at the western end where it follows the Trans European Route Numbers 8 and 10). The motorway is being constructed along a new alignment for most of its length, parallel to existing roads. In the west it is parallel to the existing main Sofia-Plovdiv road. The daily traffic on the motorway is generally between 10,000 and 17,500 (highest near Plovdiv). The ongoing construction is financed with an EIB loan and state budget allocations. When the current construction is completed there will still be a further 118 km costing USD318 million to be completed.
- (v) *Cherno – More motorway:* The Cherno-More motorway forms part of the TEN-T Corridor XIII, and links Burgas and Varna on the Black Sea coast line. This area has recently experienced rapid development primarily to cater to the seasonal tourism and service industries. The government has reported a significant increase in traffic volumes, particularly during the tourism season. The economic feasibility of the project will need to be confirmed following a thorough economic analysis of the proposed motorway. Initial estimates of construction costs are about €360 million, in addition to €90 million that may be required if a tunnel is constructed through severe mountainous terrain along part of the proposed alignment.

5.34 Rehabilitation of Transit Road Sections on Motorways: In the context of the Transit Road Programs I and II, 364 km along the constructed parts of Trakia, Hemus, Maritsa and Cherno More were rehabilitated in 1994–1998. These road sections have attained, or are about to exceed, the rehabilitation design life and will therefore require additional rehabilitation expenditure estimated at €91 million (€250,000 per km). As expected for rehabilitation expenditure, the estimated economic return from such investments would be high (well above the minimum 5 percent threshold required for EU projects). They could therefore be undertaken within the first Cohesion Fund budgeting period (2007–2009), as further delays will increase the cost of future rehabilitation works. The government should however, program these taking into account the final schedules for the construction of the motorway program.

5.35 Expansion and Upgrade of Class I Roads along the European Corridors: REA has identified several Class I candidate road investment projects that form part of the TEN-T roads within Bulgaria. Priority should be given to those investments committed under the Accession Treaty. However, detailed design and feasibility studies have been conducted for only three such road sections. Preliminary economic analyses for the proposed Class I road projects show high economic returns for all of them except for Gabrovo–Shipka, for which cost estimates are relatively high. The high economic return projects are considered to be of strategic importance, and the government has made a commitment to complete these within the Cohesion Fund budgeting period 2007–2013. With the exception of Gabrovo–Shipka, all of these could be considered for financing under the Cohesion Fund.

- (i) *Vidin-Montana (I-1/E79):* This proposed road project involves the reconstruction of 21 km of the Dimovo–Rushentsi section of the E79. The road links Vidin with Sofia and provides the Northern extension of the Struma motorway from Sofia to Vidin along Corridor IV. The total project cost is estimated at €30 million, or €1.4 million per km. REA has prepared the design study for the required works. As the construction of a

motorway along this section is currently not justified by traffic, preliminary economic analysis shows that upgrading of the existing road would have significant economic benefits. While this project is not included in the Accession Treaty, it should be given high priority due to its strategic importance. Construction could start in 2007 to provide one carriageway of the proposed future motorway.

- (ii) *Kardjali-Podkova-Makaza (I-5/E85)*: Given the importance of this road along corridor IX, rehabilitation of the existing road and new construction of an access road and a bypass around Kardjali are proposed. Phase I of the project (new access road Podkova-Makaza) is under construction. Phase II includes rehabilitation of 12 km on the existing road and 16.5 km of new construction on the two subsections of Kardjali-Djebel and Djebel-Podkova. Phase III would involve the construction of a new 15 km bypass around Kardjali. The total project cost for Phases II and III are estimated at €20 million. Detailed designs have been carried out for Phase II and a pre-feasibility study has been done for Phase III. Economic returns are adequate to justify this project. Bulgaria has committed to opening this project to traffic by 2008, implying that construction should start in 2006.
- (iii) *Romanian border–Balchik (I-9/E87)*: This project involves the rehabilitation of 60 km on the E87 between Balchik and the Romanian border at a total estimated cost of €12 million. The project has been estimated to have high economic returns, and should therefore be assigned high priority. In the Accession Treaty, the road it is expected to be opened to traffic in 2009, and therefore rehabilitation works should begin in 2008 at the latest.
- (iv) *Gabrovo-Shipka (I-5/E85)*: This proposed project includes the construction of a 10.1 km bypass around Gabrovo with a proposed tunnel to be constructed under the Shipka peak of 3.2 km. Total project costs are estimated at €180 million, which include an estimated €130 million for the tunnel. A preliminary design study was carried out for the bypass, but no economic analysis has been undertaken for the bypass and tunnel. The Gabrovo bypass is expected to yield savings in time and vehicle operating costs, as it will divert international transit traffic from the current road that passes through several villages. Despite the very high cost of the tunnel, Bulgaria has made a commitment in the Accession Treaty to open this road section to traffic by 2014. Construction would have to start by 2012 in order to comply with this. Consequently, it is recommended that detailed feasibility studies should be carried out to investigate appropriate low cost alternatives that would have acceptable economic return.

Other High Class Roads Not Eligible for Financing under the Cohesion Fund

5.36 Other Class I Road Investment Projects: In the Accession Treaty, Bulgaria has also committed to the rehabilitation and reconstruction of 182 km of other Classes I roads that are not designated as European roads. All of these have been estimated to have high economic returns, particularly the rehabilitation of road sections along the I-6 and I-7, which would make them priority projects for an estimated cost of €56.4 million. In the Accession Treaty, they are expected to be opened to traffic in 2009-2011.

5.37 Expansion and upgrade of Class II Roads: Bulgaria will need to rehabilitate and reconstruct selected Class II roads (possibly re-designating these as Class I), and the border crossing points with Serbia and Montenegro, as committed under the Accession Treaty. Since none of these roads are designated as European roads, they are expected to be proposed for financing under the ERDF.

Notwithstanding the preference by the EU DG-TREN for roads on the TEN-T corridors, these commitments will have to be financed by the government within the time period stipulated in the Accession Treaty.

5.38 *Improved cross-border links with neighboring countries:* Bulgaria has embarked on a PHARE financed program for improving border crossing points. A number of projects have already been completed under this program. Additional projects at the Greek and Romanian borders are expected to be completed before EU accession in 2007. The remaining border crossing road projects that are planned during the Cohesion Fund budget period 2007-2009 include the construction of a new border crossing point with Turkey (€3.4 million), and the rehabilitation of 15 km of Class III roads at a new border crossing point with Macedonia (€8.7 million). These projects will be financed under the PHARE program.

5.39 *Improvement of Transit through Cities and Towns (bypasses):* Demand on certain corridors may not be sufficient to justify the construction of four lane motorways at the outset. However, several of the corridors that pass through urban centers experience significant traffic congestion and poor road safety, and are therefore bottlenecks that impact both long distance and local traffic. Interventions aimed at removing such bottlenecks generally require relatively small investments to increase capacity at critical points. A range of cost-effective measures can be considered such as traffic management measures (for example parking controls and junction improvements to reduce peak hour congestion on roads near urban centers), the construction of bypasses around towns and villages, upgrading of main intersections or railway crossings, and climbing lanes on steep road sections. Consequently, Bulgaria has proposed the construction of several bypasses on Class I roads such as the E79, E83, E85, E87 and E871. Their total cost is estimated at €40 million, and all projects would be justified based on the preliminary economic analyses. Construction of these road sections would have to be phased between 2008–2010. However, since these projects are not along the TEN-T corridors, they would not be eligible for financing under the Cohesion Fund, and therefore the government would have to finance these.

5.40 *Maintenance and rehabilitation of national roads:* It is estimated that €109-145 million would be required to finance the accumulated backlog of road maintenance and rehabilitation of national roads over the period 2005 to 2009. These figures were estimated by a study commissioned in 2004 by the World Bank at the request of the Government of Bulgaria, and carried out by an international consultant using HDM-4⁵⁵. This level of investment would be necessary to restore the condition of the national road network in line with EU standards, and would significantly reduce the proportion of the road network that is currently in poor condition from about 60 percent to less than 20 percent over the period 2005–2009. After 2009, around €48 million/year would be required to maintain the road network in a steady state condition close to the average condition in most EU countries. The economic benefits that would be derived from the investment in road rehabilitation (in net present value terms) over a 15-year period are estimated to be in the region of €6.5 billion, that is, about 8 times the total investment required.

5.41 *Strengthening institutional capacity:* It would be necessary to strengthen the capacity of REA to plan and manage any increased financing for the roads sub-sector. A review of the institutional arrangements for management and financing of road network rehabilitation and maintenance in Bulgaria showed that REA needs to adopt European standards and practices for road management. It is essential that modern management information systems are introduced for purposes of prioritizing road investments, and for managing the expected increase in civil works contracts. In addition, the capacity of

⁵⁵ A 15-year road maintenance and rehabilitation strategy was prepared using the HDM-4 model for the national road network. The data used by the consultant was obtained from the Bulgarian Central Road and Bridges Laboratory with assumptions made when necessary. It was noted that the results of the study is indicative and will need to be confirmed when more accurate data becomes available.

REA would need to be strengthened particularly with regard to project management, conducting feasibility studies, and monitoring the quality of construction. Bulgaria should also consider the introduction of performance based maintenance and management contracts done by private sector contractors in order to improve the efficiency of utilization of any additional funds for road maintenance and rehabilitation. The above could be achieved through a well planned technical assistance and training aimed at a staff development program and modernization of REA facilities.

III. FINANCING OF PRIORITY ROAD PROJECTS

Summary of Investment Program

5.42 **Recommended road sector investment program:** In view of the significant fiscal impact of the motorway construction program, four financing scenarios have been analyzed as follows:

- *Scenario I* - Both Trakia and Struma motorways are PPP financed, with Struma scheduled for construction from 2007–2009.
- *Scenario II* - PPP Financing for Trakia motorway only, with Struma motorway publicly financed in 2007–2009.
- *Scenario III* - PPP Financing for Trakia motorway only, with Struma motorway publicly financed in 2010–2013.
- *Scenario IV* - Both Trakia and Struma motorways are publicly financed with Struma scheduled for 2010–2013.

5.43 Based on the analysis presented in the previous chapter, the recommended total road sector expenditures would amount to €4.5 billion during the EU post accession period 2007–2013, equivalent to €638 million p.a. on average representing about 2.7 percent of GDP annually. If PPP financing is used for construction of the Struma motorway in 2007–2009, there would be a long term contingent liability of about €70 million per year based on estimated availability payments over the life of the concession. Given the advanced stage of the concession for the Trakia motorway, it is assumed that the government will have included any long term contingent liabilities in future budget forecasts. Table 5.5 summarizes the proposed road sector investment program for the three scenarios (see Annex I for details), that includes:

- (i) Completion of ongoing road investment projects, such as the ISPA-financed Liulin motorway, with two options for financing the Trakia motorway either through a PPP concession or through public funds, and other ongoing projects financed under the PHARE border crossing program.
- (ii) The road sector investment program based on the priorities identified in the previous section.
- (iii) Ongoing rehabilitation and maintenance expenditures, required investments in road safety, and immediate maintenance and repair needs for bridges.
- (iv) Debt service payments, and long term contingent liabilities for scenario I with PPP financing of the two motorways (Trakia and Struma).

Table 5.5: Proposed Road Investment Program

	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Total Recurrent expenditure	199	199	199	184	169	153	138	1,241	177
Debt service payments	34	34	34	34	34	34	34	237	34
Scenario I									
Total Capital Investments	471	654	595	209	367	376	318	2,989	427
PPP Contingent Liabilities				70	70	70	70	280	
Total Program with PPP liabilities	704	887	828	497	639	633	560	4,747	678
percent of GDP	3.3	4.0	3.7	2.1	2.5	2.4	2.0		2.9
Scenario II									
Total Capital Investments	471	654	595	209	367	376	318	2,989	427
Total Investment Program	704	887	828	427	569	563	490	4,467	638
percent of GDP	3.3	4.0	3.7	1.8	2.3	2.1	1.8		2.7
Scenarios III & IV									
Total Capital Investments	271	454	395	359	517	526	468	2,989	427
Total Investment Program	504	687	628	577	719	713	640	4,467	638
percent of GDP	2.4	3.1	2.8	2.4	2.9	2.7	2.3		2.6

Assumptions:

1. Figures are in constant 2004 values
2. Motorways Hemus, Chernomorets, Ruse-Veliko Tarnovo will not be built during the period 2007-2013 as they would not be economically justified before 2013.
3. Maintenance and rehabilitation requirements are based on World Bank analysis of 2004.
4. Contingent liabilities estimated for PPP financing of Struma motorway only.

Financing Sources

5.44 The sources from which the government can finance the proposed road investment program, in addition to the national budget for road transport (which at 1.1 percent of GDP in 2005 roughly corresponded to the sum of road user charges and excise taxes), include EU grants, sovereign borrowing, domestic borrowing and private sector financing.

5.45 **User charges—fuel tax:** Bulgaria charges road users through fuel (excise) tax, which flows directly to the government budget. Until 2004, a proportion of the revenues collected from fuel tax were made available to REA for road maintenance and rehabilitation. However, since the introduction of the vignette charge in 2004, direct transfers to REA from the fuel tax ceased. It is noted that the excise charges on fuel are currently below levels recommended by the EU, which are designed to reflect the marginal cost of road usage. Following EU accession, Bulgaria will have to increase its excise and fuel tax rates to EU levels. According to the budget projections from the Ministry of Finance for 2006–2008, increased excise duty on petrol, diesel and kerosene will result in additional revenues of €65 million in 2006, €81 million in 2007, and €69 million in 2008. For the purposes of this analysis, it has been assumed that REA will continue to receive fuel tax revenue in the magnitude of €114 million p.a. (2003 level).

5.46 **User charges—vignette:** Bulgaria introduced a vignette charge in 2004 for vehicles using the national road network. Following a delay in vignette introduction and collection in 2004, revenues from vignette charges have increased significantly in 2005. Based on actual collection rates during the first nine months of 2005, income from vignette charges is projected to total €67 million in 2005 and increase

to €86 million by 2007. Currently, vignette prices differentiate between domestic and foreign road users and vehicle classes. Vignette prices will be aligned between domestic and foreign road users by the time of EU accession in 2007. The vignette system is intended to cover the cost of routine maintenance of the road network. Vignette charges are collected through a Republican Road Infrastructure Fund in the Ministry of Regional Development and Public Works which is administered by REA.

5.47 EU grant funds—Cohesion Fund: Following EU accession, Bulgaria will be eligible for EU grants from the Cohesion Fund, which provides financial support for environment and trans-European transport infrastructure projects in member countries with a Gross National Income (GNI) of less than 90 percent of the Community average. All projects proposed for EU grant funding need to undergo a standardized cost-benefit analysis, and must meet the minimum economic rate of return of 5 percent. About 50 percent of the Cohesion Fund is available for transport projects, although the total amount of grant financing for Bulgaria has yet to be finalized for the whole budget period 2007-2013. For 2007-2009, the European Commission has indicated that €2.3 billion will be available for structural and cohesion grant funds to Bulgaria, one third of which (€767 million) is expected to be available under the Cohesion Fund. For the period 2007-2013, the Ministry of Transport and Communications expects to receive about €1.2 billion from the Cohesion Fund for the transport sector. Based on the distribution of ISPA transport funds between road and rail investments as a guide, 50 percent of the transport sector allocation is assumed to be available for road investment projects (i.e. about €600 million). This will require about €90 million in co-financing from the national budget (based on an average 15 percent for the entire program). However, co-financing requirements may be higher where total project costs exceed eligible costs for Cohesion Fund financing, such as sizeable land acquisition. Unlike EU structural funds, the Cohesion Fund is not subject to the principle of additionality, and therefore can substitute national expenditure in the transport sector to alleviate fiscal pressures.

5.48 EU grant funds—European Regional Development Fund (ERDF): EU grants would also be available for projects that foster regional competitiveness and employment within the EU (objective 2 of European cohesion policy under the proposed reform), as well as European territorial cooperation (objective 3). These initiatives include investments in regional economic and social development, such as productive investments in infrastructure, sustainable employment and cross-border cooperation. Investments in class II, III and IV roads are eligible for funding under the ERDF, if they fulfill these objectives. This is the case for the class II, III and IV roads that Bulgaria committed to upgrade to be able to carry EU axle load standards of 11.5 tons per axle. Upgrading these roads will allow heavy trucks to pass through Bulgaria, facilitating transit traffic and trans-European trade flows. In addition, none of the required upgrades on class II, III and IV roads would be eligible for financing from the Cohesion Fund. For this reason, it is recommended to seek financing for these investments under the ERDF. Unlike the Cohesion Fund, the ERDF is subject to the additionality principle, and therefore the funds must be incremental to government budget resources.

5.49 Sovereign borrowing: Given its prudent fiscal management and low Debt / GDP ratio, Bulgaria has flexibility for additional sovereign borrowing for their road sector investments, while still meeting the required Debt / GDP limit of 60 percent under the EU Stability and Growth Pact. However, like all sources of financing, it carries with it costs. Bulgaria could also finance part of its needs in road transport from domestic capital markets.

5.50 PPP financing: Where economically justified and financially feasible, private participation in Bulgaria's road investment program should be considered. Public-private partnerships bring significant advantages, as they (i) provide access to incremental financing to resource-constraint governments; (ii) create efficiency gains due to economies of scale, improved risk allocation and improved quality of service; (iii) allow for earlier implementation of infrastructure projects, as government expenditures can be phased over time rather than requiring significant upfront investments; and (iv) can be recorded off-

balance sheet as long as the construction risk and either the availability or demand risk is shifted to the private sector (EuroStat guidelines). PPPs should only be entered into if they provide value for money. This should be evaluated using the framework of a public sector comparator. In general, PPPs provide value as long as efficiency gains outweigh increased financing costs and risks retained by the government. Adequate risk sharing is therefore paramount to avoid unnecessary burdens on the government budget. In addition, PPPs require a stable and transparent legal framework and competitive bidding to ensure value for money and avoid overspending by government. The Government has a choice of structuring PPPs in the road sector as toll roads, shadow tolls, or availability payment schemes. Toll roads require sufficient traffic to generate adequate financial returns for private sector participation, and a free parallel road which can create significant traffic diversion. In addition, EU regulations forbid the concurrent charging of direct tolls and vignettes particularly for heavy trucks (greater than 12 tonnes). Given these risks and the potential need for considerable government support, toll roads may not be feasible for most roads in Bulgaria due to the relatively low traffic flows. The government should therefore consider alternative options. Shadow tolls avoid traffic diversion at the cost of fluctuating government support. Availability payments based on performance criteria have proven a successful model for PPPs in the road sector in other Eastern European countries such as Hungary. Availability payments also have the benefit of predictable government obligations, which would need to be recorded on the government's balance sheet. Given their limited size relative to total project costs, however, the costs of government obligations could be outweighed by the benefits of access to private financing and early project realization.

Recommended Financing Strategy

5.51 Utilization of EU Grant Funds: The available EU Cohesion funds could be utilized either for construction of new motorways, or for improvements to Class I roads that form parts of the TEN-T corridors. All proposed projects are required to have the minimum acceptable economic rate of return for EU grant funds. The Trakia and Struma have been shown to have the highest economic rate of return from among the proposed motorway projects, and at the same time, are also assessed to have the highest potential for PPP financing. However, experience in the region has shown that for PPP financing to be successful, some form of government support (such as minimum traffic guarantees, or availability payments) would be necessary. The resulting long term contingent liabilities would therefore need to be factored into the budget by the government. At the same time, it should be considered that EU grant funds are not completely free either given the 15 percent co-financing requirement. Assessing the fiscal impacts of PPPs versus EU grant absorption would therefore require comparing potential fiscal liabilities created by the PPP to the 15 percent co-financing requirement.

5.52 Consequently, two alternatives have been considered for utilizing the estimated €600 of EU grant funds: (i) for the rehabilitation and new construction of Class I road investments along the TEN-T corridors; or (ii) for the two motorways (Trakia, Struma) if these are financed by the government and therefore would be eligible for the EU grant funds. While the rehabilitation of Class I roads along the TEN-T corridors would yield higher economic returns when compared to new motorway construction, the government considers the motorways to have much higher strategic importance. It is expected that co-financing requirements will average 15 percent of the total investments in projects selected for EU Cohesion Grant financing. In addition to the candidate motorway projects described in paragraph 5.35, the following Class I roads could be considered for financing from the Cohesion grant:

- (i) *Vidin-Montana (I-1/E79):* The total project cost is estimated at €30 million and sections of this could constitute one carriageway of the proposed future motorway.
- (ii) *Kardjali-Podkova-Markaza (I-5/E85):* The total project cost for Phases II and III are estimated at €20 million. Detailed designs have been carried out for Phase II and a pre-feasibility study has been done for Phase III. In the Accession Treaty, Bulgaria has

committed to opening this project to traffic by 2008, implying that construction should start in 2006.

- (iii) *Romanian border–Balchik (I-9/E87)*: The total estimated cost of for this is €12 million. In the Accession Treaty, this road is expected to be opened to traffic in 2009, and therefore rehabilitation works should begin in 2008 at the latest.
- (iv) *Gabrovo-Shipka (I-5/E85)*: The total project costs are estimated at €180 million, which include an estimated €130 million for the tunnel. A preliminary design study was carried out for the bypass, but no economic analysis has been undertaken for the bypass and tunnel. Bulgaria has made a commitment in the Accession Treaty to open this road section to traffic by 2014. The proposed financing plan does not include the construction of the tunnel.
- (v) *Proposed Improvements on Class I roads*: Construction of bypasses on the E79, E83, E85, E87 and E871. Their total cost is estimated at €40 million and all projects would be justified based on the preliminary economic analyses. Construction of these road sections would have to be phased in 2008–2010.

5.53 PPP financing: Under Scenario I, private sector financing would be sought for the construction of both Trakia and Struma motorways, as these are currently the only two motorways with the potential for attracting private financing. However, as noted above, PPP financing would incur long term contingent liabilities for the government that are estimated to be in the region of €70 million per year starting from 2010 after completion of the Struma motorway⁵⁶. The Vidin-Sofia motorway could become economically justified after the completion of the Danube Bridge. Depending on projected traffic levels, private sector participation could be sought at that stage. Financing for the construction of the Maritsa motorway should therefore be considered only after completion of the second Danube Bridge and a detailed feasibility study. Current projected traffic levels would indicate that the project would not attract private sector investment without significant public sector support.

5.54 Sovereign borrowing: Table 5.6 summarizes existing and projected sources of financing for Bulgaria's road investment program (see Annex I for details). Under Scenario IV with no PPP financing, user charges would be expected to cover 31 percent of the 2007-13 road expenditures, and EU grants would provide 19 percent of these expenditures, requiring 5 percent in co-financing. If private sector financing is used for both the Trakia and Struma motorways (Scenario I), this would account for 25 percent of the total road expenditures. Under all financing scenarios (with or without PPP financing), there will be a need for increased allocations from the government budget and/or sovereign borrowing to finance the road investments. Alternatively, the government may consider extending the period over which the investments will be made, thereby stretching out the investment horizon beyond 2013, or dropping some of the projects, keeping only those with the highest economic returns within the budget constraint.

5.55 The total co-financing required for ISPA, PHARE, Cohesion Fund and ERDF grants in the road sector are estimated at €200 million over the 2007-2013 period. To the extent that road maintenance cannot be fully financed from the government budget, sovereign borrowing (from IFIs, international and domestic capital markets) could be considered. Loan financing could also be used for the construction of new roads that are not suitable for private sector financing, but only if the projected economic rate of return is high enough to justify the investment. Depending on the development of future traffic levels,

⁵⁶ Estimates of the contingent liability are based on availability payments over a 25 year concession period.

this could include the Vidin-Sofia and Maritsa motorways in the medium long term, provided that the economic rate of return would be high enough.

Table 5.6: Sources of Financing for the Bulgaria Road Investment Program
(constant 2004 Euros)

Sources of Finance	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Budget Expenditures/Roads sector revenues	200	200	200	200	200	200	200	1,400	200
<i>Vignette</i>	86	86	86	86	86	86	86	602	86
<i>Historical Budget Allocation (Fuel Excise Tax)</i>	114	114	114	114	114	114	114	798	114
Total EU grants (for roads)*	77	114	102	129	131	138	157	847	121
Scenario I – with PPP financing for Trakia and Struma Motorways									
Total private sector financing	372	372	372	0	0	0	0	1,116	159
Total Available Funds	649	686	674	329	331	338	357	3,364	481
% GDP	3.1 %	3.1 %	3.0 %	1.4 %	1.3 %	1.3 %	1.3 %		2.1 %
Total Roads Expenditures	704	887	828	497	639	633	560	4,748	678
Financing Gap	55	201	154	165	306	295	205	1,382	
% GDP	0.3 %	0.9 %	0.7 %	0.7 %	1.2 %	1.1 %	0.7 %		0.8 %
Scenario II – PPP financing for Trakia, public financing for Struma in 2007-2009									
Total private sector financing	172	172	172	0	0	0	0	516	74
Total Available Funds	449	486	474	331	333	338	355	2,765	395
% GDP	2.1 %	2.2 %	2.1 %	1.4 %	1.3 %	1.3 %	1.3 %		1.6 %
Total Roads Expenditures	704	887	828	427	569	563	490	4,467	638
Financing Gap	255	401	354	95	236	225	135	1,702	
% GDP	1.2 %	1.8 %	1.6 %	0.4 %	0.9 %	0.8 %	0.5 %		1.0 %
Scenario III – PPP financing for Trakia, public financing for Struma in 2010-2013									
Total private sector financing	172	172	172	0	0	0	0	516	74
Total Available Funds	449	486	474	331	333	338	355	2,765	395
% GDP	2.1 %	2.2 %	2.1 %	1.4 %	1.3 %	1.3 %	1.3 %		1.7 %
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638
Financing Gap	55	201	154	245	386	375	285	1,702	
% GDP	0.3 %	0.9 %	0.7 %	1.0 %	1.5 %	1.4 %	1.0 %		1.0 %
Scenario IV – No PPP financing for both Trakia and Struma Motorways									
Total Available Funds	277	314	302	331	333	338	355	2,249	321
% GDP	1.3 %	1.4 %	1.3 %	1.4 %	1.3 %	1.3 %	1.3 %		1.3 %
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638
Financing Gap	227	373	326	245	386	375	285	2,218	
% GDP	1.1 %	1.7 %	1.4 %	1.0 %	1.5 %	1.4 %	1.0 %		1.3 %
(*) Implied co-financing needed for EU Grants									
	23	32	26	33	33	34	39	221	32
% GDP	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %		0.1 %

Assumptions:

1. REA will continue to receive fuel tax revenues at 2003 levels for recurring maintenance and rehabilitation. Alternatively, to avoid earmarking, one can assume that the current road transport budget of 1.1% of GDP (which roughly equals the sum of current vignette and fuel excise tax revenues) would continue for the foreseeable future.
2. 50 percent of the total cohesion fund for transport (Euro 600 m) is allocated to the road sector. Committed fund are assumed to be paid based on the N+2 rule.
3. Contingent liabilities are included for the Struma concession if constructed between 2007 – 2009.
4. Investments for upgrading of class II, III + IV roads are funded by the ERDF.

IV. CONCLUSION

5.56 Road infrastructure assets were negatively affected by the sharp decline in public investments during the period 1998-2002. This led to severe backlogs in road asset maintenance and rehabilitation, which in turn has adversely affected the movement of goods and people in Bulgaria, and added extra costs to the Bulgarian economy.

5.57 To clear the maintenance and rehabilitation backlog and to bring existing infrastructure to EU technical and safety standards, Bulgaria would have to use all of its current 1.1 percent of GDP road budget envelope (about EUR 250) on priority interventions with the highest economic returns, including maintenance programs, rehabilitation, upgrading and improvement of critical links as well as bypasses and bridges.

5.58 EU grants for the road sector are estimated to be about 0.5 percent of GDP per year, and it is suggested that these funds should be used for interventions that meet the economic return criterion, and are critical for economic growth and EU accession. Available data on traffic and construction costs suggest that rehabilitation and improvement of Class I roads on European corridors would yield the highest economic returns. Previous economic analyses have indicated that the Trakia and Struma proposed motorway projects would have the required minimum economic rate of return. However, the estimated construction cost of these two motorways is close to twice the estimated amount of EU cohesion grants expected to be made available to Bulgaria's road sector over the period 2007-2013. The government will therefore need to decide on the utilization of the Cohesion grants either for the Class I roads, or for the two motorways, or a combination of these roads. As soon as new traffic data becomes available, REA would be well advised to update the economic analyses and fine-tune the investment strategy, since other Class I roads and motorways on the TEN-T corridors may then be included if they pass the economic test.

5.59 In addition, the Government also has a program to construct over 1,000 kilometers of other motorways with an estimated total cost of more than Euro 3 billion excluding taxes. Implementing such a program over a 10-15 year period using public funds would consume an additional 1.0 to 1.5 percent of GDP annually running the risk of crowding out investment in other sectors. Therefore, the government is exploring the potential for involving private investors in the program implementation. However, given the significant cost of the proposed program, the uncertainty of the demand for this type of infrastructure, and the willingness and ability to pay for them by Bulgarian motorists, there is likely to be a call for government support. Experience in the region has shown that for PPP financing to be successful, some form of government support (such as minimum traffic guarantees, or availability payments) has been necessary. The resulting long term contingent liabilities would therefore need to be factored into the budget by the government. Hence, it is recommended to carry out a careful feasibility study of every single motorway project to ensure public funds, if at all available, are used efficiently.

ANNEX I

Proposed Road Investment Program Scenario I ^{1,2} (2004 Euro)

	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Proposed Capital Investments									
Motorway construction ³	200	245	245	128	306	306	230	1,662	237
Class I road upgrading	32	165	152	36	17	19	17	437	62
Class II road upgrading	4	10	10	35	25	16	33	133	19
Class III & IV road improvements	3	3	6	10	19	35	38	114	16
Committed Capital Investments									
Liulin motorway	49	49	0	0	0	0	0	99	14
Trakia motorway incl. Sofia ring	172	172	172	0	0	0	0	516	74
PHARE border crossing	10	9	9	0	0	0	0	28	4
Total Capital Investments	471	654	595	209	367	376	318	2,989	427
Recurrent expenditure									
Bridge Repair & maintenance	10	10	10	10	10	10	10	70	10
Rehabilitation & periodic maint. ⁴	109	109	109	94	79	63	48	611	87
Routine & winter road maintenance	60	60	60	60	60	60	60	420	60
Road safety & Technical assistance	20	20	20	20	20	20	20	140	20
Total Recurrent expenditure	199	199	199	184	169	153	138	1,241	177
Debt service	34	34	34	34	34	34	34	237	34
Total Roads Expenditures	704	887	828	427	569	563	490	3,907	638
% of GDP	3.3 %	4.0 %	3.7 %	1.8 %	2.3 %	2.1 %	1.8 %		2.7 %
PPP Contingent Liabilities ⁵				70	70	70	70	280	
Total with PPP liabilities	704	887	828	497	639	633	560	4,747	678
% of GDP	3.3 %	4.0 %	3.7 %	2.1 %	2.5 %	2.4 %	2.0 %		2.9 %

Assumptions:

1. Both Trakia and Struma financed with PPP, and Struma scheduled for 2007 – 2009.
2. Figures are in constant 2004 values
3. Motorways Hemus, Chernomore, Ruse-Veliko Tarnovo will not be built during the period 2007-2013 as they would not be economically justified before 2013.
4. Maintenance and rehabilitation requirements are based on World Bank analysis of 2004.
5. Contingent liabilities estimated for PPP financing of Struma motorway only.

ANNEX I

Proposed road investment program Scenario II ^{1,2} (2004 Euro)

	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Proposed Capital Investments									
Motorway construction ³	200	245	245	128	306	306	230	1,662	237
Class I road upgrading	32	165	152	36	17	19	17	437	62
Class II road upgrading	4	10	10	35	25	16	33	133	19
Class III & IV road improvements	3	3	6	10	19	35	38	114	16
Committed Capital Investments									
Liulin motorway	49	49	0	0	0	0	0	99	14
Trakia motorway incl. Sofia ring	172	172	172	0	0	0	0	516	74
PHARE border crossing	10	9	9	0	0	0	0	28	4
Total Capital Investments	471	654	595	209	367	376	318	2,989	427
Recurrent expenditure									
Bridge Repair & maintenance	10	10	10	10	10	10	10	70	10
Rehabilitation & periodic maint. ⁴	109	109	109	94	79	63	48	611	87
Routine & winter road maintenance	60	60	60	60	60	60	60	420	60
Road safety & Technical assistance	20	20	20	20	20	20	20	140	20
Total Recurrent expenditure	199	199	199	184	169	153	138	1,241	177
Debt service	34	34	34	34	34	34	34	237	34
Total Roads Expenditures	704	887	828	427	569	563	490	4,467	638
% of GDP	3.3 %	4.0 %	3.7 %	1.8 %	2.3 %	2.1 %	1.8 %		2.7 %

Assumptions:

1. Only the Trakia motorway is PPP financed, and Struma is publicly financed in 2007 – 2009.
2. Figures are in constant 2004 values
3. Motorways Hemus, Cherno More, Ruse-Veliko Tarnovo will not be built during the period 2007-2013 as they would not be economically justified before 2013.
4. Maintenance and rehabilitation requirements are based on World Bank analysis of 2004.

ANNEX I

Proposed road investment program - Scenario III^{1,2} (2004 Euro)

	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Proposed Capital Investments									
Motorway construction ^{3,4}	0	45	45	278	456	456	380	1,662	237
Class I road upgrading	32	165	152	36	17	19	17	437	62
Class II road upgrading	4	10	10	35	25	16	33	133	19
Class III & IV road improvements	3	3	6	10	19	35	38	114	16
Committed Capital Investments									
Liulin motorway	49	49	0	0	0	0	0	99	14
Trakia motorway incl. Sofia ring	172	172	172	0	0	0	0	516	74
PHARE border crossing	10	9	9	0	0	0	0	28	4
Total Capital Investments	271	454	395	359	517	526	468	2,989	427
Recurrent expenditure									
Bridge Repair & maintenance	10	10	10	10	10	10	10	70	10
Rehabilitation & periodic maint. ⁴	109	109	109	94	79	63	48	611	87
Routine & winter road maintenance	60	60	60	60	60	60	60	420	60
Road safety & Technical assistance	20	20	20	20	20	20	20	140	20
Total Recurrent expenditure	199	199	199	184	169	153	138	1,241	177
Debt service	34	34	34	34	34	34	34	237	34
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638
% of GDP	2.4 %	3.1 %	2.8 %	2.4 %	2.9 %	2.7 %	2.3 %		2.6 %

Assumptions:

1. Trakia motorway financed with PPP and Struma without PPP in 2010–2013.
2. Figures are in constant 2004 values
3. Motorways Hemus, Cherno More, Ruse-Veliko Tarnovo will not be built during the period 2007-2013 as they would not be economically justified before 2013.
4. Maintenance and rehabilitation requirements are based on World Bank analysis of 2004.
5. Contingent liabilities estimated for PPP financing of Struma motorway would commence in 2014.

ANNEX I

Proposed road investment program - Scenario IV ^{1,2} (2004 Euro)

	2007	2008	2009	2010	2011	2012	2013	Total	Avg.
Proposed Capital Investments									
Motorway construction ³	0	45	45	278	456	456	380	1,662	237
Class I road upgrading	32	165	152	36	17	19	17	437	62
Class II road upgrading	4	10	10	35	25	16	33	133	19
Class III & IV road improvements	3	3	6	10	19	35	38	114	16
Committed Capital Investments									
Liulin motorway	49	49	0	0	0	0	0	99	14
Trakia motorway incl. Sofia ring	172	172	172	0	0	0	0	516	74
PHARE border crossing	10	9	9	0	0	0	0	28	4
Total Capital Investments	271	454	395	359	517	526	468	2,989	427
Recurrent expenditure									
Bridge Repair & maintenance	10	10	10	10	10	10	10	70	10
Rehabilitation & periodic maint. ⁴	109	109	109	94	79	63	48	611	87
Routine & winter road maintenance	60	60	60	60	60	60	60	420	60
Road safety & Technical assistance	20	20	20	20	20	20	20	140	20
Total Recurrent expenditure	199	199	199	184	169	153	138	1,241	177
Debt service	34	34	34	34	34	34	34	237	34
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638
% of GDP	2.4 %	3.1 %	2.8 %	2.4 %	2.9 %	2.7 %	2.3 %		2.6 %

Assumptions:

1. Both Trakia and Struma motorways financed without PPP. Struma construction scheduled for 2010–2013.
2. Figures are in constant 2004 values
3. Motorways Hemus, Cherno More, Ruse-Veliko Tarnovo will not be built during the period 2007-2013 as they would not be economically justified before 2013.
4. Maintenance and rehabilitation requirements are based on World Bank analysis of 2004.

ANNEX I

Sources of Financing – Scenario I¹ (2004 Euro)

Sources of Finance	2007	2008	2009	2010	2011	2012	2013	Total	Avg.	(%)
Vignette	86	86	86	86	86	86	86	602	86	
Excise tax (historical budget allocation)	114	114	114	114	114	114	114	798	114	
Total user charges	200	200	200	200	200	200	200	1,400	200	29 %
ISPA	37	37	0	0	0	0	0	74	11	
PHARE	7	7	7	0	0	0	0	20	3	
Cohesion Fund (for roads)	26	59	82	93	96	98	100	555	79	
ERDF - Roads	6	11	13	36	35	40	57	198	28	
Total EU grants	77	114	102	129	131	138	157	847	121	18 %
% GDP	0.4 %	0.5 %	0.4 %	0.6 %	0.5 %	0.5 %	0.6 %		0.5 %	
PPP Financing:										
Trakia Concession	172	172	172	0	0	0	0	516	74	
Struma Concession	200	200	200	0	0	0	0	600	86	
Total private sector financing	372	372	372	0	0	0	0	1,116	159	24 %
% GDP	1.7 %	1.7 %	1.6 %	0.0 %	0.0 %	0.0 %	0.0 %		0.7 %	
Total Available Funds	649	686	674	331	333	338	355	3,365	481	71 %
% GDP	3.1 %	3.1 %	3.0 %	1.4 %	1.3 %	1.3 %	1.3 %		1.9 %	
Total Roads Expenditures	704	887	828	497	639	633	560	4,747	678	100 %
Financing Gap	55	201	154	165	306	295	205	1,382	197	29 %
% GDP	0.3 %	0.9 %	0.7 %	0.7 %	1.2 %	1.1 %	0.7 %		0.8 %	
Co-financing ISPA	12	12	0	0	0	0	0	25	4	
Co-financing PHARE	3	3	3	0	0	0	0	8	1	
Co-financing Cohesion Fund	5	11	15	18	18	18	18	104	15	
Co-financing ERDF	2	3	4	11	11	13	18	62	9	
Total implied co-financing needed for EU Grants	22	29	22	29	29	31	36	199	28	6 %
% GDP	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %		0.1 %	

Notes:

1. Both Trakia and Struma financed with PPP, and Struma scheduled for 2007 – 2009.

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Sources of Financing – Scenario II ¹ (2004 Euro)

Sources of Finance	2007	2008	2009	2010	2011	2012	2013	Total	Avg.	(%)
Vignette	86	86	86	86	86	86	86	602	86	
Excise tax (historical budget allocation)	114	114	114	114	114	114	114	798	114	
Total user charges	200	200	200	200	200	200	200	1,400	200	31 %
ISPA	37	37	0	0	0	0	0	74	11	
PHARE	7	7	7	0	0	0	0	20	3	
Cohesion Fund (for roads)	26	59	82	93	96	98	100	555	79	
ERDF - Roads	6	11	13	36	35	40	57	198	28	
Total EU grants	77	114	102	129	131	138	157	847	121	19 %
% GDP	0.4 %	0.5 %	0.4 %	0.6 %	0.5 %	0.5 %	0.6 %		0.5 %	
PPP Financing:										
Trakia Concession	172	172	172	0	0	0	0	516	74	
Total private sector financing	172	172	172	0	0	0	0	516	74	12 %
% GDP	0.8 %	0.8 %	0.8 %	0.0 %	0.0 %	0.0 %	0.0 %		0.3 %	
Total Available Funds	449	486	474	331	333	338	355	2,765	395	62 %
% GDP	2.1 %	2.2 %	2.1 %	1.4 %	1.3 %	1.3 %	1.3 %		1.6 %	
Total Roads Expenditures	704	887	828	427	569	563	490	4,467	638	100 %
Financing Gap	255	401	354	95	236	225	135	1,702	243	38 %
% GDP	1.2 %	1.8 %	1.6 %	0.4 %	0.9 %	0.8 %	0.5 %		1.0 %	
Co-financing ISPA	12	12	0	0	0	0	0	25	4	
Co-financing PHARE	3	3	3	0	0	0	0	8	1	
Co-financing Cohesion Fund	5	11	15	18	18	18	18	104	15	
Co-financing ERDF	2	3	4	11	11	13	18	62	9	
Total implied co-financing needed for EU Grants	22	29	22	29	29	31	36	199	28	7 %
% GDP	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %		0.1 %	

Notes:

1. Only the Trakia motorway is financed with PPP, and Struma is publicly financed in 2007 – 2009.

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Sources of Financing – Scenario III ¹ (2004 Euro)

Sources of Finance	2007	2008	2009	2010	2011	2012	2013	Total	Avg.	(%)
Vignette	86	86	86	86	86	86	86	602	86	
Excise tax (historical budget allocation)	114	114	114	114	114	114	114	798	114	
Total user charges	200	200	200	200	200	200	200	1,400	200	31 %
ISPA	37	37	0	0	0	0	0	74	11	
PHARE	7	7	7	0	0	0	0	20	3	
Cohesion Fund (for roads)	26	59	82	93	96	98	100	555	79	
ERDF - Roads	6	11	13	36	35	40	57	198	28	
Total EU grants	77	114	102	129	131	138	157	847	121	19 %
% GDP	0.4 %	0.5 %	0.4 %	0.6 %	0.5 %	0.5 %	0.6 %		0.5 %	
PPP Financing:										
Trakia Concession	172	172	172	0	0	0	0	516	74	
Total private sector financing	172	172	172	0	0	0	0	516	74	12 %
% GDP	0.8 %	0.8 %	0.8 %	0.0 %	0.0 %	0.0 %	0.0 %		0.3 %	
Total Available Funds	449	486	474	331	333	338	355	2,765	395	62 %
% GDP	2.1 %	2.2 %	2.1 %	1.4 %	1.3 %	1.3 %	1.3 %		1.7 %	
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638	100 %
Financing Gap	55	201	154	245	386	375	285	1,702	243	38 %
% GDP	0.3 %	0.9 %	0.7 %	1.0 %	1.5 %	1.4 %	1.0 %		1.0	
Co-financing ISPA	12	12	0	0	0	0	0	25	4	
Co-financing PHARE	3	3	3	0	0	0	0	8	1	
Co-financing Cohesion Fund	5	11	15	18	18	18	18	104	15	
Co-financing ERDF	2	3	4	11	11	13	18	62	9	
Total implied co-financing needed for EU Grants	22	29	22	29	29	31	36	199	28	7 %
% GDP	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %		0.1 %	

Note:

1. PPP Financing for Trakia motorway with Struma motorway publicly financed in 2010 - 2013

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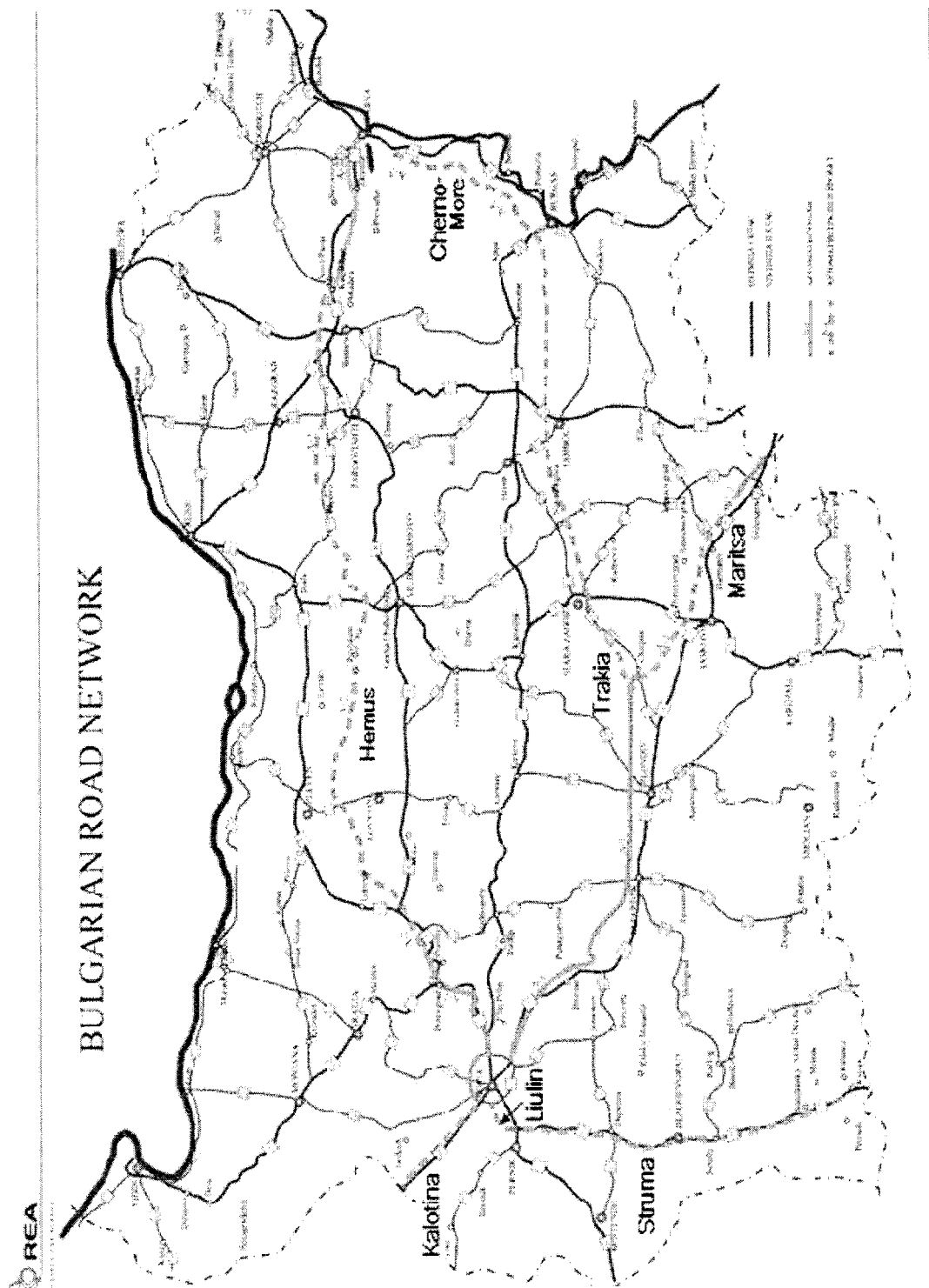
Sources of Financing – Scenario IV ¹ (2004 Euro)

Sources of Finance	2007	2008	2009	2010	2011	2012	2013	Total	Avg.	(%)
Vignette	86	86	86	86	86	86	86	602	86	
Excise tax (historical budget allocation)	114	114	114	114	114	114	114	798	114	
Total user charges	200	200	200	200	200	200	200	1,400	200	31 %
ISPA	37	37	0	0	0	0	0	74	11	
PHARE	7	7	7	0	0	0	0	20	3	
Cohesion Fund (for roads)	26	59	82	93	96	98	100	555	79	
ERDF - Roads	6	11	13	36	35	40	57	198	28	
Total EU grants	77	114	102	129	131	138	157	847	121	19 %
% GDP	0.4 %	0.5 %	0.4 %	0.6 %	0.5 %	0.5 %	0.6 %		0.5 %	
Total private sector financing	0	0	0	0	0	0	0	0	0	0 %
% GDP	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %		0.0 %	
Total Available Funds	277	314	302	331	333	338	355	2,249	321	50 %
% GDP	1.3 %	1.4 %	1.3 %	1.4 %	1.3 %	1.3 %	1.3 %		1.3 %	
Total Roads Expenditures	504	687	628	577	719	713	640	4,467	638	100 %
Financing Gap	227	373	326	245	386	375	285	2,218	317	50 %
	1.1 %	1.7 %	1.4 %	1.0 %	1.5 %	1.4 %	1.0 %		1.3 %	
Co-financing ISPA	12	12	0	0	0	0	0	25	4	
Co-financing PHARE	3	3	3	0	0	0	0	8	1	
Co-financing Cohesion Fund	5	11	15	18	18	18	18	104	15	
Co-financing ERDF	2	3	4	11	11	13	18	62	9	
Total implied co-financing needed for EU Grants	22	29	22	29	29	31	36	199	28	8 %
% GDP	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %	0.1 %		0.1 %	

Note:

1. Both Trakia and Struma motorways publicly financed with Struma scheduled for 2010 - 2013

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6. ENVIRONMENTAL ISSUES

6.1 The Government of Bulgaria concluded the accession negotiations for the environmental chapter with the European Union in 2004. During the negotiations the government made several commitments to meet a number of directives, the most demanding of which in budgetary terms are those relating to water and wastewater and to solid waste⁵⁷. The programs have been worked out at varying levels of detail, and in all cases there are still some open questions.

6.2 This section reviews the current government plans relating to the directives dealing with water and solid waste, as well as all anticipated environmental expenditures from state sources, paying special attention to the budgetary implications of these plans. The realism of these plans, in particular in terms of the projected financing, will be assessed.

I. EU DIRECTIVES RELATED TO WATER SUPPLY, URBAN WASTEWATER AND SEWERAGE

6.3 Under various directives Bulgaria is obliged to: (a) upgrade 42 water supply treatment plants and 3,500 pumping stations in the country, and eliminate water shortages in the supply system; (b) build an additional 16,000 km. of sewers to connect 2.4 million people to a sewerage network, to increase the number of municipalities where waste water is treated and to upgrade the treatment of waste water in other municipalities; and (c) rehabilitate the water distribution network, which has not been adequately maintained over the last 15 years. The total cost of the agreed program under these directives has been estimated by the Ministry of Regional Development and Public Works (MRDPW) to be €6.9 billion over the period to 2014. Details of the figures by area of investment are given in Table 6.1.⁵⁸

6.4 Total funds needed from public sources for the water programs is projected to rise from €88 million euros in 2005 to €192 million in 2007, and €295 million in 2008. Table 6.1 shows the expected sources for the water-related directives for Bulgaria as estimated by MRDPW. The proposed demands on direct budgetary resources go up from about €8 million in 2005 to a maximum of €25 million in 2011. Since municipalities are currently subsidized from the central budget, the municipal contribution shown in Table 6.1 will also have to come from the budget, at least for the next two to three years. Finally, the Enterprise for Management of Environmental Protection is a self financed fund that is part of the consolidated budget. These would be substantial public spending increases that need to be analyzed carefully in terms of the available domestic resources. In new EU member states, in addition to special

⁵⁷ The water-related directives of importance are 91/271 on urban wastewater and sewerage; 98/83 on drinking water quality, 91/676 on water protection against agricultural nitrates, and 76/464 on unloading of hazardous substances into water. During negotiations with the EU, commitments were made only on Directives 91/271 and 76/464, and the latter will not create financing burdens until after 2014. In due course the Water Framework Directive will have major implications for local government expenditures, but the new EU member states have a long period over which to comply and the short term impacts are not likely to be important. The significant solid waste related directives are 99/31 on the storage of solid waste, and 94/62 on packaging and wastes from packaging. Experience from new EU member states shows that while both these have budgetary implications, it is the storage of solid waste that has the biggest impact.

⁵⁸ These figures will be updated in an operational environmental program that should be ready by end 2005. Data from this program was not available. Also, there are some differences between MRDPW and the Ministry of Environment and Water (MOEW) figures. MOEW's implementation program for sewerage and wastewater puts the investment cost at about €2.2 billion, whereas the MRDPW estimates are close €2.9 billion. Other differences arise in the proposed financing of the programs (see below). In both cases expenditures are spread out over a period extending to 2014. The detailed calculations provided below are based on MRDPW figures.

EU funds, the state budget, municipality budgets, and environmental funds, the private sector has been a source of environmental investment finance.

6.5 These figures are based on what the government considers realistic assumptions about the availability of EU funds, yet there remains considerable uncertainty about such funds, access to which requires well-prepared documents for individual projects. Actual disbursement from ISPA funds, for example, was short of expectations in 2002 and 2003, because of a lack of projects that had been prepared in accordance with EU requirements. Capacity for project preparation is being increased but further enhancements are needed, especially at the municipal level, although there is some disagreement between Ministries as to the adequacy of human resources available for the preparation and implementation of the program.⁵⁹

6.6 The plans include annual financial contributions from the municipalities of about 30 percent over the planning period (2005-2014), but in the short term (2006-2008) these funds would likely need to come from the state budget. Present average tariffs for water and wastewater services cover at most the operating and maintenance costs, and do not leave any surplus for investment to upgrade facilities. In 2002 the average tariff for water and wastewater was around €0.51/M³ (about one BGN/ M³). The MRDPW report estimates that an increase in tariffs of about 10 percent would generate an additional BGN 50-70 million for investment. The required increases for financing of the plan in Table 6.1, however, are much higher. A recent Bank study has estimated that for a program with 25 percent of investment for both water supply and wastewater coming from private sources, tariffs would have to rise to €0.67/M³ in 2007, going up to €2.28/M³ in 2014⁶⁰. Although such tariffs are estimated to be less than four percent of average household income in respective years, they could nevertheless place hardship on households in the lower quintiles of the income distribution. To avoid this, a social protection scheme that included support for water charges would have to be in place. At present there is, however, no work ongoing to set up such a scheme. Also, the political appetite for increasing tariffs at the present time is low, making the proposed contribution from the operators unrealistic. For the purposes of the short term analysis—that is, 2006-2008—it is reasonable to assume that almost all the municipal contribution identified in Table 6.1 would have to come from the state budget.

⁵⁹ MOEW and the EU office in Sofia took the view that the implementation of adequately prepared programs was likely to be a constraint, especially for 2009-2010, preparation for which should start now. MRDPW, on the other hand is more optimistic, arguing that the technical assistance funds available should be sufficient to increase capacity in time for the implementation of the program laid out in Table 1. Both agree that the greatest shortage in capacity is in the municipalities, especially the small ones.

⁶⁰ Bulgaria: Financing the Water and Wastewater Sector, World Bank South Central Europe Country Unit, Europe and Central Asia Region, February, 2005. The figures in Table 6.1 for municipal financing do not show the impacts of any tariff increases. They simply state what funds will come from municipal channels. Most of those funds, however, will most likely be transfers from the budget. The role of tariff increases in filling this funding gap needs further analysis.

Table 6.1: Agreed Program for the Implementation of All Directives Related to Water Supply and Wastewater

	Unit	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Total
Population	Number	7,740,829	7,696,568	7,652,307	7,608,045	7,563,784	7,519,522	7,475,261	7,437,373	7,399,484	7,361,596	7,323,708	-
GDP	mln. BGN	38,008	41,614	45,615	49,540	53,584	55,714	59,893	63,486	67,295	71,333	75,613	-
GDP	mln. Euro	19,433	21,276	23,322	25,329	27,397	28,486	30,623	32,460	34,407	36,472	38,660	-
GDP growth	%	5.56	5.80	5.50	5.90	5.90	4.50	4.50	3.50	3.50	3.50	3.50	-
Inflation	%	6.1	4.7	5.8	3.1	2.8	3.0	3.0	2.5	2.5	2.5	2.5	-
Real income annual increase	%	4.53	3.95	3.94	3.66	3.35	4.91	4.76	4.42	3.88	3.86	3.78	-
Water Supply	mln. Euro	3	4	12	98	152	157	208	180	155	132	38	1,139
Water Efficiency	mln. Euro	6	6	33	266	390	446	540	420	352	272	101	2,832
Sewerage and Wasteshed	mln. Euro	51	102	210	310	329	377	402	350	378	287	166	2,962
Total	mln. Euro	60	112	255	674	871	980	1150	950	885	691	305	6,933
Total Expenditure as % of GDP		0.31 %	0.52 %	1.11 %	2.73 %	3.29 %	3.44 %	3.76 %	2.93 %	2.57 %	1.89 %	0.79 %	-
Exchange rate													
Financing of which:													
	Leva/Euro	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	1.95583	
	Unit	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Total
ISPA-Cohesion Funds 1/	mln. Euro	5	12	65	427	506	560	670	445	420	281	0	3,391
IFIs	mln. Euro	0	10	30	55	70	70	80	90	50	50	0	505
State Budget	mln. Euro	5	8	10	12	15	20	20	25	25	20	15	175
Enterprise for Management of Environmental Protection	mln. Euro	20	50	70	80	80	80	80	90	90	90	90	820
Municipalities	mln. Euro	30	30	80	100	200	250	300	300	300	250	200	2,040
Private	mln. Euro	0	0	0	0	0	0	0	0	0	0	0	0
Total	mln. Euro	60	110	255	674	871	980	1150	950	885	691	305	6,931
External Funds as % of Total		8 %	20 %	37 %	72 %	66 %	64 %	65 %	56 %	53 %	48 %	0 %	56 %
Direct State Budget as % of Total		8 %	7 %	4 %	2 %	2 %	2 %	2 %	3 %	3 %	3 %	5 %	3 %
Total State Exp.as % of Total		92 %	80 %	63 %	28 %	34 %	36 %	35 %	44 %	47 %	52 %	100 %	44 %

6.7 Government's plans leave open possible involvement of the private sector, and there is some interest from potential investors in this sector in Bulgaria. Sofia already has a concession involving a public-private partnership operating the water and wastewater system for the city (although there are reported to be some difficulties in implementing the investment program and coordinating it with MOEW). Discussions are ongoing with prospective Dutch and German partners and a Dutch-Bulgarian Public Private Partnership revolving fund has been set up with €280,000. In spite of these developments, however, a significant role for the private sector is unlikely in the short to medium term for the following reasons:

6.8 The government still has to address the issue of the ownership of municipal assets. Of the 50 or so water companies, 13 are entirely owned by the state, 20 are owned by the municipalities, 16 are jointly owned, and one is jointly owned by a municipality and private investor (Sofia is excluded from this list as it has special status of a PPP). Where ownership is shared with the municipalities, what assets belong to whom is unclear. The question of what will be transferred to any new operators/owners of the water companies needs to be resolved. In September 2005, a Water Management Law was submitted for coordination to the Ministry of Finance. The draft envisages changes to the ownership of assets that will address the questions raised here about the transfer of some of the assets, and to clarify questions of ownership, but these are still waiting Parliament approval.

6.9 As noted above any private sector involvement would require significant increases in tariffs.⁶¹ The body responsible for the regulation of water charges has been established under the Water Regulation Services Act, its administrative structure is being set up, and it should be operational by 2006. It mirrors in many respects the energy sector regulator with whom it shares an overall regulatory structure. With this body in place some increase in tariffs is inevitable, notwithstanding the low political will for such increases. Furthermore, if the energy regulator is any guide, the increases could be substantial. This suggests that municipal financing of water-related investments may be possible some time in the future, possibly under a number of PPPs. However, given the time lag between setting up the institutions, announcing any increases in tariffs, and getting to the stage at which tariffs meet private sector expectations, serious private sector involvement for water and sewerage will take some time to be realized.

6.10 To attract the private sector, there would also need to be clarity about the responsibility for management of the sector between MRDPW and MOEW. At present the division of responsibilities is not clear, and as a result there is lack of clarity of policy and actions.

6.11 In view of the above, a major role for the private sector in financing the implementation of the urban wastewater directive is unlikely for at least 2-3 years though private sector involvement would likely increase cost efficiency. To put this in perspective, it should be noted that in none of the new EU member states has the private sector played a major role in the financing of the urban wastewater directive.

6.12 The other important source of finance is the state agency Initiative for the Management of Environmental Protection, (also referred to as the Enterprise for the Management of Environment Protection Related Activities or EMPERA), which is the former Environment Fund that is now part of the consolidated national budget. Since it still derives revenues from earmarked sources, such as environmental charges and fees for special services provided, expenditures from the fund are not fully

⁶¹ The government is also hopeful of increasing its tax base as a result of the increase in property values in Bulgaria. For example, Varna has seen an increase in property tax receipts from BGN 7-8 million in 2002 ago to BGN 22 million in 2005. Such taxes will, however, apply to few municipalities and demands on those revenues in terms of improved services will come from many quarters.

comparable to those from the budget in general. MRDPW shows a significant role for this source in the financing plan—up from €50 million in 2005 to €80 million in 2007-2010, and to €90 million in 2011-2014. This is in contrast with the estimates of MOEW, which show allocations declining from about €20 million in 2005, to €2 million in 2008, and to zero by 2011. Some further reconciliation of these two set of estimates is required. According to the Initiative Fund, amounts available for water sector investments in 2005 would be BGN 22 million⁶² (€11.3 million), and unlikely to exceed BGN 30 million (€15.4 million) in 2006. Beyond that increases in expenditures depend on the scope for increasing revenues from abstraction fees and water usage permits. In the view of the Fund, the maximum revenue from these sources is about BGN 40 million (€20.5 million). All these amounts fall short of the planned financing from this source from 2005 and beyond. It should be noted that the new EU member states have relied quite heavily on environmental funds, and the potential for these funds to provide finance in the short to medium term is considerable.

6.13 The implementation plan assigns a small role for International Financial Institutions as well. The plan envisages about €30 million IFI financing in 2006, going up to €70 million in 2007-2008, €90 million in 2011, and declining to €50 million in 2012-2013. Most of these funds are slated to come from EBRD and EIB. The EBRD credit for €11 million would be for Burgas water and waste water company and to cover 14 municipalities. The credit would not require a sovereign guarantee, which MRDPW and MoF consider important, given the restrictions on sovereign guarantees imposed by the Currency Board. Another five municipalities are planned to be co-financed in the same way, with total credits of €60 million. EIB would finance another €60 million, but with sovereign guarantees. The realization of these plans may be problematic, however, as the credits would imply increased tariffs, which remain politically difficult and also problematic to collect.

6.14 The demand being made on the state budget from these directives is comparable to experience of new EU member states. For example, in Poland total investment of about €14 billion euros is being financed roughly as follows (with Bulgaria's figures in parenthesis): local governments 31 percent (*Bulgaria 29 percent*), environmental funds 25 percent (*Bulgaria 12 percent*), EU funds 19 percent (*Bulgaria 49 percent*); and state budget 3 percent. The much higher dependence on EU funds in Bulgaria than in Poland suggests that perhaps Bulgaria is being optimistic about how much it can raise from these sources, and that a more careful reflection of these numbers is warranted.

6.15 In sum, the figures above indicate that, while direct budgetary pressures from the water related directives are not large, the overall dependence on public resources will be substantial, in particular from municipalities. It is, however, unlikely that municipalities will generate the surpluses required to meet the targets set for them. Second, expectations about EU forthcoming funds are likely to be too optimistic; apart from the fact that they need co-financing, which is uncertain, there are also likely to be delays in planning and implementation of the projects. Third, the amounts available from the former environment fund are likely to be less than the amounts given in Table 6.1. All these factors point to the need for a more realistic appraisal of plans for the water supply, urban wastewater and sewerage sub-sector.

II. EU DIRECTIVE 99/31/EC ON LANDFILL OF WASTE

6.16 Bulgaria has agreed a program of investment in upgrading some landfills, closing of dangerous ones, and establishing several new ones, including special sites for the disposal of hazardous waste. The program involves a total of 54 projects, covering 264 municipalities, and entails an investment of €478 million in 2005-2015. Details of the financing plan are given in Table 6.2.

⁶² An exchange rate of BGN 1.95583 to the euro has been used throughout.

6.17 The state budget is estimated to provide about €20-25 million annually up to 2007, and €13-16 million a year in 2008-2015 to finance the program. Municipalities are expected to finance €4-6 million a year to 2007, and €9-10 million a year in 2008 and 2009. The state environment fund would be expected to provide a relatively small amount (one to two million euros per annum), as well as the private sector.

6.18 The landfill costs seem to be rather low. Poland, for example, has a program of €8.3 billion over a similar period, which costs 17 times as much, while Poland's population is only 4.5 times larger than that of Bulgaria.

6.19 Some of the observations made on planned financing of the urban wastewater directive apply also to the planned financing of landfills:

- a) Proper and timely project preparation is expected to be a constraint on the access to EU funds. Further assistance to build capacity for project preparation would be helpful in ensuring that the program is implemented according to schedule. There is also a long time lag in taking projects from the planning stage to implementation--it takes about four years to obtain the EU funds, and site selection for new landfills is difficult in Bulgaria.
- b) The program envisages some contribution, albeit modest, from municipalities towards the financing of the program. Although present the legislation does allow the recovery of construction costs of landfills from waste fees, in practice tariff levels are likely to be too low to support the investment needs in this sector. Other mechanisms for addressing the problem may hold more promise, such as more effective use of the waste fees that are collected, and an increase in selective/separate waste collection activities, which should reduce the amount of waste generated.
- c) If charges were raised for waste services, and other reforms in asset ownership were made as outlined in the previous section on urban wastewater, there would be scope for higher private sector financing than assumed (at least in the later years). Initiatives for building waste incinerators are underway in Sofia and Plovdiv, and at an early stage in Varna. Other East European countries that have joined the EU have had a more substantial role for public-private partnerships in the waste sector than planned in Bulgaria.⁶³

⁶³ The role of municipalities to finance the expansion of waste services has been substantial in new EU member states. In Poland, for example, while local governments plan to raise about 10 percent of the required finance, private-public partnerships are raising 18 percent. In the Czech Republic about 90 percent of the resources are being raised from the municipalities own resources and the rest from private sources.

Table 6.2: Agreed Program for the Implementation of Directive 99/31/EC Related to Landfill of Waste

	Unit	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	Total
Population	number	7,785,091	7,740,829	7,696,568	7,652,307	7,608,045	7,563,784	7,519,522	7,475,261	7,437,373	7,399,484	7,361,596	7,323,708	7,285,820	-
GDP	mln. BGN	34,547	38,008	41,614	45,615	49,540	53,584	55,714	59,893	63,486	67,295	71,333	75,613	80,150	-
GDP	mln. Euro	17,663	19,433	21,276	23,322	25,329	27,397	28,486	30,623	32,460	34,407	36,472	38,660	40,980	-
GDP growth	%	4.45	5.56	5.80	5.50	5.90	5.90	4.50	4.50	3.50	3.50	3.50	3.50	3.50	-
Inflation	%	2.3	6.1	4.7	5.8	3.1	2.8	3.0	3.0	2.5	2.5	2.5	2.5	2.5	-
Real income annual increase		5.36	4.53	3.95	3.94	3.66	3.35	4.91	4.76	4.42	3.88	3.86	3.78	4.09	-
Financing	mln. Euro														0
Proposed Expenditure	mln. Euro	23.8	45.6	54.8	54.4	53.4	34.3	35.0	28.5	28.8	31.6	33.1	28.8	25.4	477.5
<i>Of which:</i>															
ISPA-Cohesion Funds I/	mln. Euro	13.3	25.1	28.7	23.2	19.7	8.2	8.8	7.1	8.4	9.6	10.2	7.3	5.8	175.4
Other Donors	mln. Euro														
State Budget Municipal Budgets	mln. Euro	8.6	15.2	20.1	23.8	24.8	13.5	14.3	12.9	14.4	15.8	16.6	14.0	12.7	206.7
Enterprise for Management of Environmental Protection	mln. Euro	0.4	3.1	3.6	4.9	6.2	9.8	8.9	5.3	2.6	2.6	2.6	3.5	2.6	56.1
Private	mln. Euro	0.0	0.6	0.7	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.6	1.8	13.1
Total	mln. Euro	23.8	45.6	54.8	54.4	53.4	34.3	35.0	28.5	28.8	31.6	33.1	28.8	25.4	477.5
External Funds as % of Total		56 %	55 %	52 %	43 %	37 %	24 %	25 %	25 %	29 %	30 %	31 %	25 %	23 %	37 %
State Budget as % of Total		36 %	33 %	37 %	44 %	46 %	39 %	41 %	45 %	50 %	50 %	50 %	49 %	50 %	43 %
Total Expenditure as % of GDP		0.13 %	0.23 %	0.26 %	0.24 %	0.22 %	0.13 %	0.12 %	0.09 %	0.09 %	0.09 %	0.09 %	0.07 %	0.06 %	-

III. FITTING THE INVESTMENT PLANS FOR WATER, SEWERAGE AND LANDFILL INTO THE CONSOLIDATED BUDGET

6.20 The projected public financing for water, waste water, land fill significantly exceeds the projected capital budget envelope for environmental investments. Table 6.3 provides a summary of the planned capital expenditures in 2005-2008 on *all* aspects of environmental protection, including water and waste management. It covers financing from EU grants and the domestic sources, including the Environment Fund. The figures are compared with the planned financing of water and waste water and landfill from public sources, as highlighted in Tables 6.1 and 6.2. Table 6.3 shows that the plans for water and waste investment significantly exceed the expected budget allocations. In particular from 2006 onwards the planned spending on these two areas exceeds the projected budget allocation for all environmental related capital outlays by a considerable and an increasing amount.

Table 6.3: All Environmental Public Expenditure

Million BGN	2004	2005	2006	2007	2008
Capital Exp in Consolidated Budget	140	297	321	369	424
<i>Financing for Water and Waste</i>					
External Funds Inc. EU	59	99	231	199	1142
State budget	39	55	66	72	56
Environment Fund	42	101	140	160	160
Local Government	65	66	166	208	410
Total Excluding Local Government	140	255	437	431	1357
Total Including Local Government	205	321	603	638	1767

Source: Government of Bulgaria, Ministry of Finance and MOEW

IV. SETTING THE ENVIRONMENTAL FINANCING NEEDS INCLUDING WASTE AND WATER IN THE BROADER CONTEXT OF THE MTFF

6.21 The Bulgarian medium term fiscal framework (MTFF) allocates about 2.6-2.8 percent of the consolidated budget for protection of the environment (see Table 6.4). This is about one percent of GDP, which is comparable to other CEE countries and EU15. Total spending on the environment in EU15 is about two percent of GDP, but about half comes from the private sector. In the new EU member states a much smaller amount comes from the private sector.

6.22 Table 6.4 also shows the changing composition of public environmental expenditure in Bulgaria: current expenditures are expected to decline substantially, while capital expenditures increase. The share of capital expenditure to protect the environment is projected to increase from nine percent in 2004 to 16 percent in 2008. This is a welcome trend; the decline in current expenditure reflects a decreased subsidy for environment-related services, while the increase in the investment share shows the support for the initial, up-front program of upgrading environmental services in the country.

Table 6.4: Environmental Public Expenditure

Million BGN	2004	2005	2006	2007	2008
Capital	140	297	321	369	424
Current	249	120	121	125	128
Total	388	417	442	494	552
<i>As percent of Consolidated Government Expenditure</i>					
Capital (%)	0.9	1.9	1.9	2.0	2.2
Current (%)	1.7	0.7	0.7	0.7	0.6
Total (%)	2.6	2.6	2.6	2.7	2.8

V. CONCLUSIONS

6.23 The meeting of the environmental part of the *acquis communautaire* will increase demands on budgetary resources in Bulgaria over the next decade, and total spending on environmental protection is expected to increase by about 50 percent in 2004-2008 (while environment related capital expenditures are expected to increase three-fold, the current expenditures are expected to decline). At the same time, adequate waste water and waste management will help to facilitate growth by making rural areas (80 percent of the country's territory) attractive for business and local population, and the system construction might also spur new entrepreneurship.

6.24 The spending plans for the water and waste sectors are, however, inconsistent with the overall budget envelope for environment, and significantly exceed the projected medium-term budget allocations. Reconciliation of plans is required to stay within the available budget envelope.

6.25 Also, there are several problems with the details of water and wastewater sector financing plans that would need to be addressed. Perhaps the most serious is the unrealistic nature of assumptions regarding municipal financing. These are predicated on increases in tariffs that are overly optimistic. In case these tariff increases are not realized, programs will either fall behind their schedule, or the state will have to make up the shortfall. This would cause expenditure in this category of the consolidated budget to go over its planned limit. Also, the ownership of water utility companies' assets would need to be clarified to entice private sector participation in the sector.

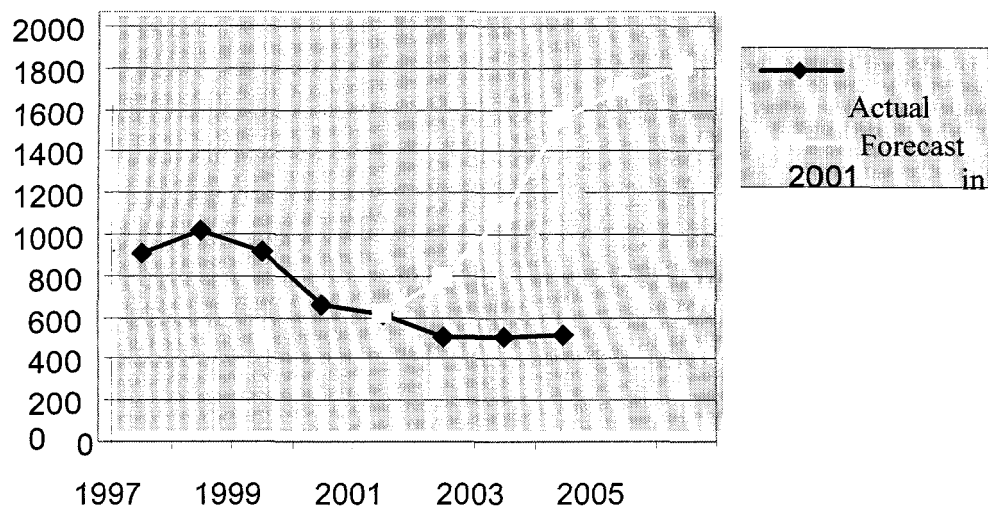
6.26 It is also quite likely that the level of funds anticipated from the EU will not be forthcoming in full. Although MOEW has increased its pipeline of prepared projects for such funding, and these should be sufficient to absorb the funds assigned for 2007-2008, the pipeline will have to be substantially further increased to absorb future allocations. As a result, the commitment made by the government to the EU over the medium term may not be realized, and it would be critical to prioritize the undertaking committed investments to ensure the most important ones get completed. Further, MRDPW, and MOEW estimates of financing flows differ, and would need to be reconciled. This may be due to different dates at which they were prepared.

6.27 In sum, though undesirable, one can expect some delay in Bulgaria in the implementation of environmental programs agreed with the EU. However, the Cohesion Group of countries (Spain, Portugal and Greece), and the recent new EU member states have also implemented their environmental programs on a delayed schedule, owing to large and unaffordable financing requirements. Figure 1 shows the actual and anticipated expenditures for Poland, from 2001 onwards. The sharp divergence between actual

and anticipated spending is notable, and is currently being investigated⁶⁴. Although Bulgaria hopefully will not experience such a large difference between actual and planned expenditures, some downward adjustment in the former may be expected.

Figure 6.1: Actual vs. Forecasted Environment Expenditures in 2001

(Million Zloty–2001 Prices)



6.28 The key recommendations for the government in the area of environment are: (i) the environmental plans would need to be scaled down to reflect reduced availability of municipal financing, and ensure their consistency with the projected budget envelope; (ii) there is a need to increase capacity for timely preparation of plans and documents to ensure access to EU funds; and (iii) to facilitate private sector participation of water and waste services delivery, there is a need to increase tariffs for these services. That would require changes in the legal framework, setting up of the regulatory framework, and ensuring adequate social protection for vulnerable groups

⁶⁴ The graph is based on work in progress.

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